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Computation of Multi-Particle Phase-Space Integrals and Their Applications in High-Energy Physics

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Resumo

A crescente precisão alcançada pelas experiências realizadas em colisores de partículas exige previsões teóricas com um grau de rigor comparável. Uma parte substancial destes processos é governada pela interação forte, descrita pela Cromodinâmica Quântica (QCD). A energias elevadas, esta teoria pode ser tratada perturbativamente, organizando as previsões físicas como uma expansão em potências da constante de acoplamento forte, α_s. Contudo, aumentar a precisão de um cálculo não corresponde apenas a acrescentar diagramas de Feynman: exige controlar uma estrutura complexa de contribuições de emissão real e virtual, bem como as divergências que surgem em cada uma delas.

As divergências infravermelhas constituem uma das principais dificuldades dos cálculos perturbativos em QCD. As contribuições de emissão real tornam-se singulares quando um partão é emitido com energia muito baixa, o limite mole, ou quando dois partões se tornam colineares. Paralelamente, as amplitudes com laços contêm polos infravermelhos associados às mesmas regiões cinemáticas. Embora estas divergências se cancelem em observáveis físicos suficientemente inclusivos, as componentes individuais do cálculo não são finitas. É, por isso, necessário reorganizar a previsão teórica de modo a isolar as singularidades e a efetuar o seu cancelamento de forma analítica e controlada.

O formalismo de subtração por antenas fornece um enquadramento adequado para esse objetivo. Neste método, a radiação não resolvida é descrita entre dois partões duros ligados por cor, designados por radiadores. As funções de antena reproduzem os limites moles e colineares relevantes de forma local no espaço de fase. Quando são subtraídas às contribuições reais, eliminam as singularidades antes da integração numérica. Depois de integradas analiticamente sobre o espaço de fase da radiação não resolvida, originam polos em regularização dimensional que cancelam os polos das contribuições virtuais. O cancelamento destas singularidades permite que cada contribuição seja integrada numericamente em quatro dimensões.

Esta dissertação apresenta o desenvolvimento de *AntCalc*, um pacote simbólico para a construção e integração automatizadas de funções de antena em QCD. O projeto procura automatizar etapas que requerem implementações específicas para cada antena, reunindo num mesmo fluxo as convenções de normalização, construção e integração. O trabalho centra-se nas antenas quark–antiquark sem massa, na configuração cinemática final–final, até à ordem next-to-next-to-leading order (NNLO). Para além deste objetivo principal, é estudada uma extensão exploratória ao regime massivo através da implementação da antena A_3^0.

A base teórica da dissertação inclui uma introdução à QCD perturbativa, aos campos de quarks e gluões e à construção de amplitudes a partir de diagramas de Feynman. São discutidas as divergências ultravioleta, tratadas por renormalização, e as divergências infravermelhas, que exigem a combinação entre várias contribuições perturbativas. Utiliza-se regularização dimensional, na qual o número de dimensões do espaço-tempo é tomado como $d = 4 - 2\epsilon$. Neste enquadramento, as divergências manifestam-se como polos em ϵ, e as quantidades integradas são expressas por séries de Laurent. O cancelamento dos polos pode, assim, ser acompanhado explicitamente ao combinar as diferentes contribuições para uma

quantidade física.

É também apresentada a estrutura das funções de antena e a sua decomposição em cor. As antenas são obtidas a partir de amplitudes físicas normalizadas e organizadas de acordo com os radiadores e as partículas emitidas. A fatorização do espaço de fase é essencial neste processo, pois permite separar o espaço de fase reduzido, associado aos radiadores após a emissão não resolvida ser removida, do espaço de fase correspondente à própria emissão. A integração pode então ser efetuada apenas sobre o espaço de fase da radiação não resolvida, independentemente do processo duro em que a antena venha a ser inserida. A dissertação introduz ainda os objetos $\mathcal{T}$, que combinam antenas integradas, fatores de cor e normalizações de forma apropriada para a construção de observáveis inclusivos.

A integração das funções de antena é uma das etapas mais exigentes do cálculo. As integrais de espaço de fase são reescritas como integrais de laço através da técnica de reverse unitarity. Em seguida, são reduzidas a um conjunto finito de integrais mestres usando identidades de integração por partes (integration by parts, IBP) e o algoritmo de Laporta. Em casos adequados, recorre-se também à redução de Passarino–Veltman. Uma vez identificados os integrais mestres e substituídas as respetivas expressões conhecidas, obtém-se a forma integrada da antena como uma expansão em ϵ. O resultado da redução fica, deste modo, expresso num conjunto limitado de integrais mestres, cujas expansões em ϵ determinam as partes singular e finita da antena integrada.

O pacote *AntCalc* foi implementado no ambiente *Wolfram Mathematica* e integra ferramentas especializadas para as diferentes fases do cálculo. FeynArts é utilizado para gerar os diagramas de Feynman relevantes, enquanto *FeynCalc* e *FeynHelpers* apoiam a construção e a manipulação das amplitudes. A redução IBP é realizada com *LiteRed2*. A arquitetura do pacote é orientada por perfis, isto é, por descrições estruturadas que especificam o processo, as amplitudes, as famílias de integrais, os fatores de normalização e os integrais mestres necessários. A informação específica de cada antena fica, assim, concentrada no respetivo perfil, enquanto as rotinas de construção e integração permanecem comuns aos diferentes casos.

O fluxo de trabalho divide-se em duas fases principais: construção e integração. Na fase de construção, o utilizador fornece a descrição do processo e o pacote gera os diagramas e amplitudes correspondentes. São depois efetuadas as manipulações algébricas necessárias para extrair a função de antena não integrada, ou as suas componentes de cor quando aplicável. Na fase de integração, essa expressão é convertida para uma forma apropriada à redução, associada à família de integrais correta e reduzida aos integrais mestres. Finalmente, a substituição dos resultados dos integrais mestres e a expansão em torno de $\epsilon = 0$ produzem a antena integrada. Ao longo deste fluxo são efetuados testes de consistência e apresentados diagnósticos do cálculo.

Como exemplo detalhado, é apresentada a derivação da antena A_3^0, que descreve a emissão de um gluão entre um quark e um antiquark sem massa ao nível de árvore. Este caso ilustra todo o percurso, desde a geração da amplitude até à expressão integrada. A validação local da antena é efetuada através dos seus limites infravermelhos. No limite mole, verifica-se que a expressão reproduz o fator eikonal esperado; nos limites colineares, confirma-se a recuperação da função de separação de Altarelli–Parisi apropriada. A concordância nestes limites constitui uma verificação local da expressão obtida, independente da comparação posterior da antena integrada com resultados da literatura.

Para além de A_3^0, *AntCalc* é aplicado a um conjunto de antenas sem massa até NNLO. São obtidas contribuições de dois, três e quatro partões, incluindo os conjuntos A_2^2, A_3^1 e A_4^0, bem como antenas com outras estruturas de cor, tais como B_4^0 e C_4^0. Os resultados não integrados e integrados são comparados com resultados estabelecidos na literatura, verificando-se concordância. A concordância com os resultados publicados verifica, em simultâneo, as convenções de regularização dimensional e normalização, bem como o tratamento da estrutura de cor e do espaço de fase adotado na implementação.

A validação global é realizada através do cálculo do rácio hadrónico sem massa, ou R-ratio. Este observável relaciona a secção eficaz de produção de hadrões em aniquilação eletrão–positrão com a secção eficaz de produção de pares de muões. Por ser inclusivo, constitui um teste rigoroso da consistência do formalismo: os polos infravermelhos provenientes das contribuições reais e virtuais devem cancelar exatamente. A montagem do R-ratio a partir dos objetos integrados produzidos por *AntCalc* confirma esse cancelamento e reproduz o resultado finito conhecido até NNLO. A reprodução do R-ratio complementa, deste modo, os testes locais das antenas, verificando o cancelamento dos polos quando as diferentes contribuições são combinadas num observável físico.

A dissertação explora ainda a extensão ao caso massivo pela construção e integração da antena massiva A_3^0. A presença de uma massa altera a cinemática, a estrutura do espaço de fase e os limites infravermelhos, eliminando, em particular, algumas singularidades colineares existentes no caso sem massa. Este cálculo fornece uma primeira implementação do fluxo massivo em *AntCalc* e identifica as alterações necessárias à construção e integração de futuras antenas massivas.

A principal limitação da versão atual do pacote é o custo computacional da execução inteiramente em Mathematica. Embora este ambiente seja particularmente útil para desenvolvimento simbólico, alguns cálculos tornam-se lentos e exigentes em memória. O trabalho futuro deverá, por isso, avançar para uma arquitetura modular e multilinguagem. Prevê-se a utilização de Python como orquestrador, FORM para manipulação algébrica eficiente e Kira para reduções IBP mais robustas, mantendo as ferramentas atuais nas tarefas em que são mais adequadas. Entre as extensões previstas encontram-se outras antenas massivas, antenas com radiadores quark–gluão e gluão–gluão, configurações inicial–final e inicial–inicial e, a mais longo prazo, cálculos a ordens perturbativas superiores, incluindo $\mathrm{N^3LO}$.

Em síntese, o trabalho desenvolvido mostra que as etapas de construção, redução e integração das funções de antena consideradas podem ser reunidas num único procedimento automatizado. O pacote reproduz resultados conhecidos para antenas sem massa até NNLO, valida os limites infravermelhos locais e recupera corretamente o R-ratio hadrónico. A implementação obtida pode servir de base à inclusão de novas classes de antenas sem exigir que todo o procedimento de construção e integração seja novamente implementado de raíz.

Palavras chave: QCD; Subtração por Antenas; NNLO; Singularidades Infravermelhas; *Mathematica*

Abstract

High-precision collider measurements require theoretical predictions that are accurate to their same level. In perturbative Quantum Chromodynamics (QCD), these predictions are organised as an expansion of the strong coupling constant, α_s. At higher orders, the calculation is complicated by infrared singularities associated with soft and collinear parton radiation. Although these singularities cancel in physical observables, the real-emission and virtual contributions are separately divergent and therefore cannot be handled numerically without first isolating and cancelling them.

Antenna subtraction provides a systematic way to do this. It uses universal antenna functions to describe unresolved radiation between colour-connected hard partons. These functions are subtracted locally from the real-emission contribution and then integrated analytically in dimensional regularisation, making the cancellation of infrared poles against the virtual terms explicit.

This thesis develops *AntCalc*, a symbolic framework in Wolfram Mathematica for constructing and integrating QCD antenna functions. The package combines *FeynArts*, *FeynCalc*, *FeynHelpers*, and *LiteRed2*, while keeping process-dependent information separate from the general calculation workflow. It generates the required amplitudes, constructs unintegrated antennae or colour components, maps the corresponding phase-space integrals onto integral families, and reduces them to master integrals through integration-by-parts identities.

The implementation reproduces the massless final-final quark-antiquark antennae through next-to-next-to-leading order. The tree-level three-parton antenna A_3^0 is used as a detailed example and is checked in its soft and collinear limits. The integrated results agree with the literature, and the full NNLO massless hadronic R-ratio is reconstructed as an additional validation. A preliminary massive A_3^0 implementation shows how the same workflow can be extended beyond the massless case. Future work will extend AntCalc to further massive antennae, additional partonic configurations, and a more modular multi-language architecture.

Keywords: QCD; Antenna Subtraction; NNLO; Infrared Singularities; *Mathematica*

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List of Abbreviations and Symbols

Abbreviations and Acronyms

Term	Meaning
QCD	Quantum Chromodynamics
LO / NLO / NNLO	Leading / Next-to-Leading / Next-to-Next-to-Leading Order
IR / UV	Infrared / Ultraviolet
MI	Master Integral(s)
RR / RV / VV	double-Real, Real-Virtual and double-Virtual NNLO contributions
AP	Altarelli–Parisi (splitting function)
CDR	Conventional Dimensional Regularisation
KLN	Kinoshita–Lee–Nauenberg (theorem)
SM	Standard Model of Particle Physics
EFT	Effective Field Theory
SUSY	Supersymmetry
MC	Monte Carlo
IBP	Integration by Parts

Antenna and Regularisation Notation

Symbol	Meaning
X_n^l	Generic antenna symbol: n is the final-state multiplicity, l the loop order
A, B, C	Quark-antiquark antenna families (implemented)
D, E	Quark-gluon antenna families (not implemented in the present work)
F, G, H	Gluon-gluon antenna families (not implemented in the present work)
X (undecorated)	Leading-colour component (from the raw “Lead” interference term)
$\tilde{X}$	Subleading-colour component (raw “Sublead” interference term)
$\hat{X}$	Quark-loop component (raw “QL” interference term)

Symbol	Meaning
$\check{X}$	Double-fermion-loop self-interference component (NNLO A_2^2 only)
S_ϵ	$(4\pi)^{-\epsilon} e^{\epsilon \gamma_E}$, the $\overline{\text{MS}}$ normalisation factor
Λ_l	$(8\pi^2)^l$, loop-order normalisation
G_k	$(8\pi^2)^k$, perturbative-order scale normalisation
$C(\epsilon, k)$	$G_k S_\epsilon$, combined normalisation factor
$\mathcal{N}_X$	Build-stage colour and coupling normalisation
$\mathcal{N}(\epsilon, k)$	Full integration normalisation, $C(\epsilon, k)/\Phi_2$
$\mathcal{T}$-objects	Process-wide colour- and kinematics-conscious objects assembled from antennae, used in the R-ratio construction
Φ_n	d-dimensional n-body phase-space volume

General Kinematics

Symbol	Meaning
α_s	Strong coupling constant
ϵ, $d = 4 - 2\epsilon$	Dimensional-regularisation parameter; spacetime dimension
s_{ij}	Mandelstam invariant, $2p_i \cdot p_j$
μ	Renormalisation/regularisation scale
q^2	Hard-process invariant mass squared
N	Number of colours (= 3)
N_f	Number of active quark flavours
C_F, C_A	Quadratic Casimir invariants (fundamental / adjoint representation)
g_s	Strong coupling, $g_s = \sqrt{4\pi\alpha_s}$

Chapter 1

Introduction

High-energy collider experiments, such as the proton-proton collisions studied at the Large Hadron Collider (LHC) at CERN, have, in recent years, provided ever-increasing precision in measurements of processes governed by the strong interaction. To be able to use these measurements fully, theoretical calculations must achieve similar levels of precision. At high energies, Quantum Chromodynamics (QCD) permits such predictions to be organised as a perturbative expansion in powers of the strong coupling constant α_s.

Obtaining higher-order perturbative predictions is not simply a matter of calculating additional Feynman diagrams. Real-emission and virtual-loop contributions develop infrared singularities in kinematic configurations involving soft (low energy) or collinear partons. These individual contributions are divergent, and their infrared singularities must be extracted and cancelled before numerical integration can be performed. For infrared-safe observables, such as the hadronic R-ratio, the singularities cancel between the real and virtual contributions, leaving a finite physical result.

The antenna subtraction formalism provides a framework for extracting and cancelling these infrared singularities, rendering the individual contributions suitable for numerical integration. The formalism describes unresolved radiation between pairs of colour-connected hard partons through universal antenna functions. These functions reproduce the relevant infrared limits locally and, after analytic integration over the unresolved phase space in dimensional regularisation, $d = 4 - 2\epsilon$, yield Laurent series whose ϵ-pole structure cancels that of the virtual contributions. Their systematic construction and integration are therefore central to higher-order QCD calculations within the antenna subtraction framework.

This thesis develops and validates *AntCalc*, a symbolic framework for the automated construction and integration of QCD antenna functions. The work focuses on massless final-final quark-antiquark antennae through next-to-next-to-leading order (NNLO), combining symbolic amplitude generation with reduction methods for phase-space integrals. The resulting implementation is tested globally through the calculation of the NNLO massless hadronic R-ratio, a well-known inclusive observable in e^+e^- annihilation that provides a benchmark for the complete implementation. Locally, individual antenna functions are tested, where appropriate, by verifying that they reproduce the expected eikonal factors and Altarelli-Parisi (AP) splitting functions in the relevant unresolved limits.

The remainder of the thesis is organised as follows. Chapter 2 introduces the theoretical foundations required throughout this work, including perturbative QCD, infrared singularities, dimensional regularisation, and the antenna subtraction formalism. It defines the massless antenna functions and discusses their colour structure, phase-space factorisation, and role in fixed-order calculations. Chapter 3 describes *AntCalc*'s framework, including its architecture, antenna construction and integration workflows, reduction procedures, and the conventions and validation tools used to obtain the results. Chapter 4 presents the antenna functions obtained using *AntCalc*, beginning with the master integrals required for their integration. The A_3^0 antenna is discussed in detail, from its unintegrated construction to its integrated form and the validation of its soft and collinear limits. Selected NLO and NNLO antenna results are then presented more concisely. The chapter concludes with an exploratory extension of the A_3^0 antenna to the massive case. Chapter 5 provides a global validation of the framework through the calculation of the massless hadronic R-ratio through NNLO. The relevant antenna contributions are combined, demonstrating the cancellation of infrared poles and agreement with the known finite result. Chapter 6 discusses future directions for *AntCalc*, including more efficient symbolic and reduction backends, completion of the massive antenna sector, general model support, and extensions to higher perturbative orders, particularly $\mathrm{N}^3\mathrm{LO}$. Finally, Chapter 7 summarises the main conclusions of the thesis.

Chapter 2

Theoretical Foundations

2.1 QCD: Fields, Interactions and Perturbative Amplitudes

QCD is the quantum field theory of the strong interaction, whose elementary particles are the quarks and the gluons. Quarks are fermions that carry colour and interact through the exchange of gluons. Owing to the non-Abelian gauge structure of QCD, gluons themselves also carry colour charge and, therefore, undergo self-interaction.

At sufficiently high energy scales, the strong coupling, α_s, becomes sufficiently small for physical observables to be calculated perturbatively, as an expansion in powers of α_s. In this framework, the interaction structure of QCD determines the amplitudes for quark and gluon radiation that are studied throughout this thesis.

2.1.1 Quark and Gluon Fields

Quarks are spin-$1/2$ fields carrying flavour and a three-valued colour charge. They are QCD's matter fields. Gluons are the eight gauge fields associated with local $SU(3)_c$ colour symmetry. The QCD Lagrangian describes their propagation and interaction. It is written as follows,

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\,\mu\nu} + \sum_f \bar{\psi}_f \left(i\gamma^\mu D_\mu - m_f \right) \psi_f . \tag{2.1}$$

In this expression, $G^a_{\mu\nu}$ is the non-Abelian field-strength tensor, D_μ is the covariant derivative, ψ_f is the quark field and m_f is the quark mass. The flavour label f is summed over the quark species included in the theory, while the adjoint colour index a runs from one to eight. This Lagrangian is cleanly divided into two pieces, the quark and the gluon pieces.

Defining the covariant derivative as,

$$D_\mu = \partial_\mu - i g_s T^a A^a_\mu \tag{2.2}$$

where T^a represents the generators of the fundamental representation of $SU(3)_c$ and A^a_μ is the gluon gauge field, the quark Lagrangian term may be expanded to,

$$\mathcal{L}_q = \sum_f \left[\bar{\psi}_f \left(i\gamma^\mu \partial_\mu - m_f \right) \psi_f + g_s \bar{\psi}_f \gamma^\mu T^a \psi_f A^a_\mu \right] . \tag{2.3}$$

Here g_s is the strong coupling. In the expanded quark term, the first piece describes free quark propagation and the second quark-gluon interaction. This second piece is also responsible for generating the $q\bar{q}g$ vertex.

We now also define the totally antisymmetric structure constants of the $SU(3)_c$ Lie algebra f^{abc} as,

$$\left[T^a, T^b\right] = i f^{abc} T^c. \tag{2.4}$$

They may be used to define the non-Abelian field-strength tensor $G^a_{\mu\nu}$ as,

$$G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g_s f^{abc} A^b_\mu A^c_\nu. \tag{2.5}$$

The first two terms of the field-strength tensor are Abelian-like and, in fact, they can be rewritten using the Abelian-like linear part of the non-Abelian field-strength tensor $F^a_{\mu\nu}$, as,

$$G^a_{\mu\nu} = F^a_{\mu\nu} + g_s f^{abc} A^b_\mu A^c_\nu, \quad F^a_{\mu\nu} \equiv \partial_\mu A^a_\nu - \partial_\nu A^a_\mu. \tag{2.6}$$

The second term arises because, as shown in Eq. 2.4, the $SU(3)_c$ generators do not commute. The non-Abelian field-strength tensor is used to write the gluon Lagrangian term. After expanding it becomes,

$$\mathcal{L}_g = -\frac{1}{4}\left(F^a_{\mu\nu}\right)^2 - \frac{g_s}{2} f^{abc} F^{a\,\mu\nu} A^b_\mu A^c_\nu - \frac{g_s^2}{4} f^{abc} f^{ade} A^b_\mu A^c_\nu A^{d\,\mu} A^{e\,\nu}. \tag{2.7}$$

Here, much like in $\mathcal{L}_q$, the gluon Lagrangian is also divided into three terms. The first term is quadratic in the gluon field and, after gauge-fixing, describes free gluon propagation as [1],

$$\Delta_{ab,\mu\nu}(k) = \delta_{ab}\frac{i}{k^2}\left(-g_{\mu\nu} + (1-\lambda)\frac{k_\mu k_\nu}{k^2}\right), \quad \mathcal{L}_{\text{gf}} = -\frac{1}{2\lambda}\left(\partial^\mu A^a_\mu\right)^2 \tag{2.8}$$

where $\mathcal{L}_{\text{gf}}$ is the gauge-fixing Lagrangian, δ_{ab} is the Kronecker delta, and λ is the gauge parameter. Throughout this text the t'Hooft-Feynman convention is used, which makes $\lambda = 1$, following the default convention from *FeynArts* [2]. It is important to note, however, that physical observables are independent of the gauge choice. Unlike the remaining terms, the first contains no non-Abelian self-interaction. The cubic and quartic terms in the gluon fields generate, respectively, the three- and four-gluon vertices, ggg and $gggg$.

Using a covariant gauge-fixing term in a non-Abelian theory like QCD requires it to be supplemented by a ghost Lagrangian, described as [1],

$$\mathcal{L}_{\text{ghost}} = \partial_\mu \eta^{a\,\dagger}\left(D^\mu_{ab}\eta^B\right). \tag{2.9}$$

This is the default ghost sector as used in *FeynArts* [2] and, as for the λ convention, is the one used for the duration of the present text and in *AntCalc*. These ghost fields cancel the unphysical timelike and longitudinal gluon polarisations that would, otherwise, propagate in covariant-gauge loop diagrams. This ensures that only the two physical transverse polarisation states contribute to physical amplitudes [1].

Together, the two pieces, $\mathcal{L}_q$ and $\mathcal{L}_g$, constitute the gauge-invariant core of QCD. They generate the three elementary interaction vertices among the quark and gluon fields: $q\bar{q}g$, ggg and $gggg$ [1, 3].

2.1.2 From the QCD Lagrangian to Feynman Diagrams

The interaction terms in $\mathcal{L}_q$ and $\mathcal{L}_g$ determine the interaction vertices of QCD, which, together with the corresponding propagators, form the Feynman rules we use to compute scattering amplitudes perturbatively. Feynman diagrams are graphical representations of the individual contributions to the perturbative expansion of an amplitude. Each process may require more than one diagram to be fully represented; in that case, the complete amplitude is given as the sum of the amplitudes associated with all contributing diagrams. Fig. 2.1 depicts the two Feynman diagrams contributing to the process $\gamma^* \to q(k_1)g(k_3)\bar{q}(k_2)$, where a virtual photon splits into a quark-antiquark pair, with the gluon emitted from either the quark or antiquark line.

(a) Gluon emitted from the quark line. **(b)** Gluon emitted from the antiquark line.

Figure 2.1: Tree-level diagrams for the process $\gamma^* \to q(k_1)g(k_3)\bar{q}(k_2)$.

The final amplitude in this case would then follow,

$$\mathcal{M}^{(0)}_{\gamma^* \to qg\bar{q}} = \mathcal{M}^{(0)}_{q-\text{emission}} + \mathcal{M}^{(0)}_{\bar{q}-\text{emission}}. \tag{2.10}$$

In the diagrams in Fig. 2.1, straight, curly, and wavy lines denote fermions (quarks), gluons and photons, respectively. These diagrams represent the reduced hadronic-current process with γ^* as an off-shell and colourless source. In the full process, $e^+e^- \to \gamma^* \to qg\bar{q}$, the photon is internal between the lepton (electron-positron) and hadron (quark-antiquark) currents. All other lines are external, meaning that they represent outgoing asymptotic partons, the quarks and gluons. Each of the diagrams features two vertices, where the lines meet. The vertices represent the interactions between the particles, which are determined by their respective Lagrangians. Each diagram contains an electromagnetic $\gamma^* q\bar{q}$ current insertion and a QCD $q\bar{q}g$ interaction vertex. The former follows from the QED sector, whereas the latter is generated by $\mathcal{L}_{\text{QCD}}$. A rate or cross section often depends on the squared amplitude $|\mathcal{M}|^2$, given as,

$$|\mathcal{M}|^2 = \sum_{D,D'} \mathcal{M}_D \mathcal{M}^*_{D'}, \tag{2.11}$$

where D and D' index the Feynman diagrams contributing to the amplitude, so that the sum includes both the diagonal $D = D'$ terms and the interference between distinct diagrams.

2.2 Divergences in Perturbative QCD

2.2.1 Infrared Divergences

Beyond leading order in perturbation theory, QCD predictions receive contributions from both real and virtual corrections. Real corrections correspond to processes with additional parton emission in the final state, while virtual corrections arise from loop contributions to the amplitude. Both are required at a given perturbative order and can develop infrared singularities, which must ultimately cancel in infrared-safe physical observables.

The two QCD infrared limits, soft and collinear, can be derived explicitly as follows. We start by defining the physical system under study: consider two massless final-state partons i and j with four-momenta p_i and p_j. By momentum conservation, the four-momentum of the internal virtual propagator is $P = p_i + p_j$. The matrix element contains this propagator and in the square amplitude it appears as a $1/P^2$ factor. Expanding it and applying the massless condition ($p^2 = 0$), we get $P^2 = 2p_i \cdot p_j = 2E_iE_j(1 - \cos\theta_{ij})$ [1]. Importantly, this quantity defines the Mandelstam invariants as $s_{ij} = 2p_i \cdot p_j = P^2$, in the massless regime[1], which will serve as the natural integration variable throughout the work. The limits are then apparent: the soft limit is hit when one of the energies goes to zero and the collinear limit, when θ_{ij} does. In both cases,

$$\lim_{E_i \to 0} s_{ij} = 0$$

$$\lim_{\theta_{ij} \to 0} s_{ij} = 0. \tag{2.12}$$

As an example, let us consider the $\gamma^* \to q(k_1)g(k_3)\bar{q}(k_2)$ process. If one of the quarks and the gluon become collinear, say $q(k_1)$ and $g(k_3)$, then $s_{13} \to 0$ and the amplitude has a single pole, $|\mathcal{M}|^2 \sim 1/s_{13} \to \infty$. If the gluon is soft, then both $s_{13}, s_{23} \to 0$, and the amplitude has two poles, $|\mathcal{M}|^2 \sim 1/(s_{13}s_{23}) \to \infty$. Finally, if only one of the quarks is soft, then the pole formed is the same as in the collinear limit, and requires no separate dedicated soft-quark counterterm [1]. Furthermore, there is also the case for the NNLO process $\gamma^* \to q(k_1)\bar{q}(k_2)q'(k_4)\bar{q}'(k_5)$ in which the $q'(k_4)$ and $\bar{q}'(k_5)$ become simultaneously soft. In that case, the amplitude has a double pole around the relevant invariant, s_{45}, which appears as $|\mathcal{M}|^2 \sim 1/s_{45}^2 \to \infty$ [4].

From an experimental point of view, a real emission that becomes soft or collinear may be unresolved, meaning that it cannot be distinguished from a configuration in which that emission is absent. For an infrared-safe observable, the two configurations yield the same value of the observable in the corresponding limit.

The virtual corrections contain infrared singularities related to those arising from unresolved real radiation. According to the Kinoshita–Lee–Nauenberg (KLN) theorem, these singularities cancel once all degenerate contributions to an infrared-safe physical observable are appropriately combined [5, 6]. The resulting observable is therefore finite, despite the presence of infrared divergences in the individual real and virtual contributions.

In practice, however, this cancellation cannot be performed directly at the integrand level, since the real and virtual contributions are separately divergent and are defined on phase spaces with different particle multiplicities. At NLO, the real contribution contains one additional final-state parton and is integrated over the $(n + 1)$-particle phase space, whereas the virtual contribution is integrated over the n-particle phase space. A subtraction method resolves this mismatch by introducing a local counterterm $d\sigma^S$ that reproduces the singular behaviour of the real contribution $d\sigma^R$ in all unresolved limits. The counterterm can then be subtracted from the real contribution, rendering their difference finite and suitable for numerical integration. The same counterterm is analytically integrated over the unresolved one-particle phase space and added to the virtual contribution, where its explicit infrared poles cancel those of the virtual correction. Schematically, at NLO, this is expressed as,

$$\sigma = \int_{n+1} \left(d\sigma^R - d\sigma^S \right) + \int_n \left(d\sigma^V + \int_1 d\sigma^S \right) \tag{2.13}$$

[1]The general definition of the Mandelstam invariant is $s_{ij} = (p_i + p_j)^2$, but as $p_k^2 = 0$ for any k, in this massless regime, then the massless Mandelstam invariant is defined as $s_{ij} = 2p_i \cdot p_j$.

where $d\sigma^R$ represents the differential real radiation cross-section, $d\sigma^V$ the differential virtual corrections cross-section and $d\sigma^S$ the differential subtraction counterterm [7]. Importantly, $d\sigma^S$ must reproduce $d\sigma^R$'s exact singular behaviour at every point in the phase space, such that the divergences cancel locally.

2.2.2 Ultraviolet Divergences

Infrared divergences are not the only kind of divergences we find in QCD. In loop diagrams, the loop momentum l is integrated over the whole of momentum space. Regions in which l becomes soft or collinear to an external massless momentum can give rise to infrared singularities, which contribute to the singular structure of $d\sigma^V$. In addition, loop integrals may become divergent when the magnitude of the loop momentum becomes arbitrarily large, $|l| \to \infty$. These are referred to as ultraviolet (UV) divergences.

The UV behaviour of a loop integral can be understood through simple power counting. In d dimensions, the loop integration measure can be written in spherical coordinates as,

$$d^d l = l^{d-1} dl \, d\Omega_{d-1}. \tag{2.14}$$

Whether the integral diverges or not depends on whether the integrand converges fast enough to counter the l^{d-1} measure term. Since each propagator contributes $\sim 1/l^2$ to the integrand, schematically, a diagram with n_p propagators in the loop and no extra powers of l in the numerator becomes,

$$\int^{\infty} l^{d-1} \cdot \frac{1}{l^{2n_p}} = \int^{\infty} l^{d-1-2n_p} dl \tag{2.15}$$

which is UV divergent whenever $d - 1 - 2n_p \geq -1 \Rightarrow d \geq 2n_p$ [3, 8].

The presence of UV divergences means that loop corrections expressed in terms of the bare coupling α_0 from the QCD Lagrangian, as shown in Eq. 2.1, are not finite beyond tree level, since their higher-order coefficients diverge [1]. This property makes α_0 unsuitable as the order-by-order perturbative series variable we require. These divergences are removed through renormalisation, in which the bare coupling is traded for a finite renormalised coupling, α_s, defined at a renormalisation scale μ [8]. This renormalisation is made explicit for the one-loop antenna sets in Subsection 3.3.2.

2.3 Dimensional Regularisation

The subtraction scheme from Eq. 2.13 separates the NLO cross-section into contributions that can ultimately be evaluated numerically. The difference $d\sigma^R - d\sigma^S$ is finite in the unresolved regions and can therefore be integrated numerically in four dimensions. The subtraction term must, however, still be integrated analytically over the unresolved one-parton phase space,

$$\int_1 d\sigma^S. \tag{2.16}$$

Since $d\sigma^S$ is designed explicitly to carry over all of $d\sigma^R$'s divergent structures, this integral blows up once integrated over four dimensions – as does the virtual contribution $d\sigma^V$, whose infrared divergences originate from the loop integration, as explored in Subsection 2.2.2. To regulate these divergent contributions we continue the number of space-time dimensions away from four, via,

$$d = 4 - 2\epsilon \tag{2.17}$$

so that infrared divergences appear as explicit analytic poles in ϵ. After analytic integration over the unresolved phase space, this leaves us with two ϵ-dependent expressions, for the virtual corrections and for the now unresolved-parton-integrated subtraction cross-section,

$$\int_n \left(d\sigma^V + d\sigma_1^S\right). \tag{2.18}$$

From the KLN theorem, as described in Section 2.2.1, we know that these poles must cancel. This combination is therefore finite as $\epsilon \longrightarrow 0$, allowing the remaining phase-space integration to be performed numerically in four dimensions.

Though the underlying principle is always the same, dimensional regularisation can be implemented using different schemes. The most common are 't Hooft-Veltman [9], Four-Dimensional Helicity [10] and Conventional Dimensional Regularisation (CDR) [1]. In CDR, all momenta and polarisation states are treated in $d = 4 - 2\epsilon$ dimensions. This is the scheme used throughout the present work, since *AntCalc*'s primary objective is the simplification and democratisation of the antenna framework established in [4], which uses CDR throughout.

The use of CDR, and in particular the complete continuation to d-dimensional space, introduces some complications, however. The move to d-dimensional spherical coordinates means that the phase-space integrals are now performed over d-dimensional solid angles, which produce characteristic factors such as powers of 4π, Euler's constant γ_E, and Gamma functions like $\Gamma(1-\epsilon)$. These factors are a consequence of performing the angular and radial integrations away from four dimensions, and depend on the conventions adopted for dimensional regularisation. To express the integrated antennae in the $\overline{\text{MS}}$ (modified minimal subtraction)-compatible convention [11] used throughout this work, we multiply the raw d-dimensional integrals by the normalisation factor $S_\epsilon = (4\pi)^{-\epsilon} e^{\epsilon\gamma_E}$. This convention removes the ubiquitous 4π and γ_E factors from the coefficients of the perturbative expansion.

A separate normalisation arises from the convention used to express the perturbative expansion in the strong coupling. In the standard QCD convention adopted here, the expansion is written in terms of $\alpha_s/(2\pi)$, instead of just α_s, since the squared matrix elements scale with the whole raw coupling constant $g^2 = 4\pi\alpha_s = 8\pi^2(\alpha_s/(2\pi))$ rather than with α_s alone. This makes a further normalisation parameter necessary. The scale normalisation,

$$\begin{aligned} G_k &= \left(8\pi^2\right)^k \\ &= \left(8\pi^2\right)^{n-2} \Lambda_l, \quad \Lambda_l \equiv \left(8\pi^2\right)^l \end{aligned} \tag{2.19}$$

is then introduced to normalise those expansion terms, and the loop-order-only normalisation Λ_l explicitly shown as the virtual-loop contribution. The combined normalisation here introduced is thus,

$$C(\epsilon, k) = G_k S_\epsilon = \left(8\pi^2\right)^k (4\pi)^{-\epsilon} e^{\epsilon\gamma_E} \tag{2.20}$$

where k represents the perturbative order as $k = 1$ for NLO, $k = 2$ for NNLO, ... and is, itself defined as,

$$k = (n-2) + l \tag{2.21}$$

for some antenna with n-final-state particles and l loops[2].

Since multi-particle phase spaces recursively factorise, we must also normalise our phase-space inte-

[2]Loops will be discussed and the k index widely used in Subsection 2.4.2.

gral by the d-dimensional two-body phase space volume Φ_2, as is convention in the antenna-subtraction literature [4]. The factorisation method used is discussed in Subsection 2.4.5, and written in Eq. 2.45. This normalisation, together with the previous combined normalisation factor make the global normalisation factor $\mathcal{N}(\epsilon, k)$, used throughout the present work,

$$\mathcal{N}(\epsilon, k) = \frac{C(\epsilon, k)}{\Phi_2}. \tag{2.22}$$

As described in Subsection 2.2.2, we need to trade the bare coupling constant α_0 for a suitable order by order perturbative series variable. That is achieved as follows [4],

$$\alpha_0 \mu_0^{2\epsilon} S_\epsilon = \alpha_s \mu^{2\epsilon} \left[1 - \frac{\beta_0}{\epsilon} \frac{\alpha_s}{2\pi} + O(\alpha_s^2) \right], \quad \beta_0 = \frac{11N - 2N_f}{6} \tag{2.23}$$

where μ_0 is the bare mass parameter that keeps α_0 dimensionless, μ is the renormalisation scale (the renormalised mass parameter) that functions equally for α_s. It is more commonly scaled as $\mu^2 = q^2$ for the massless case. β_0 is the one-loop coefficient of the QCD beta function [1, 8], that tracks how the renormalised coupling α_s changes with the renormalisation scale μ, and finally N is the number of colours, 3, and N_f the number of massless active quark flavours.

2.4 The Antenna Subtraction Formalism

In perturbative QCD, when a particle like a gluon, j, becomes unresolved as described above, its squared matrix element $|\mathcal{M}_{n+1}|^2$ factorises into a universal singular factor for this process, $\mathcal{S}_j$ and a lower-multiplicity squared matrix element (a level below), as follows [7, 4],

$$|\mathcal{M}_{n+1}(..., i, j, k, ...)|^2 \to \mathcal{S}_j \cdot |\mathcal{M}_n(..., I, K, ...)|^2. \tag{2.24}$$

Since this is a universal property, it is possible to compute the subtraction cross-section $d\sigma^S$ by summing over all possible unresolved pairs. Therefore,

$$d\sigma^S = \sum_{\text{unresolved pairs}} \mathcal{S}_j \cdot d\sigma^B \tag{2.25}$$

where $d\sigma^B$ denotes the Born process $\gamma^* \to q\bar{q}$, cross-section and $\mathcal{S}_j$ is the singular factor for each of the j particles being summed over.

2.4.1 NLO and NNLO Subtraction Terms

As described in previous sections, we use $d\sigma^S$ in the antenna subtraction formalism. We use this method to create a finite differential prediction tool (the differential cross-section) for some F_n infrared-safe quantity or observable, for n final-state particles. F_n, in this case, is a distribution. The terms of the subtraction formula must carry their respective infrared-safe quantities through the process. As an example, the differential (F_n-dependent) NLO finite cross-section is given by,

$$d\sigma^{\text{NLO}} = \int_{m+1} \left(d\sigma^R_{\text{NLO}} F_{m+1} - d\sigma^S_{\text{NLO}} F_m \right) + \int_m \left(d\sigma^V_{\text{NLO}} + \int_1 d\sigma^S_{\text{NLO}} \right) F_m \tag{2.26}$$

where, at NLO, $m = 2$ is the number of final-state particles at Born-level [1].

The observable considered throughout this work, the hadronic R-ratio, is fully-inclusive and makes no distinction between resolved and unresolved final states at any multiplicity. Its measurement distribution F_n is, therefore, equal to one for every n. The cross-section is, therefore, no longer a differential prediction in F_n, but rather reduces to the plain, fully-summed total. At NLO, the formula remains close in shape to Eq. 2.26, but at NNLO, due to the larger number of possible corrections, it must account for contributions where the two additional orders of α_s are realised as two extra real partons (RR), one extra real parton and one loop (RV), or two loops with no extra final-state partons (VV). The NLO and NNLO antenna subtraction formulae are written as,

$$
\begin{aligned}
\sigma^{\text{NLO}} &= \int_{m+1} \left(d\sigma^{R}_{\text{NLO}} - d\sigma^{S}_{\text{NLO}} \right) + \int_{m} \left(d\sigma^{V}_{\text{NLO}} + \int_{1} d\sigma^{S}_{\text{NLO}} \right) \\
\sigma^{\text{NNLO}} &= \int_{m+2} \left(d\sigma^{RR}_{\text{NNLO}} - d\sigma^{S}_{\text{NNLO}} \right) + \int_{m+1} \left(d\sigma^{RV}_{\text{NNLO}} - d\sigma^{T}_{\text{NNLO}} \right) \\
&\quad + \int_{m} \left(d\sigma^{VV}_{\text{NNLO}} + \int_{1} d\sigma^{T}_{\text{NNLO}} + \int_{2} d\sigma^{S}_{\text{NNLO}} \right).
\end{aligned}
\tag{2.27}
$$

Where the NLO formula is restricted to a single subtraction counterterm, the NNLO formula has two, $d\sigma^S$ and $d\sigma^T$, which respectively subtract the singularities of $d\sigma^{RR}$ at single- and double-unresolved level, and $d\sigma^{RV}$ at single-unresolved level [4]. This inadvertently adds another level of complexity for NNLO: the KLN theorem guarantees only that the fully inclusive sum over degenerate configurations is finite, but says nothing about any particular piece of a decomposition being finite on its own. With the split including multiple subtractions at the same time, we are implying that each of these differences is separately, pointwise finite, everywhere in its own phase space.

As hinted at throughout this text but not yet stated explicitly: the subtraction terms are exact, not merely asymptotic, matches to the original cross-section terms. This follows from two facts specific to this process. First, because the antenna functions are built as exact, unique expressions (Eq. 2.30), and the sum in Eq. 2.25 runs over all unresolved pairs, of which there is only one, since the Born process $\gamma^* \to q\bar{q}$ has a single colour-connected pair. Second, because the Born factor $|\mathcal{M}_2^0|^2$ depends only on the conserved quantity q^2, and therefore takes the same value whether evaluated on the exact real-emission momenta or on the reconstructed momenta used in the subtraction term. Together, these mean $d\sigma^S$ and $d\sigma^T$ coincide identically with $d\sigma^R$, $d\sigma^{RR}$, and $d\sigma^{RV}$ at every point in the phase space, for either perturbative order, not merely in the singular unresolved limit, as the general subtraction formalism only guarantees [7]. These subtraction cross-sections are expressed generally using the usual antenna notation, as defined in Subsection 2.4.2, as,

$$
\begin{aligned}
d\sigma^{V}_{\text{NLO}} &= 2C_F \sum_{a} f_a(N, N_f) X^1_{ij,(a)} d\sigma^B \\
d\sigma^{S}_{\text{NLO}} &= 2C_F \sum_{a,s} f_a(N, N_f) X^0_{ij,(a,s)} d\sigma^B \\
d\sigma^{VV}_{\text{NNLO}} &= 2C_F \sum_{a} f_a(N, N_f) X^2_{ij,(a)} d\sigma^B \\
d\sigma^{T}_{\text{NNLO}} &= 2C_F \sum_{a,s} f_a(N, N_f) X^1_{ij,(a,s)} d\sigma^B \\
d\sigma^{S}_{\text{NNLO}} &= 2C_F \sum_{a,s} f_a(N, N_f) X^0_{ij,(a,s)} d\sigma^B
\end{aligned}
\tag{2.28}
$$

where C_F is the Fundamental Casimir invariant, f_a is the function that describes the colour prefactor per colour component: N for the leading order, $1/N$ for subleading, etc., and s is summed over all possible extra final-state particle sets. That is, for example, for the quark-antiquark final-final antenna functions considered throughout this text, the s-sum in $d\sigma^S_{\mathrm{NNLO}}$ would produce gg, $q\bar{q}$, and $q'\bar{q}'$, relative to the A, B, C antenna types.

Each real-emission antenna function in Eq. 2.28 ($d\sigma^S$ and $d\sigma^T$, at either order) is evaluated on reconstructed, on-shell momenta $\tilde{p}_I$, $\tilde{p}_K$, built from the hard radiators and the unresolved momenta in between them, through a momentum mapping which conserves total momentum, $\tilde{p}_I + \tilde{p}_K = p_i + p_j + p_k$, and keeps $\tilde{p}_I^2 = \tilde{p}_K^2 = 0$. The pure-virtual terms, $d\sigma^V_{\mathrm{NLO}}$ and $d\sigma^{VV}_{\mathrm{NNLO}}$, need no such mapping, since they carry no unresolved parton to begin with. Without this reconstruction, the reduced matrix element inside $d\sigma^B$ would only be well-defined at the unresolved limit. The mapping allows the antenna subtraction term to act as a local counterterm over the whole unresolved-parton phase space, not merely asymptotically close to it. Using these expressions, σ^{NLO} and σ^{NNLO}, from Eq. 2.27, simplify to,

$$
\begin{aligned}
\sigma^{\mathrm{NLO}} &= \int_m \left(d\sigma^V_{\mathrm{NLO}} + \int_1 d\sigma^S_{\mathrm{NLO}} \right) = 2C_F \left[\sum_a f_a \int_m X^1_{ij,(a)} d\sigma^B + \sum_{a,s} f_a \int_{m+1} X^0_{ij,(a,s)} d\sigma^B \right] \\
\sigma^{\mathrm{NNLO}} &= \int_m \left(d\sigma^{VV}_{\mathrm{NNLO}} + \int_1 d\sigma^T_{\mathrm{NNLO}} + \int_2 d\sigma^S_{\mathrm{NNLO}} \right) \\
&= 2C_F \left[\sum_a f_a \int_m X^2_{ij,(a)} d\sigma^B + \sum_{a,s} f_a \int_{m+1} X^1_{ij,(a,s)} d\sigma^B + \sum_{a,s} f_a \int_{m+2} X^0_{ij,(a,s)} d\sigma^B \right]
\end{aligned}
\tag{2.29}
$$

Each order's antenna subtraction formula is given in the following Table 2.1 in terms of the antenna functions, for the relevant quark-antiquark final-final antennae considered in this text.

Order	**Antenna-function form**
LO	—
NLO	$\sigma^{\mathrm{NLO}} = \left(N - \frac{1}{N}\right)\left(\mathcal{A}^1_2 + \mathcal{A}^0_3\right)$
NNLO	$\sigma^{\mathrm{NNLO}} = \left(N - \frac{1}{N}\right) \Big[N\left(\mathcal{A}^1_2\mathcal{A}^0_3 + \mathcal{A}^1_3 + \mathcal{A}^0_4 + \mathcal{A}^2_2\right)$ $-\frac{1}{N}\left(\mathcal{A}^1_2\mathcal{A}^0_3 + \tilde{\mathcal{A}}^1_3 + \tilde{\mathcal{A}}^0_4 + C^0_4 - \tilde{\mathcal{A}}^2_2\right)$ $+N_f\left(\mathcal{B}^0_4 + \hat{\mathcal{A}}^1_3 + \hat{\mathcal{A}}^2_2\right) + \left(N - \frac{1}{N}\right)\breve{\mathcal{A}}^2_2 \Big]$

Table 2.1: Antenna subtraction formulae for σ^{NLO} and σ^{NNLO}, in terms of the antenna functions [4]. The σ^{LO} row has no antenna-function form: it is the un-normalised Born term itself.

2.4.2 Antenna Functions

Since the singular factor $\mathcal{S}_j$ must work smoothly across all unresolved limits and must also handle all spin correlations correctly, it should, optimally, be a function that preserves all the unresolved parton information without the noise coming from the hard parton process. This is easily achieved by dividing our n-level process squared matrix element by the fixed two-parton Born-level, common to all quark-antiquark antenna functions. The resulting quantity is gauge-invariant. It is the antenna, most usually notated by X_n^0 and defined as [4, 12, 13],

$$X_n^{0,\,\text{bare}} = \mathcal{N}_X \frac{\left|\mathcal{M}_n^0\right|^2}{\left|\mathcal{M}_2^0\right|^2} \tag{2.30}$$

where $|\mathcal{M}_n|^2$ represents the n-level squared matrix element and $X_n^{0,\,\text{bare}}$ is the antenna that represents that same n-level. $\left|\mathcal{M}_2^0\right|^2$ is the Born-level squared matrix element.

Fig. 2.2 illustrates this factorisation for the NLO quark-antiquark antenna, using the same $\gamma^* \to q(i)g(j)\bar{q}(k)$ process as Fig. 2.1. As particle j becomes unresolved, the process collapses onto its Born-level configuration, with the two hard radiators i and k mapped onto $\tilde{p}_I$ and $\tilde{p}_K$. The antenna function X_3^0, together with its own local phase space, is exactly the ratio that this factorisation leaves behind.

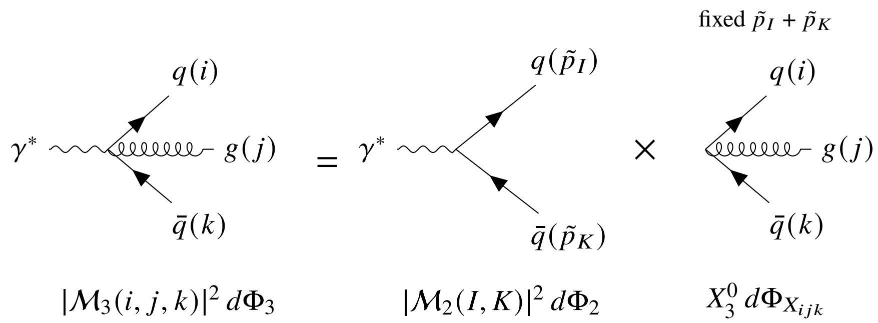

Figure 2.2: Factorisation of the squared matrix element and phase space as particle j becomes unresolved, for the NLO quark-antiquark antenna. The hard radiators i, k map onto the reduced Born momenta $\tilde{p}_I, \tilde{p}_K$; the antenna function X_3^0 depends only on the antenna's own internal invariants, evaluated at fixed total momentum $\tilde{p}_I + \tilde{p}_K$.

The same construction extends directly to A_4^0, with two unresolved partons mapping the process onto the same reduced Born configuration. The four-quark antennae B_4^0 and C_4^0, by contrast, are not built this way at all: they come from interfering flavour-flow amplitudes instead, as described in Subsection 2.4.4.

$\mathcal{N}_X$ represents the necessary normalisation over colour and coupling terms, whose general form depends on the specific antenna family under consideration. The colour-algebraic machinery required to construct it explicitly is developed in Subsection 2.4.3. It is defined, for quark-antiquark antennae (other antenna types are described in Subsection 2.4.4 and require different normalisations), using the fundamental Casimir invariant C_F and the strong coupling constant $g_s = \sqrt{4\pi\alpha_s}$, as

$$\mathcal{N}_X = \frac{1}{2 g_s^{2(n-2)} C_F}. \tag{2.31}$$

This definition is strictly correct and complete for real-emission, so called tree-level, antenna functions. These antenna functions represent loopless processes. QCD allows for the exchange of gluons in between quarks and the self-absorption (by the same quark) of a previously emitted gluon. These are the so-called virtual corrections. As described in Section 2.2.2, these corrections lead to UV divergences. Hence, loop antenna functions require normalisation and hence why the Eq. 2.30 is labelled as *bare*: it is the general definition without normalisation. This normalisation is achieved through the rescaling of the bare coupling constant α_0 to the renormalised coupling constant α_s through Eq. 2.23. This rescaling applies the Ward-Takahashi [14, 15] and Slavnov-Taylor Identities [16, 17], which guarantee the conservation of the electric and colour charges, respectively, through all topologies of the same process. They also require us to denote the number of loops their process supports. That number is the superscript in X_n^l, l, which up until now had been kept at zero for the loopless processes. n, on the other hand, represents the number of final-state particles in the process.

These antenna functions may now be ordered by perturbative order based on their k index. Let us define an antenna completely as $X^l_{n,(k)}$ where k is the perturbative order index, n is the number of final-state particles and l is the number of loops. Let us also define the perturbative expansion in Eq. 2.23 as,

$$\begin{aligned}\Delta\left(\frac{\alpha_s}{2\pi}\right) &= 1 - \frac{\beta_0}{\epsilon}\frac{\alpha_s}{2\pi} + O\left(\frac{\alpha_s}{2\pi}\right) \\ &= \Delta^{(0)} + \Delta^{(1)}\frac{\alpha_s}{2\pi} + O\left(\frac{\alpha_s}{2\pi}\right).\end{aligned} \tag{2.32}$$

We are now able to generalise Eq. 2.30 to all antenna functions as follows,

$$\begin{aligned}X^l_{n,(k)} &= X^{l,\,\text{bare}}_{n,(k)} + \sum_{m=1}^{l}\Delta^{(m)}X^{l-m,\,\text{bare}}_{n,(k-m)} \\ &= X^{l,\,\text{bare}}_{n,(k)} + \Delta^{(1)}X^{l-1,\,\text{bare}}_{n,(k-1)} + \Delta^{(2)}X^{l-2,\,\text{bare}}_{n,(k-2)} + \ldots \\ &= \frac{1}{\left|\mathcal{M}^0_2\right|^2}\left[\left|\mathcal{M}^l_n\right|^2 + \Delta^{(1)}\left|\mathcal{M}^{l-1}_n\right|^2 + \Delta^{(2)}\left|\mathcal{M}^{l-2}_n\right|^2 + \ldots\right].\end{aligned} \tag{2.33}$$

Lastly, the $\mathcal{N}_X$ normalisation factor will also need to be generalised. It becomes,

$$\mathcal{N}_X = \frac{1}{2g_s^{2(n-2+l)}C_F}. \tag{2.34}$$

2.4.3 Colour Algebra and the Antenna Colour Decomposition

The processes represented by the X^l_n antenna functions are composed of colour-carrying particles, namely quarks and gluons. When the amplitudes $\mathcal{M}^l_n$ are evaluated, the kinematics and colour are not explicitly separate algebraically, but since we require the antenna functions to be purely kinematic, we need to decompose them into inner colour structures. To be able to decompose, we must first clarify the Algebra of colour [4, 12].

Quarks carry a fundamental colour index $i \in \{1, 2, 3\}$ while gluons carry an adjoint colour index $a \in \{1, ..., 8\}$. The interactions between them are governed by the $SU(3)_c$ generators T^a. Those generators are normalised by the trace relation $\text{Tr}\left(T^aT^b\right) = T_R\delta^{ab}$, where by standard convention $T_R = 1/2$ [1, 3]. This is particularly useful since when the amplitude is squared, the two colour generators must be contracted with one another, that is T^a and $(T^a)^\dagger$, and since the generators are Hermitian, $(T^a)^\dagger = T^a$. Hence, summing over all colour we get,

$$\sum_{i_1,i_2,a}(T^a)_{i_1i_2}(T^a)_{i_2i_1} = \sum_a \text{Tr}\left(T^aT^a\right) = \sum_a T_R\delta^{aa}. \tag{2.35}$$

For a fixed a, the trace equals T_R. Summing this result over all $N^2 - 1$ adjoint indices we get,

$$\sum_a T_R\delta^{aa} = \frac{1}{2}\left(N^2 - 1\right). \tag{2.36}$$

We now also introduce the Casimir invariants: the fundamental and adjoint Casimir invariants, C_F and C_A. These represent, respectively, the total colour charge squared of the quark and of the gluon.

They are defined as,

$$C_F = \frac{N^2 - 1}{2N}$$
$$C_A = N. \tag{2.37}$$

The invariants are particularly useful to normalise the two hard quarks' colour from the antenna functions, by taking out a factor of $2C_F$. This factor equals,

$$2C_F = N - \frac{1}{N} \tag{2.38}$$

which will be a normalisation used throughout the present work.

In loop processes, colour can flow in different ways depending on the virtual particles inside the loops. So, instead of only simplifying using the fundamental Casimir invariant, we must also decompose the antenna structures into their distinct colour components. These are,

- The leading colour component (N): these come from planar Feynman diagrams (where virtual gluons are exchanged without crossing over each other). This component is represented by the base antenna in standard notation, e.g. A_3^1.
- The subleading colour component ($1/N$): these come from non-planar diagrams (where gluon lines do cross each other). This component is represented by the tilde antenna, e.g. $\tilde{A}_3^1$.
- The fermion loop (N_f): Instead of a virtual gluon, the loop can have a virtual quark-antiquark pair. This contribution depends strictly on the number of active flavours, N_f. This component is represented by the hat antenna, e.g. $\hat{A}_3^1$[4].

The direct expansion to Eq. 2.35 is to generalise to the four quark colour indices, instead of only two. That expansion is

$$\sum_a (T^a)_{i_1 i_2} (T^a)_{i_3 i_4} = \frac{1}{2}\left(\delta_{i_1 i_4}\delta_{i_2 i_3} - \frac{1}{N}\delta_{i_1 i_2}\delta_{i_3 i_4}\right). \tag{2.39}$$

It is the $SU(N)$ completeness relation, also called the *colour Fierz identity*, and is particularly useful for four-quark topologies like those described in Subsection 2.4.4 [1, 3].

The QCD conventions and colour identities used by the package were introduced above.

2.4.4 Antenna Families

Throughout this work the Born-level process under study is $\gamma^* \to q\bar{q}$. Antenna families are designations used to identify the type of process that underlies a certain antenna function; for quark-antiquark antennae these are labelled A, B and C, summarised in Table 2.2.

We here require two different families for the four-quark final state because the process may present with either like (identical) or unlike-flavour pairs with the B antenna carrying the unlike-flavour case, often notated by $\gamma^* \to q\bar{q}q'\bar{q}'$, and the C antenna carrying the like-flavour case, denoted as $\gamma^* \to q\bar{q}q\bar{q}$. This last process, due to its topology, requires symmetrisation.

To compute these two antenna functions we start by writing the four-quark tree-level amplitude as a sum of two flavour-flow sub-amplitudes as,

$$\mathcal{M}_{q\bar{q}q\bar{q}} = \mathcal{M}_D + \mathcal{M}_E \tag{2.40}$$

Final state	Tree level	One loop	Two loop
$q\bar{q}$	A_2^0	A_2^1	A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$, $\breve{A}_2^2$
$qg\bar{q}$	A_3^0	A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$	—
$qgg\bar{q}$	A_4^0, $\tilde{A}_4^0$	—	—
$qq'\bar{q}'\bar{q}$	B_4^0	—	—
$qq\bar{q}\bar{q}$	C_4^0	—	—

Table 2.2: Quark-antiquark antenna families used in this work, modelled on Table 1 of [4]. The tilde denotes the subleading-colour contribution, the hat the fermion-loop contribution, and the breve (only present at two loops) the genuine self-interference term.

where $\mathcal{M}_D$ is the direct and $\mathcal{M}_E$ is the exchanged [3] assignment amplitude.

The B_4^0 antenna, where the flavours are non-identical, so only the direct assignment amplitude is used. The antenna is built from self-interference,

$$B_4^0 = \mathcal{N}_X \frac{\mathcal{M}_D \mathcal{M}_D^*}{|\mathcal{M}_2^0|^2}. \tag{2.41}$$

On the other hand, C_4^0, where the flavours are identical, the two subamplitudes must interfere. The antenna is given as [1, 4],

$$C_4^0 = \mathcal{N}_X \frac{\mathcal{M}_D \mathcal{M}_E^* + \mathcal{M}_E \mathcal{M}_D^*}{|\mathcal{M}_2^0|^2}. \tag{2.42}$$

Though not deeply included in the present work, there are two other final-final antenna types: there are quark-gluon antennae, organised under the D and E families, and gluon-gluon antennae, organised under the F, G and H families. These types represent different Born-level processes: the first represents the boson $\rightarrow qg$ process, and the second the boson $\rightarrow gg$ process. Remarkably, these processes do not neatly fit into the Standard Model of Particle Physics (SM), as the vertices required to make a virtual photon split into either of those pairs does not exist inside the model. The implementation of those antenna functions, then, requires the use of Effective Field Theories (EFT), specifically SUSY [4, 18] and the $H \rightarrow gg$ EFT [19, 20]. The future implementation of these antenna families in *AntCalc* is discussed in Chapter 6.

2.4.5 Phase-Space Factorisation

The previous Subsections focused on the antenna functions as objects: how they are built, combined, normalised, etc. But the antenna functions as described in Subsection 2.4.2 are the singular factor $\mathcal{S}_j$, as described in Eq. 2.25, up to some normalisation factor. The antenna functions, then, as part of $d\sigma^S$ must be integrated as, schematically, as described in Eq. 2.13,

$$\int_1 d\sigma^S \tag{2.43}$$

[3] The words *direct* and *exchange* denote the way in which the like- and unlike-flavour quarks are paired amongst each other. When all flavours are identical, the pairing may be blind, since either of the quarks pairing with either of the antiquarks results in the same pair. If the flavours are not alike, however, the pairing order matters and must be considered, hence the use of the *exchanged* amplitude.

where $\int_1$ denotes the integral over the unresolved parton phase-space. This integral does not only denote the integral over the single unresolved parton, alone, on two counts: first the fact that the integral must cover all unresolved partons which, for three-final-particle antennae is only one, but for four-final-particle antennae it becomes two. Second, the hard partons and their interactions with the unresolved partons must also be integrated over. The integral becomes over the n-final-state-particle antenna phase space. This makes,

$$\int_1 d\sigma^S = \sum_{\text{unresolved pairs}} \left(\int d\Phi_{ijkl\ldots} X_n^l \right) d\sigma^B. \tag{2.44}$$

The n-final-state-particle antenna phase space $\Phi_{ijkl\ldots}$[4] is not the standard n-particle phase space as in Φ_3 or Φ_4, it is, rather, the antenna's phase space, associated with the unresolved-partons emission between the two hard radiators, it is the unresolved-parton phase space of the antenna. It has no independent parametrisation and is only accessed by factorising the known n-particle phase space [4, 21]. Since, through the recursive factorisation of the Lorentz-invariant phase space property [7], the phase space factorises, we are able to denote that the n-particle phase-space equals the $(n - n_u)$-particle, where n_u is the number of unresolved partons, phase-space times the antenna phase-space for n final-state particles[5]. For our case, two useful identities are produced for this formula [4],

$$\begin{aligned} d\Phi_3 &= d\Phi_2 d\Phi_{ijk} \\ d\Phi_4 &= d\Phi_2 d\Phi_{ijkl}. \end{aligned} \tag{2.45}$$

Since $d\Phi_n$ is perfectly factorised [7], and the X_n^l antenna functions are explicitly built to be a function of the antenna's internal invariants alone and to have no dependence on how the reduced system is oriented [4], the two phase spaces are integrated separately, with the two-particle phase-space integral producing Φ_2 directly. The antenna's own phase-space integral is then rewritten as,

$$\int d\Phi_{ijkl\ldots} X_n^l = \frac{1}{\Phi_2} \int d\Phi_n X_n^l \tag{2.46}$$

which is how the phase-space prefactor appears in $\mathcal{N}(\epsilon, k)$, as described in Section 2.3.

Hence, finally, the integrated antenna $\mathcal{X}_n^l$ is given generally[6] as, taking the aforementioned overall $\mathcal{N}$ normalisation factor [4, 21],

$$\mathcal{X}_n^l = \mathcal{N}(\epsilon, k) \int d\Phi_n X_n^l. \tag{2.47}$$

A Note On $n_u = 0$ Cases

There are two antenna types that are exempt from this rule: A_2^1 and A_2^2, as they do not present with unresolved partons, but rather only with loops, and are completely virtual antennae. These antennae are only integrated over the loop momenta rather over the phase space, thus not requiring normalisation using the two-particle phase-space [22].

[4]This phase space is notated by Φ_{ijk} for three-final-state-particle antennae like A_3^1 and Φ_{ijkl} for four-parton-final-particle antennae like A_4^0.

[5]This mechanism, as described, works at $n_u \in \{1, 2\}$. The perturbative order above, N^{3}LO, presents more complex identities for which this blueprint may not work. It also does not fit when $n_u = 0$. That case is discussed in further detail in Subsubsection 2.4.5.

[6] This definition is general for antenna functions that are already cleanly and completely given in terms of Mandelstam invariants and ϵ. This is only automatically true for antenna where $l = 0$. When $l \geq 1$, which happens for all loop antennae by definition, a prior loop-momentum integration step is necessary. This step is explored further in Section 2.5.

2.5 Phase-Space Integration and IBP Reduction

In the previous sections, antenna functions were introduced together with their role in the local subtraction of infrared singularities. Eq. 2.47 defined their integration over the unresolved phase space, leading from an unintegrated antenna X_n^l to its integrated counterpart $\mathcal{X}_n^l$. For loop-level antennae ($l \geq 1$), the loop momenta must additionally be integrated.

Both the unintegrated and integrated forms are required in a subtraction calculation, but they play different roles. The unintegrated antenna functions provide local counterterms for real-radiation matrix elements and are therefore required in numerical Monte Carlo phase-space integration. They must retain the dependence on the local kinematics of each phase-space point, which can be expressed in terms of Mandelstam invariants s_{ij}. This allows the subtraction term to reproduce the singular behaviour of the real-emission matrix element point by point in the unresolved limits. For loop-level antennae, the term *unintegrated* refers here to the unresolved phase space: the loop-momentum integration is first performed analytically, leaving an antenna function that depends on the external kinematic invariants and can be evaluated locally within the phase-space integration. The subsequent analytic integration over the unresolved phase space then produces the corresponding integrated antenna function, whose explicit infrared poles enter the cancellation against virtual contributions.

From Eq. 2.46, we have seen that through phase-space factorisation, the antennae are integrated over the antenna phase space Φ_X. If required, they are also integrated over their loop momenta. The general expression used to describe this integration setup is,

$$\mathcal{X}_n^l \sim \int \left(\prod_{r=1}^{l} d^d \ell_r \right) d\Phi_X \, \frac{N_s(\ell_r, p_a)}{\prod_{j=1}^{N} D_j^{a_j}(\ell_r, p_a)}, \quad d = 4 - 2\epsilon \tag{2.48}$$

where ℓ_r denote the loop momenta, and the set p_a consists of hard-radiator momenta $\tilde{p}_{\text{hard}}$ and the unresolved antenna momenta $p_{\text{unresolved}}$, which are integrated over Φ_X. N_s is a numerator built from scalar products, D_j are propagator or Mandelstam invariants s_{ij}'s and a_j are integer powers. An algebraic expansion of the integrand of an antenna function can generate multiple integrals with the same kinematic structure but different numerator and denominator powers. These integrals may then be organised into integral families, which fix the denominators and integration variables, and may then be systematically reduced. These families are denoted as the complete collection [21],

$$\mathcal{F}[\{D_j\}; \{\ell_r\}, d\Phi_X] = \{I(a_1, ..., a_N)\}, \quad a_j \in \mathbb{Z} \tag{2.49}$$

with,

$$I(\vec{a}) = \int \left[\prod_{r=1}^{l} d^d \ell_r \right] d\Phi_X \, \frac{1}{D_1^{a_1} ... D_N^{a_N}}, \quad \vec{a} = (a_1, ..., a_N). \tag{2.50}$$

The integrals within each family are not independent. Through integral reduction, they can be expressed in terms of a finite and substantially smaller set of master integrals,

$$I(\vec{a}) = \sum_{k=1}^{N_{\text{MI}}} c_k(\vec{a}, \epsilon, \{s_{ij}\}) M_k \tag{2.51}$$

where M_k are the master integrals and c_k are coefficients that depend on the integral indices, the dimensional regulator ϵ, and the relevant kinematic invariants. The original integration problem is therefore separated into two tasks: determining the reduction coefficients c_k and evaluating the much smaller set of master integrals M_k. Once these are known, every required integral in the family can be reconstructed from the same master-integral basis. In $\vec{a}$, positive a_j indices denote propagator powers, while non-positive indices account for absent denominators or irreducible scalar products in the numerator.

As stated previously, MC integrators require the unintegrated antenna functions in terms of the Mandelstam invariants s_{ij}, since they must evaluate the subtractions at a sampled physical external phase-space point p. The loop-level antenna required by an MC is ultimately evaluated as a function of the external invariants. Its calculation, however, involves tensor loop integrals with loop-momentum dependence, which must be reduced before the antenna can be evaluated at a sampled phase-space point. This reduction is performed through the Passarino-Veltmann (PaVe) reduction mechanism, described in detail in Subsection 2.5.1, which reduces the tensor loop integrals to linear combinations of scalar one-loop functions, usually denoted by A_0, B_0, ..., while keeping the phase-space variables and integration untouched, as [23],

$$\int d^d\ell \, \frac{\ell^{\mu_1} \ldots \ell^{\mu_R}}{\prod_j D_j^{a_j}} \xrightarrow{\text{PaVe}} \sum_k c_k\left(\{s_{ij}\}\right) \{A_0, B_0, C_0, D_0, \ldots\}_k. \tag{2.52}$$

To finally reach the integrated antenna functions $\mathcal{X}_n^l$ we do require integrating over the phase-space. For this integral we also use a reduction method: Integration By Parts (IBP) reduction, described in detail in Subsection 2.5.2. This method generates relations among scalar integrals in a common family through vanishing total derivatives, as [24],

$$0 = \int d^d\ell \, \frac{\partial}{\partial \ell^\mu}\left(v^\mu \frac{1}{\prod_j D_j^{a_j}}\right). \tag{2.53}$$

As mentioned in previous sections, emitted final-state particles are real and on shell. That condition is enforced in the phase-space through the use of Dirac-delta functions like $\delta^+\left(p_i^2\right)$. Note that these are not ordinary Feynman propagators. To bring the phase-space integration into the same algebraic framework as the loop integrals, we introduce the method of reverse unitarity to represent these Dirac-delta functions as cut propagators, through the discontinuity relation as [21],

$$2\pi i \, \delta^+\left(p_i^2\right) = \left[\frac{1}{p_i^2 - i0} - \frac{1}{p_i^2 + i0}\right]_+. \tag{2.54}$$

where both + signs, in δ^+ and $]_+$, represent the positive-energy prescription.

The family now contains ordinary propagators from virtual loops and cut propagators representing the unresolved phase-space. The IBP identities can then reduce the resulting combined cut-integral family to a finite set of MI.

After UV renormalisation and evaluation of the MI, the result is expanded as a Laurent series in ϵ, making the remaining explicit IR poles visible order by order [4, 9].

2.5.1 PaVe Reduction

At one-loop, amplitudes produce tensor integrals similar to those in Eq. 2.52 [23]. With general massive[7] denominator defined as,

$$D_0 = \ell^2 - m_0^2 + i0, \quad D_i = (\ell + r_i)^2 - m_i^2 + i0 \tag{2.55}$$

the tensor integral takes the form that follows,

$$I_N^{\mu_1 \dots \mu_R} = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{\ell^{\mu_1} \dots \ell^{\mu_R}}{D_0 D_1 \dots D_{N-1}} \tag{2.56}$$

where R is the tensor rank, r_i are the combinations of external momenta and m_i is the mass associated with the i-th propagator. The reduction is carried out in $d = 4 - 2\epsilon$ dimensions, so that,

$$g_{\mu\nu} g^{\mu\nu} = d. \tag{2.57}$$

PaVe reduces the Lorentz indices in the numerator $\ell^\mu \ell^\nu \ell^\rho \dots$ terms and expresses them in terms of external momenta p_i^μ, the metric tensor $g_{\mu\nu}$ and scalar one-loop integrals, through Lorentz covariance [23].

For a rank-one N-point integral, the only available vectors are the independent external momenta, such that,

$$I_N^\mu = \sum_{i=1}^{N-1} p_i^\mu C_i \tag{2.58}$$

where C_i are scalar coefficient functions. For rank-two, the relation becomes,

$$I_N^{\mu\nu} = \sum_{i,j=1}^{N-1} p_i^\mu p_j^\nu C_{ij} + g^{\mu\nu} C_{00} \tag{2.59}$$

and for rank-three,

$$I_N^{\mu\nu\rho} = \sum_{i,j,k=1}^{N-1} p_i^\mu p_j^\nu p_k^\rho C_{ijk} + \sum_{i=1}^{N-1} \left(g^{\mu\nu} p_i^\rho + g^{\mu\rho} p_i^\nu + g^{\nu\rho} p_i^\mu \right) C_{00i} \tag{2.60}$$

and so on for higher ranks.

The reduction mechanism works because scalar products involving the loop momentum can be rewritten using the propagator denominators as,

$$2\ell \cdot r_i = D_i - D_0 - r_i^2 + m_i^2 - m_0^2 \tag{2.61}$$

which lets us cancel denominator terms, through a method like the following example set at $i = 1$,

$$\frac{\ell \cdot r_1}{D_0 D_1} = \frac{1}{2} \left(\frac{1}{D_0} - \frac{1}{D_1} + \frac{1}{D_0 D_1} \left(-m_0^2 + m_1^2 - r_1^2 \right) \right) \tag{2.62}$$

[7] Throughout this subsection, the general and massive case will be considered for completeness. The rest of the present work, however, heavily leans towards the massless regime. In that regime the internal masses vanish and the physical external partons satisfy $p_a^2 = 0$. The shifted momenta r_i, introduced in Eq. 2.55, entering propagators are combinations of external momenta and need not be lightlike.

thereby rewriting loop-momentum scalar products in terms of denominators and scalar integrals.

While using this reduction scheme, we might also encounter Gram matrices, defined as,

$$G_{ji} = 2p_j \cdot p_i. \tag{2.63}$$

Contracting a general rank-one integral, as described in Eq. 2.58, with p_j gives,

$$p_j \cdot I_N = \sum_i (p_j \cdot p_i) C_i = \frac{1}{2} \sum_i G_{ji} C_i. \tag{2.64}$$

If we now define $R_j \equiv p_j \cdot I_N$, we are then able to compute the C_i coefficients in the Lorentz decomposition by taking,

$$G_{ji} C_i = 2R_j \Rightarrow C_i = 2 \left(G^{-1}\right)_{ij} R_j. \tag{2.65}$$

Rank-One Bubble Example

To illustrate the method, let us consider the rank-one bubble,

$$B^\mu\left(p; m_0^2, m_1^2\right) = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{\ell^\mu}{D_0 D_1} \tag{2.66}$$

where the denominators follow Eq. 2.55. By Lorentz covariance we are able to write that,

$$B^\mu = p^\mu B_1 \tag{2.67}$$

which once contracted with p_μ makes,

$$p^2 B_1 = p_\mu B^\mu = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{\ell \cdot p}{D_0 D_1}. \tag{2.68}$$

Using the relation in Eq. 2.61, where we consider $r_1 = p$, we get,

$$p^2 B_1 = \frac{1}{2} \left[A_0\left(m_0^2\right) - A_0\left(m_1^2\right) + \left(m_1^2 - m_0^2 - p^2\right) B_0\left(p^2; m_0^2, m_1^2\right)\right] \tag{2.69}$$

and therefore[8],

$$B^\mu = p^\mu \frac{A_0(m_0^2) - A_0(m_1^2) + (m_1^2 - m_0^2 - p^2) B_0}{2p^2}. \tag{2.70}$$

Here, A_0 and B_0 are the scalar one-loop integrals mentioned throughout. These two types can be defined as,

$$A_0\left(m^2\right) = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{\ell^2 - m^2 + i0}$$

$$B_0\left(p^2; m_0^2, m_1^2\right) = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{D_0 D_1}. \tag{2.71}$$

[8]The result shown in Eq. 2.70 assumes generic kinematics with $p^2 \neq 0$. Exceptional cases are obtained through an appropriate limit or alternative reduction.

2.5.2 IBP Identities and the Laporta Algorithm

IBP Identities

The integral family has already been defined in Eq. 2.50. This definition allows for the factorisation of the integral into a loop and a phase-space integral term. That factorisation results in,

$$I(\vec{a}) = \int d\Phi_X \, \mathcal{I}(\vec{a};\{D_1,...,D_N\}), \quad \mathcal{I}(\vec{a};\{D_1,...,D_N\}) = \int \prod_{r=1}^{l} d^d\ell_r \, \frac{1}{D_1^{a_1}...D_N^{a_N}}. \tag{2.72}$$

The overall normalisation factors are irrelevant to the identities and will, therefore, be suppressed throughout the following explanation. The denominators may generally be defined as,

$$D_j = \left(\sum_{r=1}^{l} c_{jr}\ell_r + r_j\right)^2 - m_j^2 + i0 \tag{2.73}$$

where r_j is built from momenta held fixed during the loop integration, including the mapped hard partons and unresolved antenna momenta. Take one loop momentum ℓ_s, for $s = 1, ..., l$, and one vector v^μ from the available loop and external momenta. Dimensional regularisation allows us to set a total derivative to zero inside $\mathcal{I}$, as [21, 24],

$$0 = \int \prod_{r=1}^{l} d^d\ell_r \, \frac{\partial}{\partial \ell_s^\mu}\left[v^\mu \frac{1}{D_1^{a_1}...D_N^{a_N}}\right] \tag{2.74}$$

as previously introduced in Eq. 2.53. In dimensional regularisation, analytic continuation justifies the vanishing of the surface term at infinity, leaving an exact relation among regulated integrals[9].

Expanding the derivative gives,

$$0 = \int \prod_{r=1}^{l} d^d\ell_r \left[\frac{\partial_{\ell_s} \cdot v}{\prod_j D_j^{a_j}} - \sum_{j=1}^{N} a_j \frac{v^\mu \partial_{\ell_s^\mu} D_j}{D_1^{a_1}...D_j^{a_j+1}...D_N^{a_N}}\right]. \tag{2.75}$$

For these quadratic denominators D_j, differentiation with respect to ℓ_s^μ is linear in momenta,

$$\partial_{\ell_s^\mu} D_j = 2c_{js}\left(\sum_r c_{jr}\ell_r + r_j\right)_\mu \tag{2.76}$$

and therefore, $v^\mu \partial_{\ell_s^\mu} D_j$ is a scalar product. As scalar products may be expressed in terms of chosen denominators and irreducible scalar products, the latter are represented by negative family indices a_j, while zero indices denote absent denominators, we are able to write schematically,

$$\sum_{\vec{b}} \rho_{\vec{b}}(\vec{a}, \epsilon, \{s_{ij}\}) \, \mathcal{I}(\vec{b}) = 0 \tag{2.77}$$

where each vector $\vec{b}$ labels a family member obtained from $\vec{a}$ through index shifts in one or more denominator powers. $\rho_{\vec{b}}$ is the corresponding coefficient in the IBP relation.

The method may be made easier to visualise with an example. Consider a one-loop bubble, akin to

[9] This method is explored further and completely in Subsubsection 4.2.1.

that studied in the previous Subsection 2.5.1. Take,

$$\mathcal{I}(a_0, a_1) = \int d^d\ell \, \frac{1}{D_0^{a_0} D_1^{a_1}} \tag{2.78}$$

where,

$$D_0 = \ell^2 - m_0^2, \quad D_1 = (\ell + p)^2 - m_1^2. \tag{2.79}$$

We now choose $v^\mu = \ell^\mu$, and build and expand the relation from Eq. 2.53 as,

$$\begin{aligned} 0 &= \int d^d\ell \, \frac{\partial}{\partial \ell^\mu} \left(\frac{\ell^\mu}{D_0^{a_0} D_1^{a_1}} \right) \\ &= d\mathcal{I}(a_0, a_1) - 2a_0 \int d^d\ell \, \frac{\ell^2}{D_0^{a_0+1} D_1^{a_1}} - 2a_1 \int d^d\ell \, \frac{\ell \cdot (\ell + p)}{D_0^{a_0} D_1^{a_1+1}}. \end{aligned} \tag{2.80}$$

We can now use,

$$\ell^2 = D_0 + m_0^2 \tag{2.81}$$

$$2\ell \cdot (\ell + p) = D_0 + D_1 + m_0^2 + m_1^2 - p^2 \tag{2.82}$$

and finally get the explicit shifted-index identity,

$$\begin{aligned} 0 = &(d - 2a_0 - a_1)\mathcal{I}(a_0, a_1) \\ &- 2a_0 m_0^2 \mathcal{I}(a_0 + 1, a_1) \\ &- a_1 \mathcal{I}(a_0 - 1, a_1 + 1) \\ &- a_1 \left(m_0^2 + m_1^2 - p^2 \right) \mathcal{I}(a_0, a_1 + 1). \end{aligned} \tag{2.83}$$

Reverse Unitarity and Cut Integrals

In the previous subsubsection, we have derived the IBP identities relative to $\mathcal{I}$. As described in Eq. 2.72, the entire integrated antenna is not merely $\mathcal{I}$, but rather,

$$I(\vec{a}) = \int d\Phi_X \, \mathcal{I}(\vec{a}). \tag{2.84}$$

The unresolved phase-space measure is not originally an ordinary loop integral, however. Instead, it has on-shell constraints, which have to be considered and transformed before $I(\vec{a})$ is reducible via the present method.

As described in Subsection 2.4.5, phase-space factorisation isolates the antenna measure through $d\Phi_{m+r} = d\Phi_m d\Phi_X$, or in the two-hard-radiator final-final case we are considering throughout the present work, $d\Phi_X = d\Phi_n / d\Phi_2$, for n final-state particles. That n-particle phase space is given generally as,

$$d\Phi_n(q : p_1, ..., p_n) = (2\pi)^d \delta^{(d)} \left(q - \sum_{i=1}^n p_i \right) \prod_{i=1}^n \frac{d^d p_i}{(2\pi)^{d-1}} \, \delta^+ \left(p_i^2 - m_i^2 \right) \tag{2.85}$$

where, as described in the start of the current section, the δ-function imposes the mass-shell condition for the i-th final-state particle, while the superscript + selects its positive-energy solution. This may be written as,

$$\delta^+\left(p_i^2 - m_i^2\right) = \theta\left(p_i^0\right)\, \delta\left(p_i^2 - m_i^2\right) \tag{2.86}$$

This δ^+-function may be reexpressed as a discontinuity as follows [21, 25],

$$2\pi i\, \delta^+\left(p_i^2 - m_i^2\right) = \left[\frac{1}{p_i^2 - m_i^2 - i0} - \frac{1}{p_i^2 - m_i^2 + i0}\right]_+ \tag{2.87}$$

where the $\pm i0$ terms specify the propagator prescriptions. The subscript + represents the positive-energy cut selection.

These objects are called *cut propagators*, because they come about from cutting a propagator line, thus putting the represented particle on-shell, as a real final-state particle. Algebraically, it also allows phase-space constraints to be written in the same denominator language as virtual propagators. A cut propagator associated with a required final-state particle must be retained during the reduction and sectors in which such a cut is absent do not represent the physical phase-space integral and thus, vanish [21]. Using the discontinuity relation to refactor the δ-functions, we are able to rewrite the integrated antenna expression schematically as,

$$\mathcal{X}_n^l \sim \int \left[\prod_{r=1}^{l} d^d \ell_r\right] \left[\prod_u d^d p_u\right] \frac{N_s(\ell_r, p_u; \tilde{p}_{\text{hard}})}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}} \tag{2.88}$$

where $D_{v,\alpha}$ are ordinary virtual propagators, $D_{c,\beta}$ are cut propagators, $\tilde{p}_{\text{hard}}$ is the hard-radiator combined momenta and p_u are integrated unresolved momenta.

After rewriting the integral in this way, the total derivative introduced in the previous Subsection 2.5.2 may now be taken with respect to any loop momentum ℓ_r^μ or integrated unresolved momentum p_u^μ. This process produces relations involving both virtual and cut denominators. This then allows us to reduce loop and phase-space integrals simultaneously using IBP reduction [21, 25].

As an example, let us now consider the massless three-particle phase-space Φ_3. It may be described as,

$$d\Phi_3(q; p_1, p_2, p_3) \propto d^d p_1 d^d p_2 d^d p_3\, \delta^+\left(p_1^2\right)\, \delta^+\left(p_2^2\right)\, \delta^+\left(p_3^2\right)\, \delta^{(d)}(q - p_1 - p_2 - p_3). \tag{2.89}$$

We start by using momentum conservation to eliminate p_3,

$$p_3 = q - p_1 - p_2. \tag{2.90}$$

This first step eliminates the rightmost δ-function. Then, using reverse unitarity, each of the remaining δ-functions may be rewritten as cut propagators. Schematically, it gives,

$$d\Phi_3 \sim d^d p_1 d^d p_2 \frac{1}{[p_1^2]_c [p_2^2]_c [(q - p_1 - p_2)^2]_c} \tag{2.91}$$

with

$$\left[\frac{1}{p^2}\right]_c \equiv \frac{1}{2\pi i}\left[\frac{1}{p^2 - i0} - \frac{1}{p^2 + i0}\right]_+ . \tag{2.92}$$

Laporta Algorithm

At NLO and NNLO these reduction systems become very complex and their solution ever more time-intensive. The Laporta algorithm is a systematic-elimination method that uses IBP relations to express more complicated integrals in terms of a smaller set of unreduced ones (master integrals) [26].

The algorithm first classifies the family members into sectors. These sectors are defined as,

$$\theta_j \equiv \theta(a_j) = \begin{cases} 1, & a_j > 0, \\ 0, & a_j \leq 0, \end{cases} \qquad \vec{\theta} = (\theta_1, ..., \theta_N). \tag{2.93}$$

After fixing an integral family and identifying its admissible sectors, the Laporta algorithm assigns an ordering to the resulting integrals according to their complexity. This hierarchy is based on the number of propagators present t, the excess denominator power r, and the numerator complexity encoded through negative indices s[10]. These quantities are defined using each integral's a-index, taken from the family's $\vec{a}$ vector, as previously defined. They are, generally,

$$t = \sum_{j=1}^{N} \theta(a_j), \tag{2.94}$$

$$r = \sum_{j=1}^{N} \max(a_j - 1, 0), \tag{2.95}$$

$$s = \sum_{j=1}^{N} \max(-a_j, 0). \tag{2.96}$$

Here, $\theta(a_j)$ records whether denominator D_j is present. The hierarchy is defined by comparing t first, then r and finally s. The algorithm uses this chosen hierarchy to select, within each IBP relation, the most complex integral to eliminate first.

As an example, consider an arbitrary family defined as $\mathcal{F}[\{D_1, D_2, D_3\}; d\Phi]$, with the denominators chosen as $D_1 = s_{12}$, $D_2 = s_{13}$ and $D_3 = s_{23}$. We may take the integrals below, all members of this family, and organise them using their respective sector vectors as follows,

$$I_1 = \int d\Phi \, \frac{s_{13}}{s_{12}} \Rightarrow I_1 = I(1, -1, 0), \tag{2.97}$$

$$I_2 = \int d\Phi \, \frac{2}{s_{12}s_{13}s_{23}} \Rightarrow I_2 = 2I(1, 1, 1), \tag{2.98}$$

$$I_3 = \int d\Phi \, \frac{1}{s_{12}^3} \Rightarrow I_3 = I(3, 0, 0) \tag{2.99}$$

where $d\Phi$ is the common phase-space measure. Importantly, two integrals may share a sector if they contain the same denominators, independently of those denominators' powers. Also, only positive indices denote denominator presence, meaning that integrals I_1 and I_3 in the example above are members of the same sector, $I_1, I_3 \in (1, 0, 0)$. In our example, only two sectors appear: $(1, 0, 0)$ and $(1, 1, 1)$. These are

[10]The present t, r and s metrics are not to be taken as unique Laporta orderings, as these vary by implementation. Instead, they are illustrative objects.

our two target sectors. The three integrals are organised in terms of the t, r and s as follows,

$$\begin{array}{c|ccc} & t & r & s \\ \hline I(1,-1,0) & 1 & 0 & 1 \\ I(1,1,1) & 3 & 0 & 0 \\ I(3,0,0) & 1 & 2 & 0 \end{array} \tag{2.100}$$

then, by order of complexity, the algorithm orders the integrals as,

$$I(1,1,1) \succ I(3,0,0) \succ I(1,-1,0). \tag{2.101}$$

After determining the integral families and sectors present, we apply the IBP identities to the seed integrals. These seed integrals are a part of a finite, bounded set of integrals chosen from each relevant sector, whose IBP identities must be sufficiently numerous to reduce every target integral and every non-zero subtopology reached along the way.

After generating IBP identities for the selected seed integrals, any relation containing these representatives is first solved for $I(1,1,1)$, the highest-ranked member, and the resulting expression is substituted recursively into the remaining system [26].

IBP reduction may produce lower sectors, but may also generate new integrals from the initially relevant sectors. We may define the relevant sectors as the target sectors plus all lower sectors produced by IBP reduction. Lastly, some sectors may be cancelled due to symmetry, if one denominator is shifted to another under loop-momentum shift or a relabelling of external momenta. If two sectors become equivalent, one is discarded from the independent reduction and the other is retained as its representative. In short, a sector is deemed relevant if it is an original target sector, or a lower sector reached from another via IBP reduction. The relevant sector must also be non-scaleless in a chosen kinematics, not redundant under a family symmetry and, for a cut family, it must retain all required cut denominators throughout.

It is important to note that cut families add a physical restriction: if an IBP reduction step cancels a cut propagator, the sector it produces is missing one of the required on-shell final-state particles and it no longer represents the intended phase-space. If that happens, the integral is set to zero [21].

The reduction is complete when the remaining non-zero family members cannot be eliminated by the available identities. These survivors form a master-integral basis.

2.5.3 Master Integrals

The IBP reduction method reduces the integral to a linear combination of master integrals as,

$$I(\vec{a}) = \sum_{k=1}^{N_{\mathrm{MI}}} c_k(\vec{a}, \epsilon, \{s_{ij}\}) M_k \tag{2.102}$$

where c_k are the reduction coefficients and represent what comes from the reduction algebra and M_k are the MI. Our remaining task is evaluating the finite set M_k [26].

Master integrals are not uniquely defined: an integral is a master only relative to the chosen family, kinematics, IBP system, and basis. This means that a different basis with different masters may be used to reduce the same integrals and span the exact same integral space [26].

Master integrals are scalar integrals with a fixed denominator/cut structure. They are generically represented as,

$$M_k = \int \left[\prod_r d^d\ell_r\right]\left[\prod_u d^d p_u\right] \frac{1}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}}. \tag{2.103}$$

For integrated antennae, the cut denominators in these master integrals encode the real unresolved phase space. Hence, a master may be a pure phase-space volume, a phase-space integral with invariant denominators, or a loop-plus-phase-space cut integral. The MI are also not necessarily finite, in fact, in dimensional regularisation they commonly contain ϵ poles [21].

For a massless fully inclusive antenna system, q^2 is often the only independent kinematic scale once the dimensional-regularisation normalisation factors have been separated. Through dimensional analysis, the MI are strongly constrained to take the form,

$$M_k(q^2, \epsilon) = \left(q^2\right)^{\lambda_k + \kappa_k \epsilon} f_k(\epsilon) \tag{2.104}$$

where λ_k is fixed by mass dimension and f_k is dimensionless, depending only on ϵ. For this reason, phase-space MI can often be written in terms of Gamma functions, as [21],

$$f_k(\epsilon) \sim \frac{\Gamma^m(1 - a\epsilon)}{\Gamma^n(1 - b\epsilon)}. \tag{2.105}$$

When evaluating master integrals, different types often call for different approaches. Simple one-scale phase-space masters often integrate directly after the phase-space is parametrised. Angular and energy-fraction integrations typically reduce to Beta functions. When a master depends on a non-trivial kinematic variable, such as $x = m^2/q^2$, the most common approach is to differentiate the master with respect to that variable. The derivative produces integrals in the same family, which IBP is then able to reduce to the master basis, as [27],

$$\frac{d}{dx}\vec{M}(x, \epsilon) = A(x, \epsilon)\vec{M}(x, \epsilon). \tag{2.106}$$

where $\vec{M} = (M_1, ..., M_{N_{\text{MI}}})^{\text{T}}$ is the vector of masters and A is a matrix of coefficient functions. Boundary conditions can then be used to determine the solution. These methods are discussed in further detail in Section 4.1.

Once the MI are evaluated, we expand them around $\epsilon = 0$, as a Laurent series [21],

$$M_k = \sum_{n=n_{\text{min}}}^{\infty} m_{k,n}\epsilon^n \tag{2.107}$$

to expose their pole and finite structure. The lowest point n_{min} can, and often is, negative, resulting in an expansion similar to,

$$M_k = \frac{m_{k,-2}}{\epsilon^2} + \frac{m_{k,-1}}{\epsilon} + m_{k,0} + O(\epsilon). \tag{2.108}$$

The required expansion depth is determined by the reduction coefficient. If $c_k \sim \epsilon^{-p}$, the M_k must be expanded through to ϵ^p to obtain the finite contribution to $c_k M_k$.

The reduction techniques introduced in this chapter provide the computational tools required for the analytic integration of the antenna functions considered in this work. In particular, the IBP identities are systematically solved through the Laporta algorithm, reducing the phase-space and loop integrals to a finite set of master integrals. As described in Chapter 3, this reduction procedure is implemented within the *AntCalc* workflow and is subsequently applied in Chapter 4 to obtain the integrated antenna functions in dimensional regularisation. The resulting Laurent expansions in ϵ provide the explicit infrared pole structure required for their use in higher-order subtraction at NLO and NNLO.

Chapter 3

Package Framework

A central contribution of this thesis is the development of *AntCalc*[1], a *Wolfram Language* package created from scratch as part of this thesis work to automate the symbolic construction and integration of QCD antenna functions. The package provides direct access to both unintegrated and integrated antenna functions and implements the computational framework described in the previous chapter. The results presented in this thesis were produced with *AntCalc*, version 0.3.0β2, corresponding to commit `4ceb991`.

AntCalc makes use of *FeynCalc* [28, 29, 30, 31] for symbolic amplitude manipulation and interfaces with *LiteRed2* [32] for IBP reduction. Its internal architecture is organised as a profile-driven route system: route-specific information is centralised in profiles, which guide the selection and configuration of the appropriate build, reduction, and integration workflows. This establishes a common contract between the package components, reducing the amount of route-specific logic that must be duplicated when existing functionality is extended or new routes are introduced.

The package supports public and experimental routes. The public antenna building and integration routes, which for the current version compute all massless quark-antiquark final-final antenna functions up to NNLO, have been validated against the literature. Experimental routes correspond to functionality that is still under development or has not yet been fully validated and should therefore not be regarded as part of the validated public implementation.

AntCalc's inner working structure can be described using the diagram in Fig. 3.1.

3.1 Design Philosophy and Scope

AntCalc was built to combine the symbolic amplitude capabilities of the FeynCalc environment with the IBP-reduction capabilities of LiteRed2, adapting and orchestrating the parts of those systems required for the construction and integration of QCD antenna functions.

The central design objective of the package is to return antenna functions in the conventional forms required for higher-order QCD calculations using the antenna subtraction method. Though it supports a wide variety of options to inspect and configure the backend operations, the default outputs of its two central functions are designed to reflect this objective. For example, the unintegrated one-loop three-parton antennae X_3^1 returned by `BuildAntenna[...]` are expressed after Passarino-Veltman reduction, in terms of scalar one-loop functions and the external kinematic invariants, following the conventional representation used in the literature. By default, `IntegrateAntenna[...]` returns

[1] Source code available at `https://github.com/HenFar/AntCalc/tree/release/0.3.x`.

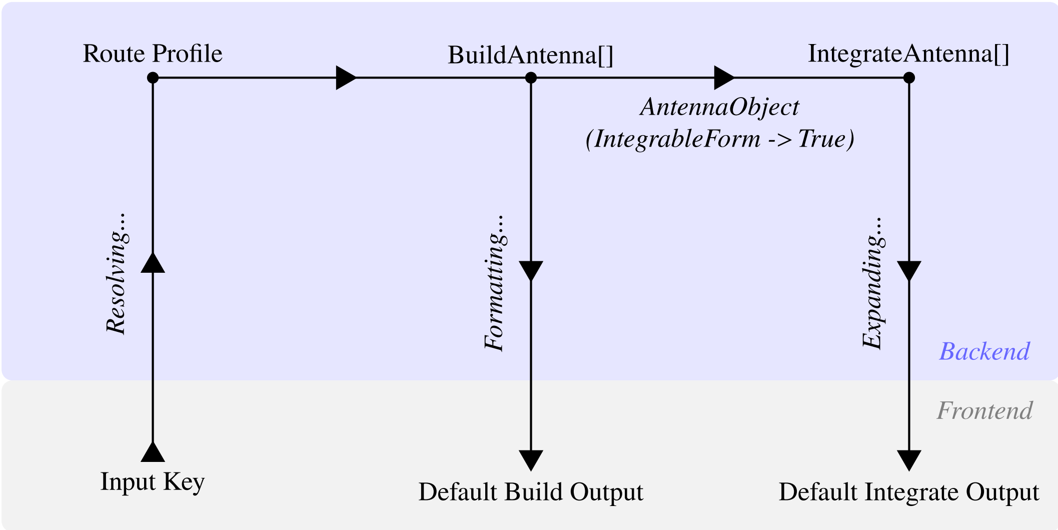

Figure 3.1: Profile-driven architecture of *AntCalc*. Route profiles configure the build and integration workflows, while the build-side route context carries the evolving calculation data. The `AntennaObject` provides the integration-ready handoff between the two stages.

the integrated antennae as Laurent series in ϵ, with the truncation order chosen to match the conventions used in the antenna literature. The NLO antennae $\mathcal{A}_2^1$ and $\mathcal{A}_3^0$ are expanded through $O(\epsilon^2)$, while the NNLO antennae $\mathcal{A}_4^0$, $\mathcal{A}_3^1$, $\mathcal{A}_2^2$, $\mathcal{B}_4^0$, and C_4^0 are returned through $O(\epsilon^0)$.

Its structure was also designed to be modular and extensible. Each of the two central functions (`BuildAntenna` and `IntegrateAntenna`) is implemented as an encapsulated public module, which communicates with the rest of the codebase using the profile system: the route profile selects and configures the appropriate workflows and the route context carries the evolving results and diagnostics between stages. Using route profiles and route contexts, as described in Section 3.2, facilitates the addition of further modules while reducing duplicated logic and direct dependencies. These new modules need not be compatible with previous implemented functions through function-specific couplings. They need only conform with the profile system, being selected and configured by route profiles, and accepting inputs from and delivering outputs to the route contexts.

Currently, *AntCalc* is organised into two tiers: public and experimental. The public tier contains all massless quark-antiquark final-final antenna functions required through NNLO (all partons are present in final-state kinematics), including the objects needed for the calculation of the massless hadronic R-ratio at this order. The antenna functions currently implemented are summarised in Table 3.1.

Perturbative Order	**Antenna**	**Status**
LO	A_2^0	Public
NLO	A_2^1, A_3^0	Public
NNLO	A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$, $\check{A}_2^2$, A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$, A_4^0, $\tilde{A}_4^0$, B_4^0, C_4^0	Public
NLO, massive	A_3^0	Experimental

Table 3.1: Overview of the antenna functions currently implemented in *AntCalc*, by perturbative order and tier.

In addition to the validated public implementation, a massive A_3^0 route is available as experimental functionality.

3.2 Profile-Driven Architecture

As described in Section 3.1, *AntCalc* relies on a profile-driven routing system. This system maps the route key to its route-specific profile and workflow.

A public antenna route key is a three-element list which describes a canonical route whose execution may be refined by options. The route key takes the form,

$$\{\text{Antenna Type}, \text{Final-State Multiplicity}, \text{Loop Order}\}. \tag{3.1}$$

As an example, the route key for the one-loop three-parton [2] antenna set is $\{\text{A}, 3, 1\}$. This key represents the complete set of colour contributions, comprising the leading-colour antenna A_3^1, the subleading-colour antenna $\tilde{A}_3^1$, and the quark-loop contribution $\hat{A}_3^1$. The default public output is therefore the ordered list $\{A_3^1, \tilde{A}_3^1, \hat{A}_3^1\}$.

The package links this key to the profile it identifies and follows that profile's directions on how the route should be constructed and/or integrated. The data produced while the route executes is also stored in the route context association. The profile system does not consist of a single profile, but rather of three main and some conventional auxiliary profiles which are routinely read while the route executes. The principal build-side profile is `AntennaProfile[key]`, which specifies information such as antenna type, multiplicity and loop order, colour normalisation, extraction rules, components, etc. `AntennaReductionProfile[key]` is separate, but still build-side. It specifies the Passarino-Veltman-reduction preferences for the appropriate routes. In this case, up to NNLO, these are the $\{\text{A}, 2, 1\}$ and $\{\text{A}, 3, 1\}$ loop routes. `AntennaIntegrationProfile[key]` is the integration-side profile, which specifies its side's backend, IBP basis families, expansion depth, etc. The specifics of each stage's profiles and options will be discussed in their respective sections, 3.3 and 3.5.

While profiles select and configure computation paths, the workflows themselves perform the calculations. The route dispatcher and workflow code consult the profiles to select and configure each stage. As an example, which will be further explained and explored in the two upcoming sections, let us consider the call `BuildAntenna[A,2,1]`. Its full route profile is,

```
AntennaProfile[{A, 2, 1}] :=
    <|
        "Key" -> {A, 2, 1},
        "Name" -> "A21",
        "AntennaType" -> A,
        "NumFinalParticles" -> 2,
        "LoopOrder" -> 1,
        "TreeAmplitude" -> AntennaAmplitude[{A, 2, 0}],
        "BornInterference" -> BornInterference[],
        "Extraction" -> "LoopScalar",
        "ColourNorm" -> colourNorm,
        "ExcludedParticles" ->
            {S[_], V[1], V[2], V[3], U[1], U[2], U[3], U[4]},
        "DropSumOver" -> True,
        "ReductionProfile" -> AntennaReductionProfile[{A, 2, 1}],
```

[2]It is understood that throughout the rest of the present work, unless explicitly mentioned, all antenna functions considered are massless quark-antiquark final-final antennae.

```
        "ConventionProfile" -> BuildAntennaConventionProfile[{A, 2, 1}]
    |>
```

In this profile, the key, name, antenna type, number of final particles (multiplicity), and loop order have been discussed before. The remaining entries configure the construction of the route: `TreeAmplitude` supplies the tree-level A_2^0 amplitude with which the one-loop amplitude is interfered, `BornInterference` selects the Born-level self-interference used to normalise the loop interference. `Extraction` selects the scalar one-loop extraction appropriate for this antenna, while `ColourNorm` provides the colour normalisation used. `ExcludedParticles` and `DropSumOver` are settings passed to the *FeynCalc*/*FeynArts* part of the build workflow, selecting what particles are not to be considered in the Feynman diagram generation step and requesting the removal of external `SumOver` structures. Lastly, `ReductionProfile` commands the workflow to complete a lookup on the selected route's antenna, A_2^1 in this case, and determines whether it requires the Passarino-Veltman reduction backend, from within its `AntennaReductionProfile` association. Since it is a one-loop antenna, it does. `ConventionProfile` selects the convention bridge so the public results follow *AntCalc*'s stated normalisation and dimensional-regularisation conventions, through its separately defined `BuildAntennaConventionPr` association.

3.3 The Build Stage

The build stage constructs the unintegrated antenna functions from the corresponding matrix elements obtained from Feynman diagrams. Its main public interface is `BuildAntenna[type, multiplicity, loop order]`. The workflow begins with a route key specified through a public call, for example `BuildAntenna[A, 3, 1]`. From this key, *AntCalc* identifies the appropriate construction workflow and the ingredients required for the corresponding antenna set. The relevant amplitudes are then constructed and combined according to the route definition, after which the resulting expressions are processed into the required unintegrated antenna functions. Internally, the intermediate results are stored in an association containing the different components of the calculation. The final step reformats this internal representation into the antenna expression, or ordered set of antenna components, returned to the user. The build workflow can therefore be summarised as follows,

```
BuildAntenna[type, multiplicity, loop order]
    -> route key;
    -> loop-order dispatcher;
    -> route-specific build workflow;
    -> build-data association / route context;
    -> public-output boundary;
    -> output: expression, component list, AntennaObject, diagnostics,
       or run record.
```

The loop-order dispatcher is a central element of this workflow. Using the route key, it directs each antenna to the appropriate computational path, since different loop orders require different computational treatments, most notably the tensor reduction required for the loop-momentum dependence of one- and two-loop antennae.

Functionally, the route-context for this stage usually contains the initial profile itself (the declarative `AntennaProfile`), the amplitude generated from it, the sectors, computed interferences, components after separation, diagnostics, and its normalised interference. Depending on the route, the context might also retain prototype components, full colour components, reference square components or loop-specific amplitudes.

A diagram in the spirit of the one in Fig. 3.1, but only relative to the build stage is shown below, in Fig. 3.2.

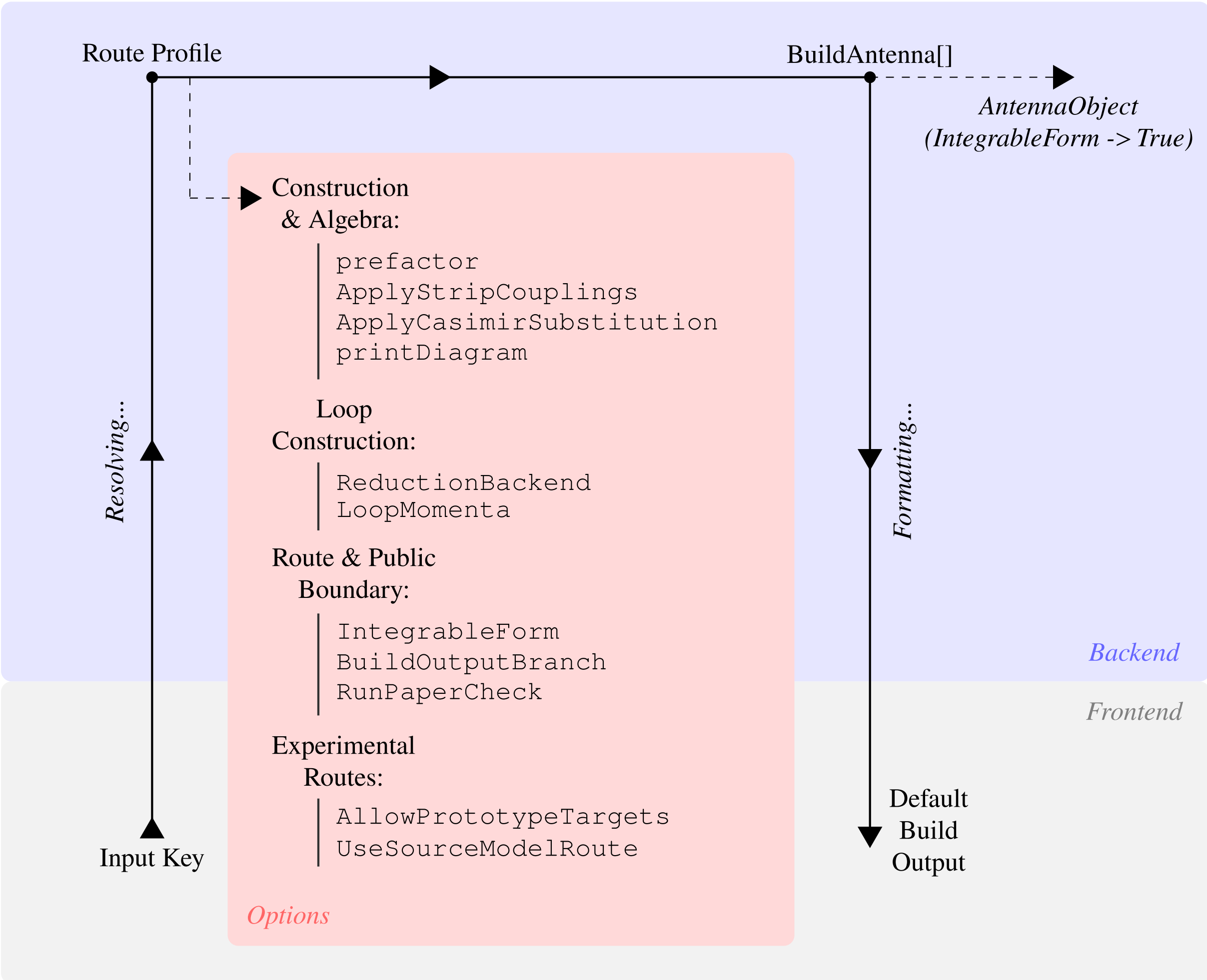

Figure 3.2: Profile-driven architecture of *AntCalc*'s Build stage. The principal build-specific options to the `BuildAntenna` function are displayed on the red card.

3.3.1 Tree-Level Routes

As mentioned, the workflow handles distinct loop-order routes differently, though the differences only appear after the dispatcher is called. Tree-level routes like those relative to the A_3^0 antenna, start by retrieving the profile and the interference type (`BornInterference` in this case), and computing the relevant process's amplitude. It uses the profile's `Production` field to select the physical construction path, and the `Extraction` and `ColourNorm` fields to select the extraction mode and the colour normalisation used for each antenna. The colour normalisation used by default has been described previously in Eq. 2.38, as $c_N \equiv N - \frac{1}{N}$. The `Extraction` mode also introduces the Born normalisation as described in Eq. 2.33 as $B_0 \equiv |\mathcal{M}_2^0|^2$. The routes selected in the `Production` and `Extraction` fields depend on the methodology required to compute the antenna functions and retrieve the components, if required. The public tree-level antenna functions are constructed using different production and extraction modes, as described in Table 3.2.

Production Mode	Extraction Mode	Current Route
`SelfInterference`	`BornScalar`	A_2^0
`SelfInterference`	`TreeColourCoefficients`	A_3^0
`ColourOrderedAntenna`	`TreeColourCoefficients`	A_4^0, $\tilde{A}_4^0$
`SectorSelfInterference`	`ColourNormScalar`	B_4^0
`SectorSymmetrisedInterference`	`MinusTwoColourNormOverSUNN`	C_4^0

Table 3.2: Overview of the tree-level production and extraction modes.

These production paths all lead to the extraction of the required antenna functions, but the way in which they do it changes with the antenna's multiplicity. `SelfInterference` starts by generating the amplitude, computing its self-interference, and then extracting the antenna. `ColourOrderedAntenna` generates the full-process amplitude and computes the full-colour interference. It then applies the colour-ordered reconstruction as specified by `ColourOrderedSpecs`, and then extracts the components, in this case the Leading and Subleading components. `SectorSelfInterference` generates the full-process amplitude and splits it into profile-defined sectors. It selects the relevant sector, computes its self-interference and extracts the antenna function. Finally, `SectorSymmetrisedInterference` starts by generating the full-process amplitude, and splitting it into direct and exchange sectors. It then computes the required symmetrised interference and extracts the antenna. A reference square is also retained for diagnostics and comparison. The extraction paths lead to the separation of the whole-process results into colour-independent antenna functions. The `BornScalar` mode is used solely in A_2^0, and it does not algebraically divide the computed interference by its Born form, establishing the antenna as the reference normalisation. It returns $A_2^0 = 1$. `TreeColourCoefficients`, as used for A_3^0, A_4^0 and $\tilde{A}_4^0$, starts by normalising the computed amplitude by the colour and the Born normalisation. It then simplifies and expands the normalised interference and searches for the N, $1/N$ and N_f coefficients, the `Leading`, `SubLead` and `QuarkLoop` coefficients in the code. If the terms are present, as described in the previous text, the interference $\mathcal{J}$ becomes,

$$\frac{\mathcal{J}}{c_N B_0} = N X_{\text{Lead}} + \frac{1}{N} X_{\text{SubLead}} + N_f X_{\text{QuarkLoop}} \tag{3.2}$$

and the X terms extracted as the antenna functions. If there are no terms of that shape, like in the A_3^0 case, then $X = \mathcal{J}/(c_N B_0)$, colour- and Born-normalised. It is important to note that, though present, no quark loop antenna should ever appear in the tree-level route. The dispatcher still considers it nonetheless since it is written to be reusable. For B_4^0, the `ColourNormScalar` mode is used, after the primary-current sector has been self-interfered. The extraction then returns the colour- and Born-normalised sector interference, as,

$$B_4^0 = \frac{\mathcal{J}_{\text{sector}}}{c_N B_0}. \tag{3.3}$$

Lastly, `MinusTwoColourNormOverSUNN` is used for C_4^0, and it takes the symmetrised interference of the direct and exchange sectors and normalises it as,

$$C_4^0 = -\frac{N \mathcal{J}_{\text{sym}}}{2 c_N B_0}. \tag{3.4}$$

The C_4^0 antenna requires a further $-N/2$ normalisation factor due to its origin as the interference of the direct and exchange flavour-flow amplitudes for identical quarks. The minus sign reflects the exchange of identical fermions, the $1/2$ factor is required when the symmetrised source contains both cross terms, as discussed in Eq. 2.42. The remaining N factor, along with the usual c_N in the denominator adjusts the interference's colour coefficient into the colour-free C_4^0 antenna.

3.3.2 One-Loop Routes

One-loop routes are dispatched to `BuildLoopAntennaData` where the primary backend reduction method is selected through the nested `ReductionProfile`, which by default is Passarino-Veltman reduction, as described in Subsection 2.5.1.

The production method consists of generating the tree-level and one-loop amplitudes, using `TreeAmplitude` and `MAmpOneLoop`, respectively, and then forming the tree-loop interference. The interference is colour- and one-loop-normalised, using c_N and $\Lambda_l = (8\pi^2)^l$, for loop order l, as described in Eq. 2.19. To this normalised interference is then applied the reduction and the components extracted. The one-loop antennae require different extraction procedures depending on their colour structure. The production and extraction modes implemented for the supported one-loop routes are summarised in Table 3.3, below.

Production Mode	Extraction Mode	Current Route
`TreeAmplitude + MAmpOneLoop`	`LoopScalar`	A_2^1
`TreeAmplitude + MAmpOneLoop`	`LoopColourCoefficients`	A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$

Table 3.3: Overview of the one-loop production and extraction modes.

Both antenna sets are produced in the same manner, as discussed. The relevant tree-level antenna amplitude, $\mathcal{M}_2^0$ and $\mathcal{M}_3^0$ respectively for A_2^1 and the A_3^1 set, is first computed. Then it is interfered with the loop amplitudes, $\mathcal{M}_2^1$ and $\mathcal{M}_3^1$. The interference is computed, before being normalised as,

$$\mathcal{J}_n^l = 2\,\mathrm{Re}\left(\left(\mathcal{M}_n^0\right)^\dagger \mathcal{M}_n^l\right) \tag{3.5}$$

where n represents the number of final-state particles, as usual.

Since A_2^1's interference does not decompose into colour components, $A_2^1 = \Lambda_1 \mathcal{J}_2^{1\prime} = \Lambda_1 \mathcal{J}_2^1/(c_N B_0)$. The A_3^1 set, on the other hand, appears closely like the A_4^0 set, but with three, rather than two components, the leading-colour, subleading-colour and quark-loop antenna functions. The interference is given as in Eq. 3.2. Lastly, as discussed in Section 2.3 and Eq. 2.23, together with Eq. 2.33, the A_3^1 set needs to be UV renormalised. Consider the notation $X_n^{l,\mathrm{bare}}$ here to denote a bare antenna, such that, for example, $A_2^{1,\mathrm{bare}} = A_2^1$. For the A_3^1 set, the interference expands to,

$$\frac{\Lambda_1}{c_N B_0}\mathcal{J}_3^1 = N T_{\mathrm{Lead}} + \frac{1}{N} T_{\mathrm{Sublead}} + N_f T_{\mathrm{QL}}. \tag{3.6}$$

Differently from other processes at this level, the A_3^1 route necessitates a further subtraction, as the interference components come in terms of a genuine one-loop three-parton antenna and an iterated lower-order contribution. That makes it so that the bare (before UV renormalisation) A_3^1 set antenna functions become,

$$
\begin{aligned}
A_3^{1,\text{bare}} &= T_{\text{Lead}} - A_2^1 A_3^0 \\
\tilde{A}_3^{1,\text{bare}} &= -\left(T_{\text{Sublead}} + A_2^1 A_3^0\right) \\
\hat{A}_3^{1,\text{bare}} &= T_{\text{QL}}.
\end{aligned}
\tag{3.7}
$$

The components are then, finally, renormalised as discussed,

$$
\begin{aligned}
A_3^1 &= A_3^{1,\text{bare}} - \frac{11}{6\epsilon} A_3^0 \\
\tilde{A}_3^1 &= \tilde{A}_3^{1,\text{bare}} \\
\hat{A}_3^1 &= \hat{A}_3^{1,\text{bare}} + \frac{1}{3\epsilon} A_3^0.
\end{aligned}
\tag{3.8}
$$

Each of these components may also be extracted alone as a single antenna, using the `Component` option. If, while using *AntCalc*, one ran `BuildAntenna[A,3,1, Component->Subleading]`, for example, only the unintegrated $\tilde{A}_3^1$ antenna would be returned.

3.3.3 Two-Loop Routes

Loop order two routes are dispatched to `BuildTwoLoopAntennaData`. There is only, formally, one set of this kind, A_2^2. It is remarkably different from the rest, however. First, these objects, alongside A_2^0 and A_2^1, are not formally antenna functions as they fail to meet the definition of a structure that captures the singular behaviour of an unresolved parton radiated between two hard, colour-connected partons [4]. They are, instead, form factors required by the antenna framework. The A_2^2 set, at two loops, is obtained from the two-loop quark form factor [22]. Second, this four-component set is not generated like A_4^0 and A_3^1 were, rather these four components come from two distinct source contributions. Their production and extraction modes are similar and handled as a bundle inside *AntCalc*, but, as shown in Table 3.4, their sources differ,

Contribution	Extraction Mode	Current Route
`TwoLoopInterf`	`TwoLoopTTermComponents`	A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$
`OneLoopSelf`	`TwoLoopTTermComponents`	$\breve{A}_2^2$

Table 3.4: Overview of the two-loop contributions and extraction modes.

The first three antenna objects are formed by interfering the LO tree-level amplitude with the two-loop amplitude. This interference follows the blueprint defined in Eq. 3.5, as the interference becomes,

$$
\mathcal{J}_2^{2,[2\times 0]} = 2\,\text{Re}\left(\left(\mathcal{M}_2^0\right)^\dagger \mathcal{M}_2^2\right).
\tag{3.9}
$$

The superscript $[2\times 0]$ denotes the nature of the operation: interfering a two-loop with a zero-loop amplitude. It is also usual to split the T-object relative to these four components, $T_{q\bar{q}}^6$, into two, based on the two contributions.

The last object, $\breve{A}_2^2$, comes from self-interfering the one-loop amplitude as,

$$
\mathcal{J}_2^{2,[1\times 1]} = \left(\mathcal{M}_2^1\right)^\dagger \mathcal{M}_2^1.
\tag{3.10}
$$

Both of these interferences are equally Born- and loop-normalised, using B_0 and Λ_2. They are also colour-normalised, but while $\mathcal{J}_2^{2,[2\times0]}$ requires only c_N, $\mathcal{J}_2^{2,[1\times1]}$ requires c_N^2 since it is computed using two one-loop amplitudes, each of which requires a c_N normalisation factor, as described in Subsection 2.4.3. They, then, become,

$$\begin{aligned}\frac{\Lambda_2}{c_N B_0}\mathcal{J}_2^{2,[2\times0]} &= N A_2^{2,\text{bare}} + \frac{1}{N}\tilde{A}_2^{2,\text{bare}} + N_f \hat{A}_2^{2,\text{bare}}\\ \frac{\Lambda_2}{c_N^2 B_0}\mathcal{J}_2^{2,[1\times1]} &= \breve{A}_2^{2,\text{bare}}.\end{aligned}\tag{3.11}$$

The antenna objects have to then be UV renormalised as follows, following the coupling renormalisation described in Section 2.3 and Eq. 2.23,

$$\begin{aligned}A_2^2 &= A_2^{2,\text{bare}} - \frac{11}{6\epsilon}A_2^1\\ \tilde{A}_2^2 &= \tilde{A}_2^{2,\text{bare}}\\ \hat{A}_2^2 &= \hat{A}_2^{2,\text{bare}} + \frac{1}{3\epsilon}A_2^1\\ \breve{A}_2^2 &= \breve{A}_2^{2,\text{bare}}.\end{aligned}\tag{3.12}$$

This route may be schematically represented as shown in Fig. 3.3, below.

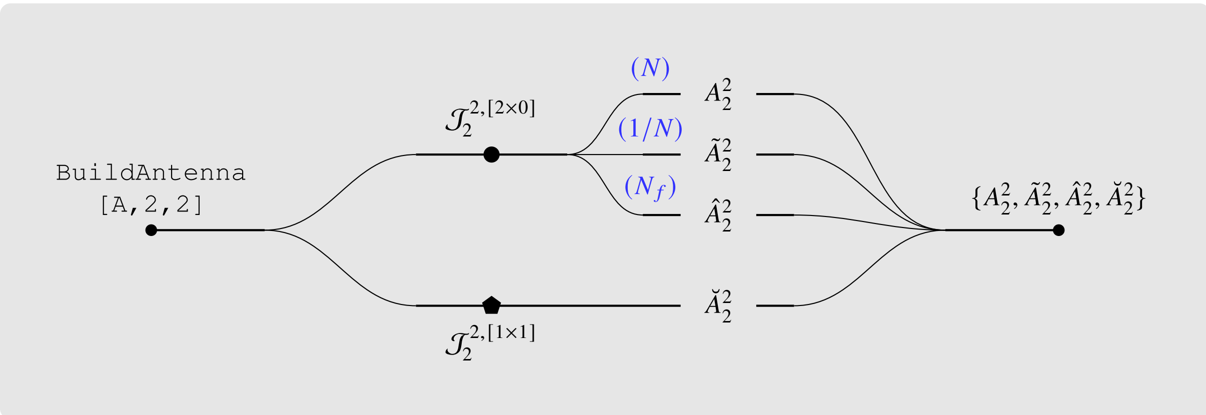

Figure 3.3: Diagram showing the build-side workflow leading to the generation of the A_2^2 antenna set. The two subsets are separated by the interferences used in their generation.

These renormalised results are now used to form the overall $T_{q\bar{q}}^6$ object. It is defined as the sum of the two contributions' sub-T-objects, $T_{q\bar{q}}^{6,[2\times0]}$ for the first, and $T_{q\bar{q}}^{6,[1\times1]}$ for the second contribution, following the interferences $\mathcal{J}$ notation as,

$$\begin{aligned}T_{q\bar{q}}^6 &= \left(N - \frac{1}{N}\right) T_{q\bar{q}}^2 \left[T_{q\bar{q}}^{6,[2\times0]} + T_{q\bar{q}}^{6,[1\times1]}\right]\\ &= \left(N - \frac{1}{N}\right) T_{q\bar{q}}^2 \left[\frac{\Lambda_2}{c_N B_0}\mathcal{J}_2^{2,[2\times0]} + \frac{\Lambda_2}{c_N^2 B_0}\mathcal{J}_2^{2,[1\times1]}\right]\\ &= \left(N - \frac{1}{N}\right) T_{q\bar{q}}^2 \left[N A_2^2 + \frac{1}{N}\tilde{A}_2^2 + N_f \hat{A}_2^2 + \left(N - \frac{1}{N}\right)\breve{A}_2^2\right].\end{aligned}\tag{3.13}$$

3.4 From the Build Stage to Integration

The build stage described above produces the unintegrated antenna functions in terms of the external kinematic invariants s_{ij}. This is the natural public representation of the antenna and is suitable for its use as a local subtraction term in a numerical phase-space integration, as discussed in Chapter 2. A standard `BuildAntenna[...]` call therefore returns either a single antenna expression or, where several colour structures are present, an ordered list of antenna components.

For the analytic calculation of the integrated antennae, however, the final expression in terms of Mandelstam invariants is not by itself sufficient. The integration stage requires additional information from the construction of the antenna, including the route definition and the integral structures needed for the subsequent reduction and phase-space integration. *AntCalc* therefore provides the option `IntegrableForm -> True`. With this option, `BuildAntenna[...]` returns an `AntennaObject` that retains both the constructed antenna and the additional information required by `IntegrateAntenna[...]`. This object provides the interface between the build and integration stages.

At this point, *AntCalc* can therefore provide the unintegrated antennae in the kinematic form required for local subtraction. The next stage addresses the complementary part of the antenna subtraction procedure: their analytic integration over the unresolved phase space in dimensional regularisation. These integrated antennae are required to combine the subtraction terms with the virtual contributions and achieve the cancellation of explicit infrared poles. The implementation of this integration stage is described in the following section.

3.5 The Integration Stage

The integration stage performs the analytic integration of the antenna functions constructed in the build stage over the corresponding unresolved phase space in $d = 4 - 2\epsilon$ dimensions, as in Eq. 2.47. Its purpose is to obtain the integrated antennae as Laurent series in ϵ, in the form required for their use in the antenna subtraction framework. The integrated antennae are denoted with calligraphic letters.

Within *AntCalc*, this stage is initiated by passing the `AntennaObject` produced by `BuildAntenna[..., IntegrableForm -> True]` to `IntegrateAntenna[...]`. The route key stored in this object determines the appropriate integration workflow, including the required reduction procedure, master integrals, and normalisation conventions. After these steps have been applied, `IntegrateAntenna[...]` returns the integrated antenna, or the corresponding set of antenna components, expanded as a Laurent series in ϵ. The integration workflow is structured as follows,

```
IntegrateAntenna[AntennaObject]
    -> recover route key and resolve profile;
    -> choose integration backend;
    -> complete IBP/Passarino-Veltman reduction;
    -> substitute master integrals when applicable;
    -> extract public integrated antennae;
    -> output: selected component, list, diagnostics, run record, or
       master-combination view.
```

It is relevant to note that `IntegrateAntenna` requires the profile to complete its function like `BuildAntenna`, since it also encodes the relevant routing information for the reduction and integration process.

Much like for `BuildAntenna`, there is no single route context object, rather the relevant information is stored around multiple objects that are accessed when required. These objects store initiation-crucial elements, like the key and the profile, the selected component and the input (unintegrated) antenna, some IBP-crucial elements, like the bases required. the master integrals to use, and the series result, along with others like the route story, or the diagnostics.

An integration-stage-specific diagram in the spirit of the one in Fig. 3.1 can also be drawn in further detail. That diagram is shown in Fig. 3.4.

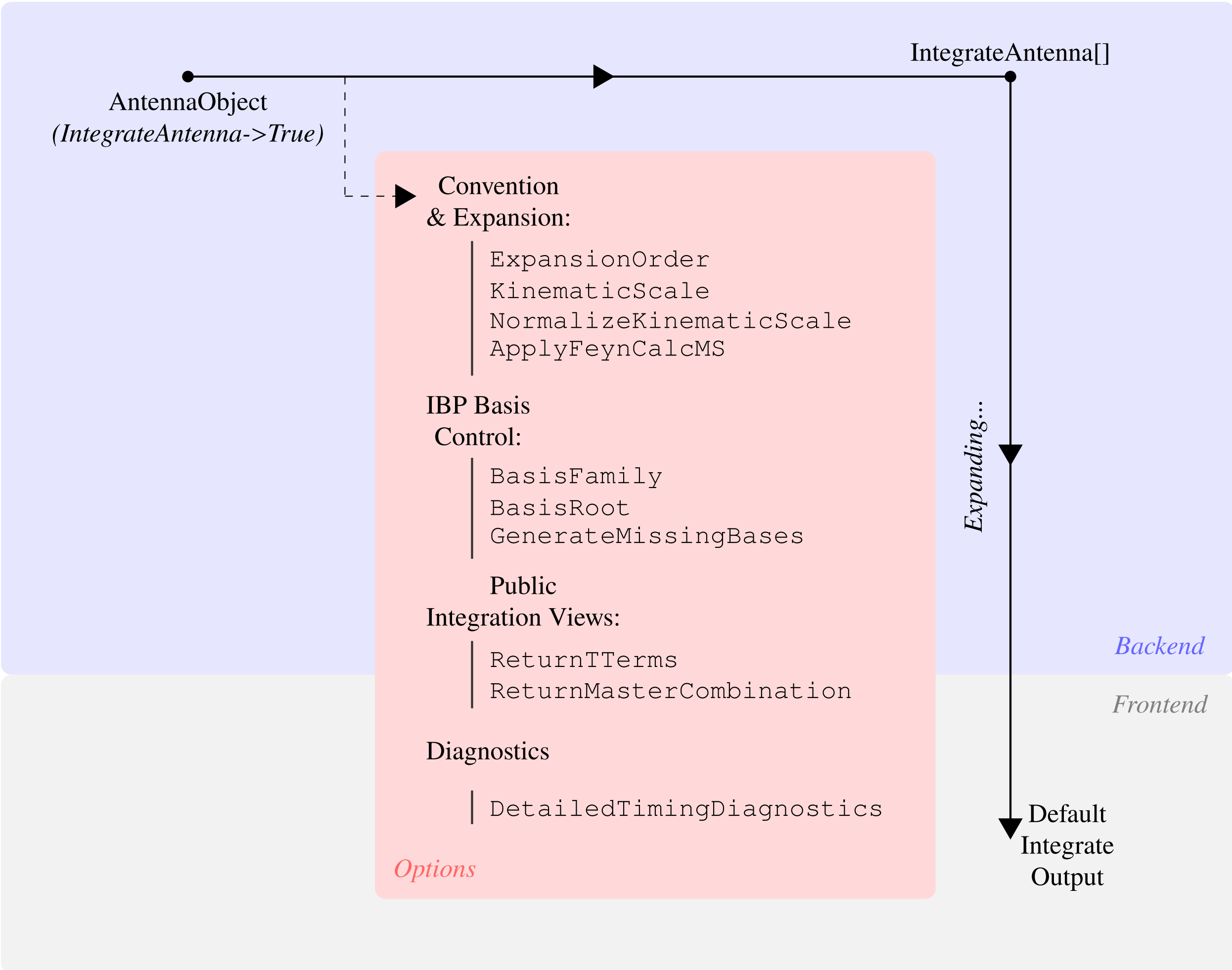

Figure 3.4: Profile-driven architecture of *AntCalc*'s Integration stage. The principal integration-only options are shown inside the red card.

3.5.1 Integration Profiles and Selection

The integration-specific profile for each route selects the kind of backend process (and external package workflow used), and the basis family required for the process. The antennae are divided by topology, that is, number of final-state particles and number of loops, and all of those that share the same topology in the current implementation, share the same basis-family set. The exception to this rule is the A_2^2 set, in which different sets are used depending on configuration. The A_2^1 antenna is also an exception to the use of integral families. Its loop integrals are reduced directly to scalar one-loop integrals using Passarino-Veltman reduction, whose known analytic expressions are then substituted and expanded as Laurent series in ϵ. An overview of these is shown in the following Table 3.5.

Default Backend	**Bases-Family Set**	**Current Route**
Passarino-Veltman / *FeynHelpers*	massless two-parton vertex	$\mathcal{A}_2^1$
IBP / *LiteRed2*	`X30`	$\mathcal{A}_3^0$
IBP / *LiteRed2*	`X40`	$\mathcal{A}_4^0, \tilde{\mathcal{A}}_4^0, \mathcal{B}_4^0, C_4^0$
IBP / *LiteRed2*	`A31`	$\mathcal{A}_3^1, \tilde{\mathcal{A}}_3^1, \hat{\mathcal{A}}_3^1$
IBP / *LiteRed2*	`A22TwoLoopTree`	$\mathcal{A}_2^2, \tilde{\mathcal{A}}_2^2, \hat{\mathcal{A}}_2^2$
IBP / *LiteRed2*	`A22OneLoopSelf`	$\breve{\mathcal{A}}_2^2$

Table 3.5: Overview of the integration-side route-profiles.

As on the build side, the integration profile does not itself perform any calculation; rather, it specifies the information required to execute the corresponding integration workflow. Consider, for example, the integration of the B_4^0 antenna. From the route key stored in the `AntennaObject`, *AntCalc* identifies the appropriate integration profile and the integral family associated with the X_4^0 phase-space topology. The unintegrated antenna is then mapped onto this integral family and reduced by IBP, using *LiteRed2*, to the corresponding set of master integrals. Their known analytic expressions are subsequently substituted to obtain the integrated antenna.

The master-integral basis used in these reductions is discussed in Chapter 4, where selected master integrals are evaluated explicitly and their role in the analytic integration of antenna functions is illustrated through representative examples.

3.5.2 Integration Routes

Passarino-Veltman Route

The A_2^1 antenna constitutes a special low-multiplicity virtual case and is the only integration route implemented without IBP reduction. As a two-parton one-loop form-factor contribution, its loop integrals can be treated directly using Passarino-Veltman reduction, which reduces the tensor loop integrals to scalar one-loop functions. *AntCalc* evaluates these scalar functions analytically using *Package-X* [33] through *FeynHelpers* [34]. The required branch and scale conventions are then applied, followed by the appropriate kinematic normalisation and Laurent expansion in ϵ, yielding the integrated $\mathcal{A}_2^1$ antenna. The workflow is summarised schematically in Fig. 3.5.

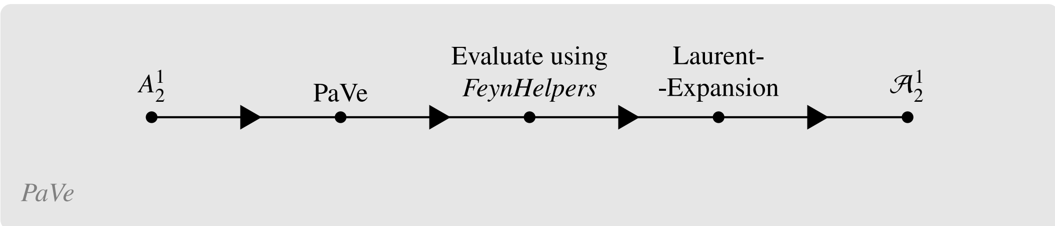

Figure 3.5: PaVe reduction for the A_2^1 antenna integration.

IBP Route

All remaining antenna functions are integrated using IBP reduction, implemented through *LiteRed2* [32], following the formalism described in Section 2.5. The antenna integrand is first expressed in terms of the momenta, invariants, and denominator structures defining the relevant integral families. The appropriate phase-space cut propagators are then introduced, allowing the phase-space integrals to be treated within the same integral-family framework as loop integrals. After expanding the antenna into individual integral terms, each term is assigned to the corresponding integral family and reduced by *LiteRed2* to a linear combination of master integrals. The reduced terms are subsequently recombined to obtain the complete integrated antenna in terms of the corresponding master-integral basis.

For the public routes, the master integrals (MIs) identified by the IBP reduction are matched to their corresponding analytic expressions known from the literature and evaluated explicitly in this work. Their expressions are then substituted by *AntCalc*. Finally, the required normalisation and dimensional-regularisation conventions are applied, and the result is expanded as a Laurent series in ϵ, yielding the integrated antenna function. Selected master integrals and explicit examples of their analytic evaluation are presented in Chapter 4, together with representative applications of this procedure to the integration of antenna functions. The complete IBP-based integration workflow is summarised schematically in Fig. 3.6.

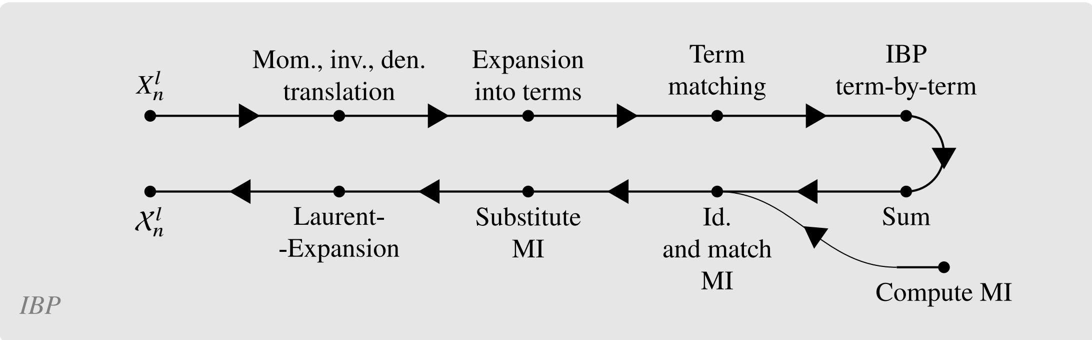

Figure 3.6: IBP reduction workflow for the remaining antennae integrations. The MI computation is external to *AntCalc*'s internal operation, and hence is represented by an external incoming arrow.

3.5.3 Normalisations and Convention Bridge

As discussed in Section 2.3, before integration, the antenna functions must be normalised using the $\mathcal{N}(\epsilon, k)$ factor, introduced in Eq. 2.22. Since in Section 3.3, where we discussed the build stage, the unintegrated antenna functions were normalised using the loop-order factor, Λ_l, in this stage, the rest of the normalisations and conventions must be applied. That begins with the application of what remains from the G_k factor and then $C(\epsilon, k)$. Finally, we reach the full $\mathcal{N}(\epsilon, k)$ by dividing by the two-particle phase-space measure. This is, schematically,

$$\begin{aligned} \mathcal{X}_n^l &= \mathcal{N}(\epsilon, k) \int d\Phi_{X_{ijk}}\, X_n^l \\ &= \frac{1}{\Phi_2} \left(8\pi^2\right)^{n-2} S_\epsilon \int d\Phi_{X_{ijk}} \left(\Lambda_l X_n^l\right). \end{aligned} \tag{3.14}$$

Note, however, that not all routes require that the rest of the G_k be applied, based on their n. There are also routes that did not require the application of the Λ_l factor in the build stage. The following Table 3.6 shows how each antenna or antenna set interacts with G_k.

Route	n, l	G_k relative to build Λ_l
A_2^1	$2, 1$	$G_1 = \Lambda_1$ (no extra multiplicity factors needed)
A_3^0	$3, 0$	$G_1 = 8\pi^2\Lambda_0 = 8\pi^2$
A_3^1 set	$3, 1$	$G_2 = 8\pi^2\Lambda_1$
A_4^0 set, B_4^0, C_4^0	$4, 0$	$G_2 = (8\pi^2)^2\Lambda_0 = (8\pi^2)^2$
A_2^2 set	$2, 2$	$G_2 = \Lambda_2$ (no extra multiplicity factors needed)

Table 3.6: Overview of the interactions of the antenna routes with the normalisation factor G_k.

Expansion Order

The final step in obtaining an integrated antenna $\mathcal{X}_n^l$, or an ordered list of its components, is to expand the result as a Laurent series in ϵ. The default truncation order depends on the perturbative order of the antenna. Following the conventions used in the literature, the NLO antennae $\mathcal{A}_2^1$ and $\mathcal{A}_3^0$ are provided through $O(\epsilon^2)$, whereas the NNLO antennae are provided through $O(\epsilon^0)$. For the currently implemented routes, the available expansion depth is also determined by the analytic master-integral expressions used by *AntCalc*, some of which have been implemented and validated only through finite order. The expansion order can nevertheless be adjusted when the required higher-order coefficients are available, using the `ExpansionOrder` option, as,

$$\begin{aligned}\texttt{IntegrateAntenna[A30]} &\xrightarrow{\text{Default Result}} \frac{1}{\epsilon^2}+\frac{3}{2\epsilon}+\frac{19}{4}-\frac{7\pi^2}{12}+\epsilon\left(\frac{109}{8}+\ldots\right)+\epsilon^2\left(\frac{639}{16}+\ldots\right)\\ \texttt{IntegrateAntenna[A30]} &\xrightarrow{\texttt{ExpansionOrder->0}} \frac{1}{\epsilon^2}+\frac{3}{2\epsilon}+\frac{19}{4}-\frac{7\pi^2}{12}\end{aligned} \tag{3.15}$$

where `A30=BuildAntenna[A,3,0,IntegrableForm -> True]`, is the A_3^0 set's `AntennaObject`. Equation 3.15 illustrates the use of `ExpansionOrder` to override the default truncation order. The option can request either lower or higher orders; however, the maximum accessible order is route-dependent and is limited by the available ϵ-expansions of the corresponding master integrals. For the example shown, all required terms are available and the result has been validated against the literature.

3.5.4 Reconstruction of Colour Components

For antenna sets containing several colour structures, the corresponding components are identified during the build stage and kept separate throughout the integration workflow. For example, the A_3^1 set contains leading-colour, subleading-colour, and closed-quark-loop components, corresponding to A_3^1, $\tilde{A}_3^1$, and $\hat{A}_3^1$, respectively. Since IBP reduction acts on the kinematic integral structure and is independent of the colour coefficients, the components can be treated separately during the reduction. *AntCalc* can therefore integrate either the complete set of components or only a selected component. The latter avoids unnecessary IBP reductions and can significantly reduce the computational cost. Keeping the components separate also allows each contribution to be mapped to the integral families and master integrals required by its particular kinematic structure. Fig. 3.7 illustrates this procedure for the A_3^1 antenna set.

It is also important to note here that the result of the A_3^1 set's integration is given, much like it was for

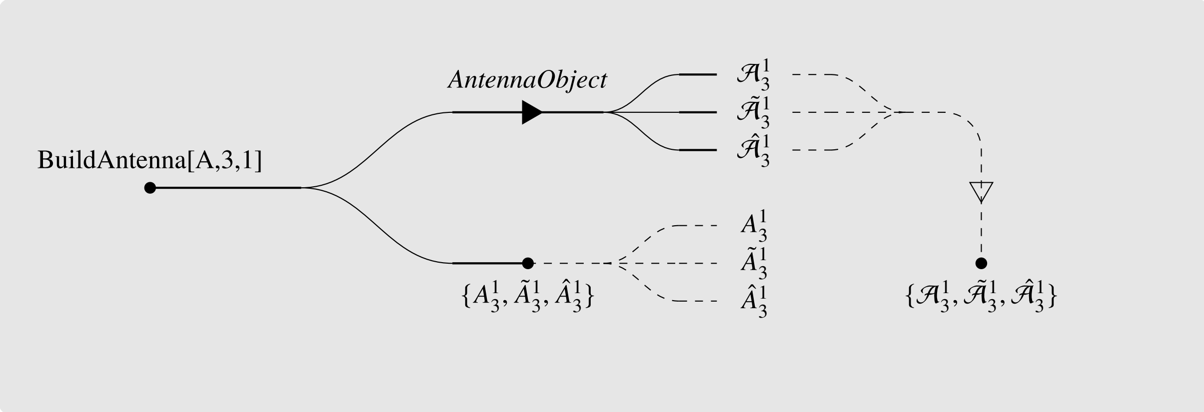

Figure 3.7: Diagram showing the component extraction steps and paths for the build-only and build-and-integrate routes. The combined route is shown above and the build-only route below it. The dashed lines represent the options to either separate the ordered lists into single components on the build-only route, or to merge the components into a single ordered list for the combined route. The A_3^1 antenna set is used as an example.

the build stage in 3.7, as,

$$\begin{aligned}
\mathcal{A}_3^1 &= \mathcal{T}_{\text{Lead}} - \mathcal{A}_2^1 \mathcal{A}_3^0 \\
\tilde{\mathcal{A}}_3^1 &= -\left(\mathcal{T}_{\text{Sublead}} + \mathcal{A}_2^1 \mathcal{A}_3^0\right) \\
\hat{\mathcal{A}}_3^1 &= \mathcal{T}_{\text{QL}}.
\end{aligned} \tag{3.16}$$

These resulting $\mathcal{A}_3^1$ and $\tilde{\mathcal{A}}_3^1$ are already UV renormalised by this stage.

Those $\mathcal{T}$-object may still be useful, however. They can be extracted instead of the antenna functions, essentially by breaking the process before that last operation, in Eq. 3.16 and returning an analogous ordered list made up of the $\mathcal{T}$-objects, as,

$$\begin{aligned}
\texttt{IntegrateAntenna[A31]} &\xrightarrow{\text{Default Result}} \{\mathcal{A}_3^1, \tilde{\mathcal{A}}_3^1, \hat{\mathcal{A}}_3^1\} \\
\texttt{IntegrateAntenna[A31]} &\xrightarrow{\texttt{ReturnTTerms -> True}} \{\mathcal{T}_{\text{Lead}}, \mathcal{T}_{\text{Sublead}}, \mathcal{T}_{\text{QL}}\}
\end{aligned} \tag{3.17}$$

where `A31=BuildAntenna[A,3,1,IntegrableForm -> True]`, is, once again, the A_3^1 set's `AntennaObject`.

3.5.5 A_2^2 Set Integration Routes

As summarised in Table 3.4, the A_2^2 antenna set is divided into two subsets, and require distinct integral families for the IBP reduction. Consequently, the two subsets follow separate integration routes in *AntCalc*. Those routes may be schematically drawn in the spirit of the diagram in Fig. 3.7, as in the diagram in Fig. 3.8, below. The workflow interprets the overall `[A,2,2]`, then separates the process into different build-side production modes as shown in the diagram in Fig. 3.3. It then merges the four components into an ordered list. That list is sent to the integrate-side workflow. This workflow starts by separating them into the subsets and then, considering that all components are selected for integration, it reduces them using the subset-specific basis families. Finally, once all four are integrated, they are once again merged onto an ordered list, which is delivered as the default output.

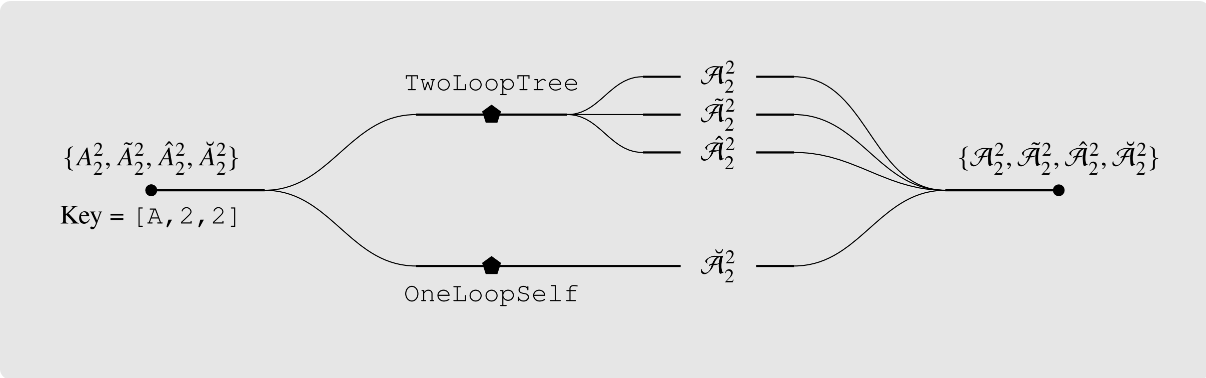

Figure 3.8: Diagram showing the component extraction steps and integration-basis families used in the integration-side workflow of the A_2^2 antenna set.

3.6 Higher-Level Functions

Besides the two canonical functions, `BuildAntenna` and `IntegrateAntenna`, *AntCalc* also supports a number of higher-level functions. This higher-level layer does not introduce new antenna-construction or integration methodology; instead it orchestrates the use of the two canonical functions to compute observables like the R-ratio, retrieve T-objects or to streamline repeated Build or Integrate evaluation.

There are a total of five higher-level functions. They are,

- `BuildAndIntegrateAntenna`
- `BuildAllAntennae`
- `BuildAndIntegrateAllAntennae`
- `TObject`
- `BuildRRatio`.

Each of these functions completes a useful higher-level task and their use can be schematically seen in Fig. 3.9. The first one streamlines the retrieval of the integrated antenna result to a single function, instead of having to build the unintegrated antenna using the `IntegrableForm` option and then to plug the intermediate result into the integration workflow. This function essentially follows the top path on the diagram in Fig. 3.7, for the A_3^1 route: it starts by calling `BuildAntenna` using the appropriate option, `IntegrableForm -> True`. It then, internally, plugs the result into `IntegrateAntenna`. It forwards the controls needed by the combined workflow to their respective stages, including component selection and Laurent-expansion order. That means that one can build and integrate only the subleading component, expanded only up to ϵ^{-1}, for example. In the diagram, this function goes all the way from its namesake node, down to where $\mathcal{X}_n^l$ leaves the line, being its schematic solution, the integrated antenna function.

There are also two bulk wrappers: `BuildAllAntennae` and `BuildAndIntegrateAllAntennae`. These two functions complete exactly what their names imply: the first one builds all public antenna functions and returns an ordered list of all antennae (and components); the second builds and integrates every public antenna function and returns an ordered list of all antennae (and components). In the diagram, the first function is separate from the rest of the workflow due to its own nature. The second

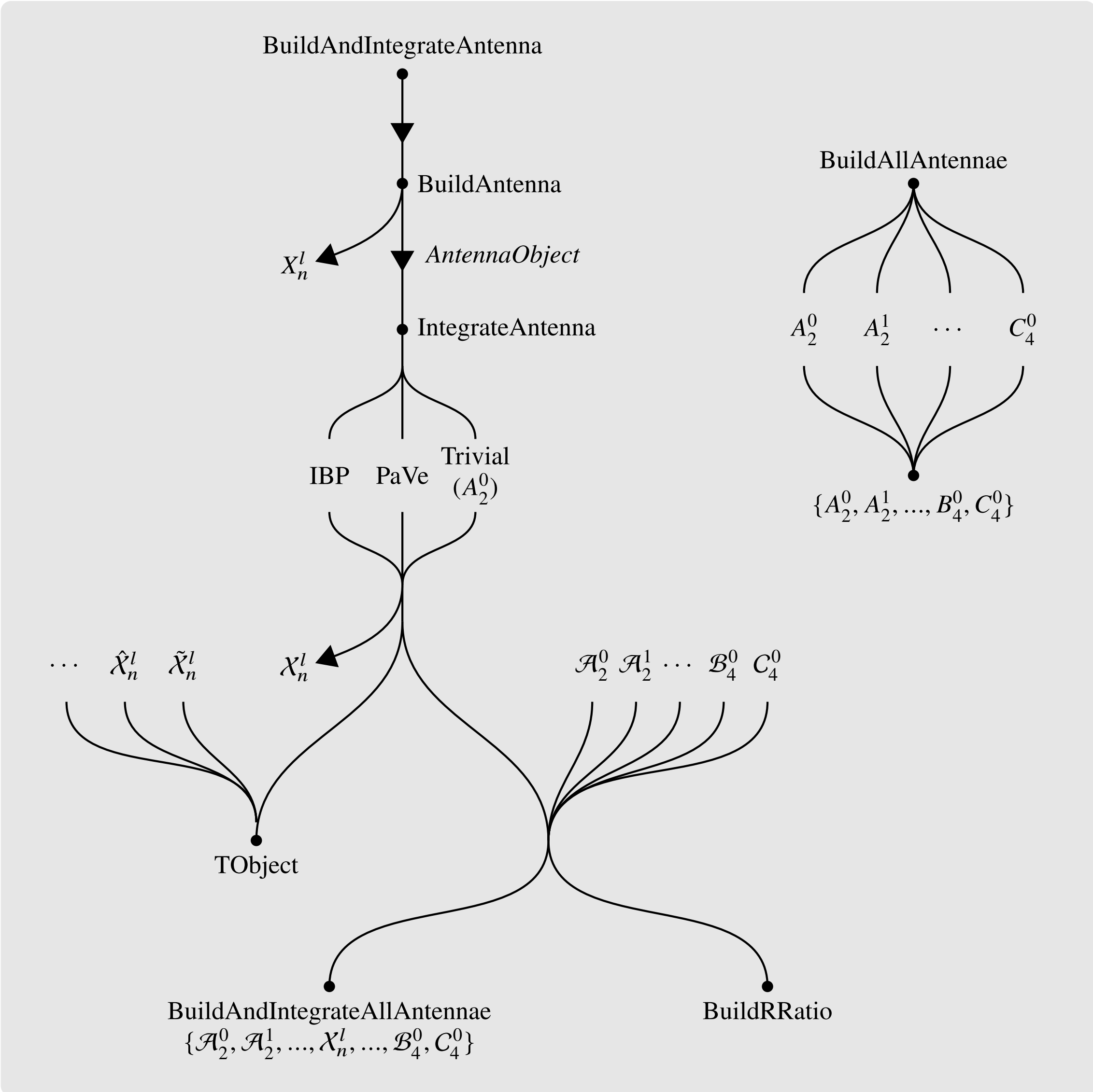

Figure 3.9: Diagram showing the complete *AntCalc* workflow including higher-level functions.

is shown at the bottom, since it calls `BuildAndIntegrateAntenna` for every public, massless, route. Their outputs can be modulated based on the requested order. That is, we can limit the wrappers to generate only antenna functions, built or integrated, up to, for example, NLO. In that case, only the list $\{A_2^0, A_2^1, A_3^0\}$ (or the same but integrated) would be returned. These functions are able to produce (build or build and integrate) all publicly available antenna functions, that is, all massless quark-antiquark final-final (SMQCD[3]) antennae, up to NNLO.

We may also retrieve a specific $\mathcal{T}$-object directly, using `TObject`. The public supported $\mathcal{T}$-objects are described in Table 3.7, below. These computation routes follow closely the methods discussed in Subsection 5.2.1 of Chapter 5. The function calls the one-shot integrated routes, using `BuildAndIntegrateAntenna` for the necessary ingredient antennae, then assembles them into the $\mathcal{T}$-object structure.

As an example, let us consider $\mathcal{T}^6_{q\bar{q}g}$. As shown in Eq. 3.7 and Table 3.7, this $\mathcal{T}$-object requires the $\mathcal{A}_3^1$ set, $\mathcal{A}_2^1$, and $\mathcal{A}_3^0$. Upon calling `BuildAndIntegrateAntenna` for those antennae, `TObject`

[3]The label SMQCD is not standard in the literature. It is used here to denote the massless QCD sector of the Standard Model.

$\mathcal{T}$-Object	Ingredients	Call
$\mathcal{T}^2_{q\bar{q}}$	Born contribution	`TObject[2,"qqbar"]`
$\mathcal{T}^4_{q\bar{q}}$	$\mathcal{A}^1_2$	`TObject[4,"qqbar"]`
$\mathcal{T}^4_{q\bar{q}g}$	$\mathcal{A}^0_3$	`TObject[4,"qqbarg"]`
$\mathcal{T}^6_{q\bar{q}}$	$\mathcal{A}^2_2, \tilde{\mathcal{A}}^2_2, \hat{\mathcal{A}}^2_2, \breve{\mathcal{A}}^2_2$	`TObject[6,"qqbar"]`
$\mathcal{T}^6_{q\bar{q}g}$	$\mathcal{A}^1_2, \mathcal{A}^0_3, \mathcal{A}^1_3, \tilde{\mathcal{A}}^1_3, \hat{\mathcal{A}}^1_3$	`TObject[6,"qqbarg"]`
$\mathcal{T}^6_{q\bar{q}q'\bar{q}'}$	$\mathcal{B}^0_4$	`TObject[6,"qqbarqprimeqprimebar"]`
$\mathcal{T}^6_{q\bar{q}q\bar{q}}$	$\mathcal{B}^0_4, C^0_4$	`TObject[6,"qqbarqqbar"]`
$\mathcal{T}^6_{q\bar{q}gg}$	$\mathcal{A}^0_4, \tilde{\mathcal{A}}^0_4$	`TObject[6,"qqbargg"]`

Table 3.7: Overview of the TObject routes.

assembles them as follows,

$$\mathcal{T}^6_{q\bar{q}g} = \left(N - \frac{1}{N}\right) T^2_{q\bar{q}} \left[N\left(\mathcal{A}^1_3 + \mathcal{A}^1_2\mathcal{A}^0_3\right) - \frac{1}{N}\left(\tilde{\mathcal{A}}^1_3 + \mathcal{A}^1_2\mathcal{A}^0_3\right) + N_f\hat{\mathcal{A}}^1_3\right]. \tag{3.18}$$

Lastly, *AntCalc* is also able to generate the massless SMQCD hadronic R-ratio, using the integrated antenna ingredients required at the selected perturbative order. The function is called based on the requested perturbative order, that is, to get the complete NNLO massless hadronic R-ratio, we use `BuildRRatio[SMQCD, maxOrder->NNLO]`, but we may also only compute the R-ratio up to NLO using `BuildRRatio[SMQCD, maxOrder->NLO]`. By default, `BuildRRatio` returns the computed finite coefficient after application of the observable convention bridge; this result has been validated against a literature reference target.

3.7 Validation and Development Diagnostics

The supported non-trivial massless public release antenna routes are validated against literature results, specifically against those in [4] and [35], by comparing their Laurent-expanded results. The same integrated antennae are used as ingredients to reproduce the known $\mathcal{T}$-objects and, with them, assemble the massless hadronic R-ratio for their antenna type. These checks test the package at increasing levels: individual integrated antenna, exclusive final-state contributions, and the inclusive observable.

As discussed in Section 3.2, the route system can retain selected diagnostic information from an evaluation, during its calculation. That makes it easy to locate any inconsistency during the route calculation that may then be inspected using the built-in diagnostics tool. One example of its use could be checking how the poles of each R-ratio order's antennae cancel each other to make a finite result. As an example, let us consider the SMQCD R-ratio up to NLO. We can run it and then, in the diagnostics association, see what antennae it uses as ingredients. We may then simply compare those against one another (in this case there are only two, $\mathcal{A}^1_2$ and $\mathcal{A}^0_3$), and see how their poles cancel. The check may go as follows. In the following example, the common colour and perturbative prefactors are omitted, since they do not affect the cancellation of the poles in $\mathcal{A}^1_2 + \mathcal{A}^0_3$.

```
In[1]:= <<AntCalc`
In[2]:= diagNLORRatio = BuildRRatio[SMQCD,maxOrder -> NLO,
```

```
ReturnDiagnostics -> True][[2]];
In[3]:= ingredients = diagNLORRatio["Ingredients"];
In[4]:= A21 = ingredients["intA21"]; A30 = ingredients["intA30"];
In[5]:= resList = {};
In[6]:=
Do[
coefA21 = Coefficient[A21,Epsilon,i];
coefA30 = Coefficient[A30,Epsilon,i];
AppendTo[resList, coefA21+coefA30]
,
{i,-2,2}
];
In[7]:= resList
Out[7]:=
```

$$\left\{0, 0, \frac{3}{4}, \frac{45}{8} - 6\zeta_3, \frac{383}{16} - \frac{7\pi^2}{16} - \frac{\pi^4}{10} - 9\zeta_3\right\}$$

The first two entries are the coefficients of ϵ^{-2} and ϵ^{-1}, respectively, and vanish as required. The remaining entries are the finite and positive-power coefficients, which do not cancel. This diagnostic inspection makes the origin of the NLO pole cancellation explicit. The release-level observable validation additionally compares the raw NNLO R-ratio series with the known pole-free and finite result.

The `Ingredients` entry used above is only one part of the diagnostic information. For a complete retained history of a calculation, *AntCalc* can instead return an `AntennaRunRecord`, which records the relevant route stages and metadata. This information was used to trace the source of discrepancies during development; the resulting validation comparisons are presented in Chapter 5.

3.7.1 Amplitude-Level and Release-Acceptance Validation

The checks described above validate an antenna function against an external reference: a published Laurent series, or a known finite observable. *AntCalc* also implements a complementary validation family that requires no such reference. The function `VerifyWardIdentity[type, numFinalParticles, loopOrder]` acts directly on the unsquared amplitude produced by the build stage, before spin summation, colour decomposition, or phase-space integration is performed. For a chosen external gluon leg, it replaces that gluon's polarisation vector by its own momentum, $\varepsilon^\mu(k) \to k^\mu$, while keeping every other external gluon polarisation explicitly transverse, and checks whether the resulting expression vanishes. This is the standard amplitude-level Ward identity, applied leg by leg to the raw generated amplitude rather than inferred from a property of the extracted or integrated antenna. It is currently applied to every public route with an external gluon leg, that is, to A_3^0, A_4^0 and the A_3^1 set, and returns a structured pass, fail, or time-out report for each checked leg. Because it depends only on the internal gauge structure of the generated amplitude, and not on comparison with any external result, it certifies a property the literature-comparison checks above cannot: that the amplitude's diagrams were assembled with the correct relative signs and couplings before extraction even begins. In principle, this makes it applicable to a route for which no literature result exists at all.

Separately from this route-level testing, *AntCalc*'s development also relies on a release-acceptance practice. Before a version is prepared, every public route is re-evaluated in a fresh kernel, with any stored or cached intermediate result disabled, and its build, integration, and one-shot outputs are compared against each other and against the literature target discussed above. Each run is recorded, together with its timing and the specific checks it satisfied, as dated evidence. This practice has already caught, and allowed the correction of, release-worker issues that were not visible from the final Laurent-series outputs alone.

3.7.2 Runtime and Memory Characteristics

To characterise the computational cost and performance of the implemented public routes, each antenna was evaluated in a fresh *Wolfram Mathematica* kernel. Package loading was excluded from the measurements. Both the build and integration stages were timed using the `AbsoluteTiming` function. Memory usage was measured using `MaxMemoryUsed[]`, relative to the initial kernel baseline. These measurements were taken using a MacBook Pro (Mac16.8), with an Apple M4 Pro 12 core (8 performance and 4 efficiency) CPU, and 24 GB unified RAM, running macOS 26.6.2 (build 25G83), *Wolfram Mathematica* 15.0.0 for Mac OS X ARM (64-bit), and AntCalc 0.3.0β2.

The lower-order routes, up to NLO, complete both stages in approximately one second, with the cost increasing substantially for the NNLO routes, in particular A_2^2, which surpasses 31 minutes. The A_4^0 route has the largest memory requirement, reaching 567 MB above the kernel benchmark. The complete results are shown in Table 3.8.

Route	**Build time**	**Integration time**	**End-to-end time**	**Peak memory above baseline**
A_2^0	0.09 s	0.49 s	0.58 s	22.7 MiB
A_3^0	0.30 s	0.85 s	1.15 s	24.8 MiB
A_2^1	0.57 s	0.40 s	0.98 s	19.3 MiB
A_3^1	74.67 s	15 min 27 s	16 min 42 s	195.3 MiB
A_2^2	21.79 s	31 min 06 s	31 min 27 s	227.7 MiB
A_4^0	3 min 03 s	13 min 51 s	16 min 53 s	567.3 MiB
B_4^0	8.65 s	17.27 s	25.92 s	143.2 MiB
C_4^0	5.56 s	5 min 26 s	5 min 32 s	174.5 MiB

Table 3.8: Representative resource measurements for the implemented massless antenna routes. The build time measures `BuildAntenna[..., IntegrableForm -> True]`, followed by `IntegrateAntenna[...]` for the integration time. Each route was evaluated in a fresh Wolfram kernel. Peak memory is the maximum kernel-memory increase relative to the initial baseline.

Chapter 4

Antenna Construction and Integration: Selected Case Studies

4.1 Master Integrals Used in *AntCalc*

As discussed in Chapter 3, *AntCalc* makes use of two reduction methods, Passarino-Veltmann reduction and integration-by-parts reduction. Both of those methods reduce a larger loop-momentum or phase-space integral into a linear combination of smaller and often less complex master integrals, as discussed in Subsection 2.5.3[1]. The MI required by each route depend on the antenna being integrated, though some antennae share the same MI, especially those of the same type and in the same perturbative order, like all tree-level antenna functions at NNLO [21]. *AntCalc* also applies integral reduction at two different stages: loop antennae are reduced before producing the public build-stage output, but only over the loop momenta, and all antennae are reduced over the whole phase space during the integration stage. This means that loop antenna functions like A_3^1, for example, may require more than one set of MI throughout a complete build-and-integrate run, and the ones required for one stage may not be required for the other. Table 4.1 shows the MI used in each stage for each antenna [4].

Antenna type(s)	Build-Side MI	Integration-Side MI
A_2^0	–	–
A_3^0	–	R_3
A_4^0, $\tilde{A}_4^0$, B_4^0, C_4^0	–	R_4, R_6, $R_{8,a}$, $R_{8,b}$
A_2^1	–	B_0, C_0
A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$	B_0, C_0, D_0	$V_{5,a}$, $V_{5,b}$, V_8
A_2^2	$A_{22,\mathrm{LO}}$, A_3, A_4, B_0, C_0	$A_{22,\mathrm{LO}}$, A_3, A_4
$\tilde{A}_2^2$	$A_{22,\mathrm{LO}}$, A_3, A_4, A_6	$A_{22,\mathrm{LO}}$, A_3, A_4, A_6
$\hat{A}_2^2$	A_4, B_0, C_0	A_4
$\breve{A}_2^2$	$A_{22,\mathrm{LO}}$	$A_{22,\mathrm{LO}}$

Table 4.1: Overview of the master integrals used in the build- and integration-stage by each antenna type [4, 21].

Both PaVe and IBP MI are shown in the table, but they are quite different in structure and integration

[1]The PaVe MI are most commonly referred to as *scalar functions*, instead of *master integrals*. They are, in effect, still master integrals, however, and the moniker will be kept throughout the rest of the present text [36].

processes and will be discussed separately. The PaVe are B_0, C_0, and D_0. All others are IBP MI.

4.1.1 PaVe Scalar Functions

As mentioned, only three of the MI considered are PaVe scalar functions. Those are B_0, C_0 and D_0 and are the PaVe two-, three- and four-point scalar integrals, respectively. The PaVe method also features the A_0 (one-point) scalar function, that does not feature in *AntCalc*. It also accounts for scalar integrals with more than four external legs, as those can be further reduced in four dimensions to combinations of D_0 functions [9, 23].

The B_0 Function

The B_0 function represents a scalar one-loop diagram with two external lines and two internal propagators. It is represented, in d dimensions as,

$$B_0\left(k^2, m_0, m_1\right) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p \frac{1}{\left(p^2 - m_0^2 + i\varepsilon\right)\left((p+k)^2 - m_1^2 + i\varepsilon\right)} \tag{4.1}$$

where k is the external momentum, and m_0 and m_1 are the internal masses. μ is the t'Hooft normalisation scale.

It is also noteworthy that this integral, unlike higher-point ones, is UV divergent and generates the pole term Δ_B,

$$\Delta_B = \frac{2}{4-d} - \gamma_E + \log(4\pi) \tag{4.2}$$

where γ_E is Euler's gamma constant. This function may also be written as an integral over the Feynman parameter x. That yields the explicit analytical form [23],

$$B_0\left(k^2, m_0, m_1\right) = \Delta_B - \int_0^1 dx \, \log\left[\frac{k^2x^2 - x\left(k^2 - m_0^2 + m_1^2\right) + m_1^2 - i\varepsilon}{\mu^2}\right] + O(d-4). \tag{4.3}$$

The C_0 Function

The C_0 function represents a scalar loop with three external lines and three internal propagators. It is a common feature in the vertex corrections of electroweak physics. It is written in d dimensions as,

$$\begin{aligned} C_0(k_1, k_2, m_0, m_1, m_2) = & \frac{(2\pi\mu)^{4-d}}{i\pi^2} \\ & \int d^d p \, \frac{1}{\left(p^2 - m_0^2 + i\varepsilon\right)\left((p+k_1)^2 - m_1^2 + i\varepsilon\right)\left((p+k_1+k_2)^2 - m_2^2 + i\varepsilon\right)}. \end{aligned} \tag{4.4}$$

This integral is UV finite and may be rewritten as a double integral over the Feynman parameters x and y as,

$$C_0 \propto \int_0^1 dx \int_0^{1-x} dy \, \left[ax^2 + by^2 + cxy + dx + ey + f\right]^{-1} \tag{4.5}$$

where the coefficients a, b, c, d, e and f are linear combinations of the external momenta and internal mass parameters [9].

The D_0 Function

The D_0 function describes a scalar loop with four external lines and four internal propagators. It is often called the scalar box integral. It is written as,

$$D_0(k_1, k_2, k_3, m_0, m_1, m_2, m_3) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p \, \frac{1}{D_0 D_1 D_2 D_3} \tag{4.6}$$

with denominator factors,

$$\begin{aligned} D_0 &= p^2 - m_0^2 + i\varepsilon \\ D_i &= \left(p + \sum_{a=1}^{i} k_a\right)^2 - m_i^2 + i\varepsilon. \end{aligned} \tag{4.7}$$

Evaluating D_0 is highly non-trivial as it features numerous physical cuts and branch points. The resulting analytical expression is expressed in terms of Spence functions (dilogarithms) [9].

4.1.2 IBP Master Integrals

The IBP MI employed in *AntCalc* can be divided into four groups, based on the antenna functions they are used with. This division goes as,

- NLO tree-level: R_3;
- NNLO tree-level: R_4, R_6, $R_{8,a}$, $R_{8,b}$;
- NNLO one-loop: $V_{5,a}$, $V_{5,b}$, V_8;
- NNLO two-loop: $A_{22,\mathrm{LO}}$, A_3, A_4, A_6.

Phase-Space Measures and Normalisations

These MI are computed over different phase-space measures, depending on the number of final-state particles. These phase-space integrals are defined over all n-particle Mandelstam invariants for each specific phase-space. For that reason, we introduce the shorthands,

$$\begin{aligned} &\mathcal{S}_2 = s_{12}, \quad d\mathcal{S}_2 = ds_{12} \\ &\mathcal{S}_3 = s_{12} + s_{13} + s_{23}, \quad d\mathcal{S}_3 = ds_{12} ds_{13} ds_{23} \\ &\mathcal{S}_4 = s_{12} + s_{13} + s_{14} + s_{23} + s_{24} + s_{34}, \quad d\mathcal{S}_4 = ds_{12} ds_{13} ds_{14} ds_{23} ds_{24} ds_{34}. \end{aligned} \tag{4.8}$$

Using the shorthands, the differential phase-spaces are defined in d dimensions as [21, 25],

$$\begin{aligned} d\Phi_2 &= (2\pi)^{2-d} s_{12}^{\frac{d-4}{2}} 2^{1-d} d\Omega_{d-1} d\mathcal{S}_2 \delta\left(q^2 - \mathcal{S}_2\right) \\ d\Phi_3 &= (2\pi)^{3-2d} 2^{-1-d} \left(q^2\right)^{2-\frac{d}{2}} \left(s_{12} s_{13} s_{23}\right)^{\frac{d-4}{2}} d\Omega_{d-1} d\Omega_{d-2} d\mathcal{S}_3 \delta\left(q^2 - \mathcal{S}_3\right) \\ d\Phi_4 &= (2\pi)^{4-3d} \left(q^2\right)^{3-\frac{d}{2}} 2^{1-2d} (-\Delta_4)^{\frac{d-5}{2}} \Theta(-\Delta_4) \\ &\qquad d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} d\mathcal{S}_4 \delta\left(q^2 - \mathcal{S}_4\right). \end{aligned} \tag{4.9}$$

Here Δ_4 is the Gram determinant given by the Källén function λ [21, 25],

$$\Delta_4 = \lambda(s_{12}s_{34}, s_{13}s_{24}, s_{14}s_{23}), \quad \lambda(x, y, z) = x^2 + y^2 + z^2 - 2(xy + xz + yz). \tag{4.10}$$

Also, Θ is the Heaviside step function and $d\Omega$ the differential solid hyperspherical angle element in d dimensions, given as,

$$\Omega_n = \int d\Omega_n = \frac{2\pi^{\frac{n}{2}}}{\Gamma(n/2)}. \tag{4.11}$$

Some MI appear normalised using the S_Γ term, to make the infrared singularity cancellations explicit and absorb common dimensional factors. The normalising term is given as [35, 21],

$$S_\Gamma = \left(\frac{(4\pi)^\epsilon}{16\pi^2\Gamma(1-\epsilon)}\right)^2. \tag{4.12}$$

It is also common to normalise with

$$S_{\Gamma,2} = P_2 S_\Gamma, \quad P_2 = \int d\Phi_2 = \frac{2^{-3+2\epsilon}\pi^{-1+\epsilon}\Gamma(1-\epsilon)}{\Gamma(2-2\epsilon)}\left(q^2\right)^{-\epsilon}. \tag{4.13}$$

The R_3 Integral

This MI is used only on the integration of the massless A_3^0 antenna. It is the direct integral over the three-final-state-particle phase-space. It is written as [36],

$$\begin{aligned} R_3 &= \int d\Phi_3 \\ &= \frac{1}{4}\frac{\Omega_{d-1}\Omega_{d-2}}{(2\pi)^{2d-3}} \iiint dE_1 dE_2 d\theta_{12} \left(E_1 E_2 \sin\theta_{12}\right)^{d-3} \delta\left(q^2 - \mathcal{S}_3\right) \\ &= \frac{1}{128 q^2 \pi^3} \int d\mathcal{S}_3\, \delta\left(q^2 - \mathcal{S}_3\right). \end{aligned} \tag{4.14}$$

After analytical evaluation the integrated result, which is the one substituted in the final integrated antenna expression after IBP reduction becomes [21],

$$R_3 = 2^{-7+4\epsilon}\pi^{-3+2\epsilon}\frac{\Gamma^3(1-\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}\left(q^2\right)^{1-2\epsilon}. \tag{4.15}$$

The R_4 Integral

The first of the tree-level NNLO MI is R_4, which corresponds to the direct integral over the four-final-state-particle phase-space. It is written as,

$$\begin{aligned} R_4 &= \int d\Phi_4 \\ &= (2\pi)^{4-3d}\left(q^2\right)^{3-\frac{d}{2}} 2^{1-2d} \int d\mathcal{S}_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \\ &\qquad (-\Delta_4)^{\frac{d-5}{2}}\Theta(-\Delta_4)\delta\left(q^2 - \mathcal{S}_4\right). \end{aligned} \tag{4.16}$$

To make the integral boundary limits integrable, we start by factorising the phase space into a two-particle phase space and a tripole phase space factor, $d\Phi_4 = d\Phi_2 d\Phi_T$ and then we map the six physical invariants s_{ij} into five dimensionless variables that range strictly between 0 and 1, making a five-dimensional hypercube. We start by defining $y_{ij} = s_{ij}/q^2$ and $y_{ijk} = s_{ijk}/q^2$. Our five dimensionless variables become [21]:

- The rescaled triple-invariant of the emitter jet, $y_{134} = \frac{s_{134}}{q^2}$;
- The longitudinal momentum fraction of the unresolved parton 3, $z_1 = \frac{p_3 \cdot \tilde{p}_2}{\tilde{p}_{134} \cdot \tilde{p}_2}$;
- The rescaled longitudinal momentum fraction of unresolved parton 4, $z_2 = (1-z_1)t = \frac{p_4 \cdot \tilde{p}_2}{\tilde{p}_{134} \cdot \tilde{p}_2}$;
- The rescaled invariant, $y_{14} = \frac{s_{14}}{q^2} = y_{134}(1-z_1)v$;
- The alignment variable χ, that parametrises the physical boundaries of the Gram determinant for y_{13} as, $y_{13} = (y_{13,b} - y_{13,a})\chi + y_{13,a} = \Delta y_{13}\chi + y_{13,a}$.

After applying the mapping, the MI becomes

$$\begin{aligned} R_4 = P_2 \frac{16^{-2}\pi^{-5+2\epsilon}}{\Gamma(1-2\epsilon)} \left(q^2\right)^{2-2\epsilon} \int_{[0,1]^5} dy_{134} dz_1 dt dv d\chi \left[y_{134}(1-y_{134})(1-z_1)\right]^{1-2\epsilon} \\ \left[z_1 t(1-t) v(1-v)\right]^{-\epsilon} \left[\chi(1-\chi)\right]^{-\frac{1}{2}-\epsilon}. \end{aligned} \tag{4.17}$$

We get [4, 21],

$$\begin{aligned} R_4 &= P_2 \left(q^2\right)^{2-2\epsilon} \frac{2^{-11+8\epsilon}\pi^{-\frac{7}{2}+2\epsilon}\Gamma^3(1-\epsilon)}{\Gamma(3-3\epsilon)\Gamma\left(\frac{5}{2}-2\epsilon\right)} \\ &= S_{\Gamma,2}\left(q^2\right)^{2-2\epsilon} \frac{\Gamma^5(1-\epsilon)\Gamma(2-2\epsilon)}{\Gamma(3-3\epsilon)\Gamma(4-4\epsilon)} \\ &= S_\Gamma \left(q^2\right)^{2-2\epsilon} \left[\frac{1}{12} + \frac{59}{72}\epsilon + O\left(\epsilon^2\right)\right]. \end{aligned} \tag{4.18}$$

The R_6 Integral

The R_6 MI is written as [21],

$$\begin{aligned} R_6 &= \int d\Phi_4 \, \frac{1}{s_{134}s_{234}} \\ &= (2\pi)^{4-3d}\left(q^2\right)^{3-\frac{d}{2}} 2^{1-2d} \int dS_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \, \frac{(-\Delta_4)^{\frac{d-5}{2}}}{s_{134}s_{234}} \Theta(-\Delta_4)\delta\left(q^2 - S_4\right). \end{aligned} \tag{4.19}$$

Because the s_{234} is non-linear in the tripole parametrisation, we may not continue using direct integration [21]. We may, however, use the Optical Theorem as a cut of the three-loop vacuum polarisation diagram I_6 as,

$$2\,\mathrm{Im}\, I_6 = -2P_2\,\mathrm{Re}\, A_4 + 2R_6 \tag{4.20}$$

with I_6 and A_4 being known from the literature and written as,

$$\begin{aligned} I_6 = \frac{(4\pi)^{3\epsilon}}{(16\pi^2)^3} \frac{\Gamma^6(1-\epsilon)\Gamma^3(1+\epsilon)}{3\epsilon^3\Gamma^3(2-2\epsilon)} \left(-q^2\right)^{-3\epsilon} \\ \left[1 + \epsilon + \epsilon^2 + \epsilon^3(14\zeta_3 - 7) + \epsilon^4\left(-67 + 14\zeta_3 + \frac{21\pi^4}{90}\right) + O\left(\epsilon^5\right)\right] \end{aligned} \tag{4.21}$$

$$
A_4 = \left(\frac{(4\pi)^\epsilon}{16\pi^2}\frac{\Gamma(1+\epsilon)\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}\right)^2\left(-q^2\right)^{-2\epsilon}\left[-\frac{1}{2\epsilon^2}-\frac{5}{2\epsilon}-\left(\frac{19}{2}+\frac{\pi^2}{6}\right)\right.
$$
$$
\left.-\epsilon\left(\frac{65}{2}+\frac{5\pi^2}{6}-2\zeta_3\right)-\epsilon^2\left(\frac{211}{2}+\frac{19\pi^2}{6}-10\zeta_3\right)+\mathcal{O}\left(\epsilon^3\right)\right]. \tag{4.22}
$$

Note that A_4 is described in Subsubsection 4.1.2.

We then apply the Minkowski boundary prescription [21] $\left(q^2\right)^{-\epsilon} = \left(q^2\right)^{-\epsilon} e^{i\pi\epsilon}$, isolate the imaginary and real parts, and get, by using $\mathcal{A}_4$ and $\mathcal{I}_6$

$$
\begin{aligned}
R_6 &= S_\Gamma\left(q^2\right)^{-2\epsilon}\frac{\Gamma^6(1-\epsilon)\Gamma^2(1+\epsilon)}{6}\left(\frac{6\cos(2\pi\epsilon)}{\Gamma^2(1-2\epsilon)}\mathcal{A}_4+\frac{\Gamma(1-\epsilon)\Gamma(1+\epsilon)\sin(3\pi\epsilon)}{\epsilon^3\pi\Gamma^2(2-2\epsilon)}\mathcal{I}_6\right)\\
&= S_{\Gamma,2}\left(q^2\right)^{-2\epsilon}\left[-1+\frac{\pi^2}{6}+\epsilon\left(-12+\frac{5\pi^2}{6}+9\zeta_3\right)+\mathcal{O}\left(\epsilon^2\right)\right].
\end{aligned} \tag{4.23}
$$

The $R_{8,a}$ Integral

The $R_{8,a}$ is the first of two MI with four denominator invariants. Its complete integral form is written as [21],

$$
\begin{aligned}
R_{8,a} &= \int d\Phi_4\,\frac{1}{s_{13}s_{23}s_{14}s_{24}}\\
&= (2\pi)^{4-3d}\left(q^2\right)^{3-\frac{d}{2}}2^{1-2d}\int dS_4 d\Omega_{d-1}d\Omega_{d-2}d\Omega_{d-3}\frac{(-\Delta_4)^{\frac{d-5}{2}}}{s_{13}s_{23}s_{14}s_{24}}\Theta(-\Delta_4)\,\delta\left(q^2-S_4\right)\\
&= P_2\left(q^2\right)^{-2-2\epsilon}\frac{16^{-2+\epsilon}\pi^{-5+2\epsilon}}{\Gamma(1-2\epsilon)}\int_{[0,1]^5} dy_{134}dz_1dtdvd\chi\\
&\qquad\left[(1-y_{134})(1-z_1)\right]^{-1-2\epsilon}\left[tvz_1^{-\epsilon}(\Delta y_{13}\chi+y_{13,a})\right]^{-1}\left[\chi(1-\chi)\right]^{-\frac{1}{2}-\epsilon}.
\end{aligned} \tag{4.24}
$$

This is the most complex MI integration of the set. As such, its computation will be written in detail here as most of the methods used are transferable to other MI. We start by defining the integration order. The computation will advance as,

$$
\chi \to v \to y_{134} \to t \to z_1
$$

with the inclusion of a u integral, also over the $[0,1]$ interval in between the t and z_1 integrals, which will be introduced later.

The first integral is over χ, and gives,

$$
I_\chi = \int_0^1 d\chi\left[\chi(1-\chi)\right]^{-\frac{1}{2}-\epsilon}(\Delta y_{13}\chi+y_{13,a})^{-1} = \frac{\Delta y_{13}^{-2\epsilon}}{y_{13,a}}\frac{\Gamma^2\left(\frac{1}{2}-\epsilon\right)}{\Gamma(1-2\epsilon)}{}_2F_1\left(1,\frac{1}{2}-\epsilon;1-2\epsilon;-\frac{\Delta y_{13}}{y_{13,a}}\right) \tag{4.25}
$$

where ${}_2F_1(\ldots)$ is a hypergeometric function. We now consider the definition $y_{13,(a,b)} = y_{134}(A \mp B)$, $A = \sqrt{(1-t)(1-v)}$, $B = \sqrt{z_1tv}$, and $Z = -B/A$, and get,

$$
-\frac{\Delta y_{13}}{y_{13,a}} = \frac{4Z}{(1+Z)^2}, \quad (1+Z)^2 = \frac{y_{13,a}}{y_{134}A^2}. \tag{4.26}
$$

Finally, by taking,

$$ {}_2F_1\left(1, b; 2b; \frac{4Z}{(1+Z)^2}\right) = (1+Z)^2 {}_2F_1\left(1, \frac{3}{2}-b; b+\frac{1}{2}; Z^2\right) \tag{4.27} $$

and

$$ 16^{\epsilon}\Delta y_{13}^{-2\epsilon} = \left[t(1-t)v(1-v)y_{134}^2 z_1\right]^{-\epsilon} \tag{4.28} $$

we get,

$$ I_\chi = \frac{\pi\Gamma(1-2\epsilon)}{y_{134}\Gamma^2(1-\epsilon)}\left(tvy_{134}^2 z_1\right)^{-\epsilon}\left[(1-t)(1-v)\right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 tv}{(1-t)(1-v)}\right). \tag{4.29} $$

The $R_{8,a}$ expression becomes,

$$ \begin{aligned} R_{8,a} = P_2\left(q^2\right)^{-2-2\epsilon}\frac{16^{-2+\epsilon}\pi^{-4+2\epsilon}}{\Gamma^2(1-\epsilon)}\int_{[0,1]^4} dy_{134}dz_1dtdv\left[y_{134}(1-y_{134})z_1(1-z_1)\right]^{-1-2\epsilon} \\ \left[t(1-t)v(1-v)\right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 tv}{(1-t)(1-v)}\right). \end{aligned} \tag{4.30} $$

We now continue to the v integral. To start, we restate the hypergeometric function as a series, which enables us to write,

$$ \begin{aligned} I_v &= \int_0^1 dv\left[v(1-v)\right]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 tv}{(1-t)(1-v)}\right) \\ &= \sum_{n\geq 0}\frac{(1+\epsilon)_n}{(1-\epsilon)_n}\left(\frac{z_1 t}{1-t}\right)^n \frac{\Gamma(-\epsilon-n)\Gamma(-\epsilon+n)}{\Gamma(-2\epsilon)} \\ &= -\frac{2\Gamma^2(1-\epsilon)}{\epsilon\Gamma(1-2\epsilon)} {}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right). \end{aligned} \tag{4.31} $$

Plugging this expression back in and computing the trivial y_{134} integral we get,

$$ \begin{aligned} R_{8,a} = -P_2\left(q^2\right)^{-2-2\epsilon}\frac{2^{-7+8\epsilon}\pi^{-\frac{7}{2}+2\epsilon}}{\epsilon^2\Gamma\left(\frac{1}{2}-2\epsilon\right)}\int_{[0,1]^2} dz_1dt\,(1-z_1)^{-1-2\epsilon} \\ \left[z_1 t(1-t)\right]^{-1-\epsilon} {}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right). \end{aligned} \tag{4.32} $$

We now restate the hypergeometric function as an Euler integral as,

$$ {}_2F_1\left(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\right) = -\epsilon\int_0^1 du\,u^{-1-\epsilon}\left[1+\frac{uz_1 t}{1-t}\right]^{-1} \tag{4.33} $$

and after computing the direct t, u and z_1 integrals, in that order, we get to the final expression,

$$ \begin{aligned} R_{8,a} = P_2\left(q^2\right)^{-2-2\epsilon}2^{-7+4\epsilon}\pi^{-3+2\epsilon}\csc(\pi\epsilon)\Gamma(-2\epsilon)\left[\frac{2^{8\epsilon}\pi}{\epsilon^2\Gamma^2\left(\frac{1}{2}-2\epsilon\right)} {}_3F_2\left(\begin{matrix} -2\epsilon, -2\epsilon, -2\epsilon \\ 1-2\epsilon, -4\epsilon \end{matrix}; 1\right) - \right. \\ \left. -\frac{2\Gamma^3(-\epsilon)}{\Gamma(1-4\epsilon)\Gamma(1-\epsilon)\Gamma(1+\epsilon)\Gamma(-3\epsilon)} {}_4F_3\left(\begin{matrix} 1, -\epsilon, -\epsilon, -\epsilon \\ 1-\epsilon, 1+\epsilon, -3\epsilon \end{matrix}; 1\right)\right] \end{aligned} \tag{4.34} $$

which after a final normalisation becomes [21, 36],

$$
\begin{aligned}
R_{8,a} =& S_\Gamma\left(q^2\right)^{-2-2\epsilon}\left[\frac{6}{\epsilon^2}\frac{\Gamma^5(1-\epsilon)\Gamma(1-2\epsilon)}{\Gamma(1-3\epsilon)\Gamma(1-4\epsilon)}{}_4F_3\left(\begin{matrix}1,-\epsilon,-\epsilon,-\epsilon\\ 1-\epsilon,1+\epsilon,-3\epsilon\end{matrix};1\right)-\right.\\
&\left.-\frac{1}{\epsilon^4}\frac{\Gamma^3(1-\epsilon)\Gamma(1+\epsilon)\Gamma^3(1-2\epsilon)}{\Gamma^2(1-4\epsilon)}{}_3F_2\left(\begin{matrix}-2\epsilon,-2\epsilon,-2\epsilon\\ -4\epsilon,1-2\epsilon\end{matrix};1\right)\right]\\
&= S_{\Gamma,2}\left(q^2\right)^{-2-2\epsilon}\left[\frac{5}{\epsilon^4}-\frac{20\pi^2}{3\epsilon^2}-\frac{126\zeta_3}{\epsilon}+\frac{7\pi^4}{18}+O(\epsilon)\right].
\end{aligned}
\tag{4.35}
$$

The $R_{8,b}$ Integral

The second of the two four-denominator-invariant MI is $R_{8,b}$. It is written as [21],

$$
R_{8,b} = \int d\Phi_4\,\frac{1}{s_{13}s_{134}s_{23}s_{234}}. \tag{4.36}
$$

Due to the presence of s_{234}, this MI must be computed using the Optical Theorem, much like R_6 was. For this case, we are able to map $R_{8,b}$ as one of the cuts of the three-loop vacuum polarisation diagram I_8 as,

$$
2\,\mathrm{Im}\,I_8 = -2P_2\,\mathrm{Re}\,A_6 + 4\,\mathrm{Re}\,V_8 + R_{8,a} + 4R_{8,b}. \tag{4.37}
$$

with,

$$
I_8 = S + O(\epsilon^0), \quad \mathrm{Im}\,S = -\frac{13\pi^4}{90}; \tag{4.38}
$$

$$
V_8 = -i\int d\Phi_3\,\frac{1}{s_{12}}\,\mathrm{Box}(s_{13},s_{13},s_{12}); \tag{4.39}
$$

$$
A_6 = \left(\frac{(4\pi)^\epsilon}{16\pi^2}\frac{\Gamma(1+\epsilon)\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}\right)^2\left(q^2\right)^{-2-2\epsilon}\left[-\frac{1}{\epsilon^4}+\frac{5\pi^2}{6\epsilon^2}+\frac{23\zeta_3}{\epsilon}+\frac{103\pi^4}{180}+O(\epsilon)\right]. \tag{4.40}
$$

Note that the A_6 and V_8 integrals are described in Subsubsections 4.1.2 and 4.1.2. In those Subsubsections, the integrals are given normalised using the usual S_Γ prefactor.

Using $\mathcal{V}_8$, $\mathcal{A}_6$ and $\mathcal{R}_{8,a}$ to denote the series parts of $\mathrm{Re}\,V_8$, $P_2\,\mathrm{Re}\,A_6$ and $R_{8,a}$, respectively, we get [21],

$$
\begin{aligned}
R_{8,b} &= \frac{1}{2}\,\mathrm{Im}\,S + S_\Gamma\left(q^2\right)^{-2-2\epsilon}\left(\frac{1}{2}\frac{\cos(2\pi\epsilon)\Gamma^6(1-\epsilon)\Gamma^2(1-\epsilon)}{\Gamma^2(1-2\epsilon)}\mathcal{A}_6-\mathcal{V}_8-\frac{1}{4}\mathcal{R}_{8,a}\right)\\
&= S_\Gamma\left(q^2\right)^{-2-2\epsilon}\left[\frac{3}{4\epsilon^4}-\frac{17\pi^2}{12\epsilon^2}-\frac{44\zeta_3}{\epsilon}-\frac{61\pi^4}{60}+O(\epsilon)\right].
\end{aligned}
\tag{4.41}
$$

The $V_{5,a}$ Integral

$V_{5,a}$ is a one-loop bubble integrated over a three-particle phase space, written as,

$$
\begin{aligned}
V_{5,a} &= \mathrm{Re}\left[-i\int d\Phi_3\int\frac{d^dp}{(2\pi)^d}\frac{1}{p^2(p-k_1-k_2-k_3)^2}\right]\\
&= \mathrm{Re}\left[\int d\Phi_3\,B_0\left(q^2;0,0\right)\right].
\end{aligned}
\tag{4.42}
$$

Since $q^2 = (p_1 + p_2 + p_3)^2$, this MI factors to the three-particle phase-space integral of the $B_0\big(q^2; 0, 0\big)$ scalar one-loop function, discussed in Subsubsection 4.1.1. After parametrising the phase space and applying the methods introduced in Subsubsection 4.1.2 we get [4],

$$\begin{aligned} V_{5,a} &= S_{\Gamma,2}\Big(q^2\Big)^{1-2\epsilon} \cos(\pi\epsilon)\left(-\frac{1}{\epsilon}\right) \frac{\Gamma^6(1-\epsilon)\Gamma(1+\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)} \\ &= S_{\Gamma,2}\Big(q^2\Big)^{1-2\epsilon}\left[-\frac{1}{2\epsilon} - \frac{13}{4} + O(\epsilon)\right]. \end{aligned} \tag{4.43}$$

The $V_{5,b}$ Integral

$V_{5,b}$ is very similar in structure to $V_{5,a}$, also being the three-particle phase-space integral of a B_0 scalar one-loop function. The integral is also computed in the same manner and yields [4],

$$\begin{aligned} V_{5,b} &= \int d\Phi_3\, B_0(s_{13}; 0, 0) \\ &= S_{\Gamma,2}\Big(q^2\Big)^{1-2\epsilon} \cos(\pi\epsilon)\left(-\frac{1}{\epsilon}\right) \frac{\Gamma^5(1-\epsilon)\Gamma(1-2\epsilon)\Gamma(1+\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-4\epsilon)} \\ &= S_{\Gamma,2}\Big(q^2\Big)^{1-2\epsilon}\left[-\frac{1}{2\epsilon} - 4 + O(\epsilon)\right]. \end{aligned} \tag{4.44}$$

The V_8 Integral

The V_8 MI is the most complex of the three V-type MI. It is given as a four-denominator propagator loop and three-particle phase-space integral. That makes it be restatable as a Box function which, itself, may be restated as a D_0 function, as discussed in Subsubsection 4.1.1. It is written as [4],

$$\begin{aligned} V_8 &= \mathrm{Re}\left[-i\int d\Phi_3 \int \frac{d^d p}{(2\pi)^d}\, \frac{1}{p^2(p-k_1)^2(p-k_1-k_3)^2(p-k_1-k_2-k_3)^2}\right] \\ &= \mathrm{Re}\left[-i\int d\Phi_3\, \frac{1}{s_{12}}\, \mathrm{Box}(s_{13}, s_{23}, s_{12})\right] \\ &= \left(\frac{(4\pi)^\epsilon \mu^{-2\epsilon}}{16\pi^2}\right) \frac{\Gamma^2(1-\epsilon)\Gamma(1+\epsilon)}{\Gamma(1-2\epsilon)} \int d\Phi_3\, \frac{1}{s_{12}}\, \mathrm{Re}\left[D_0\Big(0, 0, 0, q^2; s_{13}, s_{23}; 0, 0, 0, 0\Big)\right] \end{aligned} \tag{4.45}$$

which, after normalisation and evaluation gives,

$$V_8 = S_{\Gamma,2}\Big(q^2\Big)^{-2-2\epsilon}\left[-\frac{5}{2\epsilon^4} + \frac{9\pi^2}{2\epsilon^2} + \frac{89\zeta_3}{\epsilon} + \frac{13\pi^4}{180} + O(\epsilon)\right]. \tag{4.46}$$

The $A_{22,\mathrm{LO}}$ Integral

$A_{22,\mathrm{LO}}$ is the first of the four A-type MI, and represents a double bubble product. These are loop momenta integrals, not phase-space integrals. This first MI is given as [4],

$$\begin{aligned} A_{22,\mathrm{LO}} &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 (p-k_1-k_2)^2 \ell^2 (\ell-k_1-k_2)^2} \\ &= -\frac{\mu^{-4\epsilon}}{\left(16\pi^2\right)^2} \left[B_0\left(q^2;0,0\right)\right]^2 \\ &= S_\Gamma \left(-q^2\right)^{-2\epsilon} \frac{\Gamma^2(1+\epsilon)\Gamma^6(1-\epsilon)}{\Gamma^2(2-2\epsilon)} \left(-\frac{1}{\epsilon^2}\right) \\ &= S_\Gamma \left(-q^2\right)^{-2\epsilon} \left[-\frac{1}{\epsilon^2} - \frac{4}{\epsilon} - 12 + O(\epsilon)\right]. \end{aligned} \tag{4.47}$$

The A_3 Integral

A_3 represents a three-propagator two-loop sunset vacuum bubble, with $q^2 = s_{12}$. It is given as [4],

$$\begin{aligned} A_3 &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p-\ell-k_1-k_2)^2} \\ &= \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2} B_0\left((p-k_1-k_2)^2;0,0\right) \\ &= S_\Gamma \left(-q^2\right)^{1-2\epsilon} \frac{\Gamma(1+2\epsilon)\Gamma^5(1-\epsilon)}{\Gamma(3-3\epsilon)} \left(-\frac{1}{2(1-2\epsilon)\epsilon}\right) \\ &= S_\Gamma \left(-q^2\right)^{1-2\epsilon} \left[-\frac{1}{4\epsilon} - \frac{13}{8} + O(\epsilon)\right]. \end{aligned} \tag{4.48}$$

The A_4 Integral

A_4 is a four-propagator two-loop vacuum planar double bubble. It is given as a triangle diagram containing a bubble insertion on one of its legs. That gives [4],

$$\begin{aligned} A_4 &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p-k_1-k_2)^2 (p-\ell-k_1)^2} \\ &= \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2 (p-k_1-k_2)^2} B_0\left((p-k_1)^2;0,0\right) \\ &= S_\Gamma \left(-q^2\right)^{-2\epsilon} \frac{\Gamma(1-2\epsilon)\Gamma(1+\epsilon)\Gamma^4(1-\epsilon)\Gamma(1+2\epsilon)}{\Gamma(2-3\epsilon)} \left(-\frac{1}{2(1-2\epsilon)\epsilon^2}\right) \\ &= S_\Gamma \left(-q^2\right)^{-2\epsilon} \left[-\frac{1}{2\epsilon^2} - \frac{5}{2\epsilon} - \left(\frac{19}{2} + \frac{\pi^2}{6}\right) + O(\epsilon)\right]. \end{aligned} \tag{4.49}$$

The A_6 Integral

Lastly, the A_6 MI represents the two-loop two-point vertex. It is given as [4],

$$A_6 = \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p-k_1-k_2)^2 (p-\ell)^2 (p-\ell-k_2)^2 (\ell-k_1)^2}. \tag{4.50}$$

Unlike the other A-type antennae, this one cannot be given in terms of the scalar one-loop functions B_0, C_0 or D_0. To compute this integral we have to start by expressing it as a Mellin-Barnes contour integral, then evaluate the residues, and finally expand as a Laurent series around ϵ. This process gives,

$$\begin{aligned} A_6 &= \frac{S_\Gamma\left(-q^2\right)^{-2-2\epsilon}}{(2\pi i)^2}\int_{-i\infty}^{i\infty}\int_{-i\infty}^{i\infty} dz_1 dz_2\,\frac{\Gamma(-z_1)\Gamma(-z_2)\Gamma(2+2\epsilon+z_1+z_2)\Gamma^2(-\epsilon-z_1)\Gamma^2(-\epsilon-z_2)}{\Gamma(-2\epsilon-z_1-z_2)\Gamma(2+\epsilon+z_1+z_2)} \\ &= S_\Gamma\left(-q^2\right)^{-2-2\epsilon}\left[-\frac{1}{\epsilon^4}+\frac{5\pi^2}{6\epsilon^2}+\frac{27}{\epsilon}\zeta_3+\frac{23\pi^4}{36}+O(\epsilon)\right]. \end{aligned} \tag{4.51}$$

4.1.3 Exploratory Massive Master Integrals

AntCalc features an exploratory, but complete, massive Implementation of the A_3^0 route. To integrate it, two new MI are required. These two massive MI are,

$$\begin{aligned} I_1^{(m_q,0,m_q)} &= \int d\Phi_X^{(m_q,0,m_q)} \\ I_2^{(m_q,0,m_q)} &= \int d\Phi_X^{(m_q,0,m_q)}\, s_{13} \end{aligned} \tag{4.52}$$

where $\Phi_X^{(m_q,0,m_q)}$ is the massive tripole phase space and is defined as,

$$d\Phi_X^{(m_q,0,m_q)} = \frac{d\Phi_3^{(m_q,0,m_q)}(q;k_1,k_3,k_2)}{d\Phi_2^{(m_q,m_q)}(q;\tilde{k}_I,\tilde{k}_K)}, \quad p\to q(k_1,m_q)+g(k_3,0)+\bar{q}(k_2,m_q) \tag{4.53}$$

and m_q is the quark mass. Note that the two- and three-particle massive phase spaces are described as in Eq. 4.9 with $k_1^2=k_2^2=m_q^2$, $k_3^2=0$, and $q^2>4m_q^2$.

These two integrals evaluate to [37],

$$\begin{aligned} I_1^{(m_q,0,m_q)} &= \left(q^2\right)^{1-\epsilon} r^{2-2\epsilon}2^{-2\epsilon}\pi^{-2+\epsilon}\frac{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}{\Gamma(6-6\epsilon)}{}_2F_1\left(\frac{1}{2},2-2\epsilon;\frac{7}{2}-3\epsilon;r\right) \\ I_2^{(m_q,0,m_q)} &= \left(q^2\right)^{2-\epsilon} r^{3-2\epsilon}2^{1-2\epsilon}\pi^{-2+\epsilon}\frac{\Gamma(3-2\epsilon)\Gamma(4-3\epsilon)}{\Gamma(8-6\epsilon)}{}_2F_1\left(\frac{1}{2},3-2\epsilon;\frac{9}{2}-3\epsilon;r\right) \end{aligned} \tag{4.54}$$

where,

$$r = 1-\frac{4m_q^2}{q^2}. \tag{4.55}$$

4.2 NLO Antenna Results

The simplest non-trivial SMQCD antenna is A_3^0. As such, it may be used as an example of how AntCalc handles the physics involved in building and then integrating it over the phase space. In this chapter, we are going to go over how this antenna is mathematically computed by hand, and how AntCalc handles those same calculations on its end.

4.2.1 The Tree-Level Three-Parton Antenna A_3^0

Analytic Construction of the Unintegrated A_3^0 Antenna

The A_3^0 is the tree-level final-final quark-antiquark-gluon antenna. It represents the normalised $\gamma^*(p) \to q(k_1)g(k_3)\bar{q}(k_2)$ process. This process, in the current model, is able to produce two distinct topologies which differ by what quark line the gluon is emitted from. Those diagrams are shown below in Fig. 4.1.

(a) Gluon emitted from the quark line. **(b)** Gluon emitted from the antiquark line.

Figure 4.1: Feynman diagrams depicting the two topologies described under the A_3^0 antenna, corresponding to the process $\gamma^* \to q(k_1)g(k_3)\bar{q}(k_2)$.

The amplitude is given as follows directly from the diagrams,

$$\mathcal{M}_3^0 = e_q g_s (T^{a_3})_{i_1 i_2} \left[\bar{u}(k_1) \not{\epsilon}^*(k_3) \frac{\not{k}_1 + \not{k}_3}{s_{13}} \not{\epsilon}(p) v(k_2) - \bar{u}(k_1) \not{\epsilon}^*(p) \frac{\not{k}_2 + \not{k}_3}{s_{23}} \not{\epsilon}(k_3) v(k_2) \right] \tag{4.56}$$

where $(T^{a_3})_{i_1 i_2}$ represents the colour matrix for the quark and antiquark in positions 1 and 2 and gluon in position 3.

The amplitude is now squared as follows,

$$\begin{aligned} |\mathcal{M}_3^0|^2 &= e_q^2 g_s^2 (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_1 i_2} \left(M_A M_A^\dagger + M_B M_B^\dagger - M_A M_B^\dagger - M_B M_A^\dagger \right) \\ \text{with } M_A &= \bar{u}(k_1) \not{\epsilon}^*(k_3) \frac{\not{k}_1 + \not{k}_3}{s_{13}} \not{\epsilon}(p) v(k_2) = \bar{u}(k_1) \Gamma_A v(k_2), \\ M_B &= \bar{u}(k_1) \not{\epsilon}^*(p) \frac{\not{k}_2 + \not{k}_3}{s_{23}} \not{\epsilon}(k_3) v(k_2) = \bar{u}(k_1) \Gamma_B v(k_2). \end{aligned} \tag{4.57}$$

which in total becomes

$$\begin{aligned} = \sum_{a_3, i_1, i_2} & (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_1 i_2} \\ & \sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) \left[\Gamma_A \bar{\Gamma}_A + \Gamma_B \bar{\Gamma}_B - \Gamma_A \bar{\Gamma}_B - \Gamma_B \bar{\Gamma}_A \right] v(k_2, s_2) \bar{v}(k_2, s_2) \end{aligned} \tag{4.58}$$

Each of the Γ terms becomes, using Dirac Algebra,

$$\begin{aligned}
\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} & \bar{u}(k_1,s_1)u(k_1,s_1)\left(\Gamma_A\bar{\Gamma}_A\right)v(k_2,s_2)\bar{v}(k_2,s_2) = \\
&= \frac{1}{s_{13}^2}\sum_{\lambda_g,\lambda_\gamma}\varepsilon_\mu^*\varepsilon_\nu\varepsilon_\rho\varepsilon_\sigma^*\,\mathrm{Tr}\left(\not{k}_1\gamma^\mu(\not{k}_1+\not{k}_3)\gamma^\rho\not{k}_2\gamma^\sigma(\not{k}_1+\not{k}_3)\gamma^\nu\right) \\
&= \frac{1}{s_{13}^2}g_{\mu\nu}g_{\rho\sigma}\,\mathrm{Tr}\left(\not{k}_1\gamma^\mu(\not{k}_1+\not{k}_3)\gamma^\rho\not{k}_2\gamma^\sigma(\not{k}_1+\not{k}_3)\gamma^\nu\right) \\
&= \frac{1}{s_{13}^2}\,\mathrm{Tr}\left(\not{k}_1\gamma^\mu(\not{k}_1+\not{k}_3)\gamma^\rho\not{k}_2\gamma_\rho(\not{k}_1+\not{k}_3)\gamma_\mu\right) \\
&= \left(\frac{2-d}{s_{13}}\right)^2\mathrm{Tr}(\not{k}_1\not{P}\not{k}_2\not{P}),\quad \not{P}=(\not{k}_1+\not{k}_3) \\
&= 4\left(\frac{2-d}{s_{13}}\right)^2\left(2(k_1\cdot P)(k_2\cdot P)-(k_1\cdot k_2)P^2\right) \\
&= 2(d-2)^2\frac{s_{23}}{s_{13}}
\end{aligned} \tag{4.59}$$

$$\begin{aligned}
\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} & \bar{u}(k_1,s_1)u(k_1,s_1)\left(\Gamma_B\bar{\Gamma}_B\right)v(k_2,s_2)\bar{v}(k_2,s_2) = \\
&= 2(d-2)^2\frac{s_{13}}{s_{23}}
\end{aligned} \tag{4.60}$$

$$\begin{aligned}
\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} & \bar{u}(k_1,s_1)u(k_1,s_1)\left(\Gamma_A\bar{\Gamma}_B\right)v(k_2,s_2)\bar{v}(k_2,s_2) = \\
&= 2(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}
\end{aligned} \tag{4.61}$$

$$\begin{aligned}
\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} & \bar{u}(k_1,s_1)u(k_1,s_1)\left(\Gamma_B\bar{\Gamma}_A\right)v(k_2,s_2)\bar{v}(k_2,s_2) = \\
&= 2(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}
\end{aligned} \tag{4.62}$$

and computing the colour matrix sum as

$$\sum_{a_3,i_1,i_2}(T^{a_3})_{i_1i_2}(T^{a_3})_{i_1i_2}=\frac{1}{2}(N^2-1) \tag{4.63}$$

we are finally able to reach the final result, as follows, using dimensional regularisation as $d\to 4-2\epsilon$,

$$
\begin{aligned}
|\mathcal{M}_3^0|^2 &= e_q^2 g_s^2 (N^2-1)\left(2(d-2)^2\left(\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}\right)-4(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}\right)\\
&= 2(d-2)e_q^2 g_s^2 (N^2-1)\left((d-2)\left(\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}\right)-2\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}\right)\\
&= \frac{8}{s_{13}s_{23}}\left(2q^2 s_{12}+s_{13}^2+s_{23}^2-2\epsilon\left(q^2 s_{12}+s_{13}^2+s_{23}^2+s_{13}s_{23}\right)+\epsilon^2\left(s_{13}+s_{23}\right)^2\right)
\end{aligned}
\tag{4.64}
$$

The A_3^0 antenna also requires the $\mathcal{M}_2^0$ amplitude. This amplitude corresponds to the $\gamma^* \to q\bar{q}$ Born process. The amplitude is

$$
\mathcal{M}_2^0 = e_q \delta_{i_1 i_2} \bar{u}(k_1)\not{\epsilon}(p) v(k_2). \tag{4.65}
$$

Applying the same spin, polarisation and colour sums as above, and considering that, for a two final-state particles system $s_{12} = q^2$, we get

$$
\begin{aligned}
|\mathcal{M}_2^0|^2 = 4e_q^2(d-2)Ns_{12} &= 4e_q^2(d-2)Nq^2\\
&= 8e_q^2(1-\epsilon)Nq^2
\end{aligned}
\tag{4.66}
$$

The antenna is now finally ready to be built. That is accomplished using the following formula,

$$
A_3^0 = \frac{1}{2g_s^2 C_F}\frac{|\mathcal{M}_3^0|^2}{|\mathcal{M}_2^0|^2}. \tag{4.67}
$$

By taking $C_F = (N^2-1)/(2N)$, the formula yields, using Eqs. 4.64 and 4.66,

$$
\begin{aligned}
A_3^0 &= \frac{2q^2 s_{12}+s_{13}^2+s_{23}^2}{q^2 s_{13}s_{23}} - \epsilon\frac{(s_{13}+s_{23})^2}{q^2 s_{13}s_{23}}\\
&= \frac{1}{q^2}\left[\left(\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}+2\frac{s_{12}q^2}{s_{13}s_{23}}\right)-\epsilon\left(2+\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}\right)\right]
\end{aligned}
\tag{4.68}
$$

Phase-Space Integration through R_3

We now integrate the A_3^0 antenna over the appropriate (3 final-state particle, in this case) phase-space. This integral is computed in d dimensions and over the phase-space measure in terms of the Mandelstam invariants (s_{13}, q^2, etc.). The integral takes the form,

$$
\begin{aligned}
\mathcal{A}_3^0 &= C(\epsilon,1)\int d\Phi_{ijk}\, A_3^0\\
&= \mathcal{N}(\epsilon,1)\,(4\pi)^{3-2d}\left(q^2\right)^{1-\frac{d}{2}}\int ds_{12}ds_{13}ds_{23}d\Omega\,(s_{12}s_{13}s_{23})^{\frac{d-4}{2}}\delta\left(q^2-s_{12}-s_{13}-s_{23}\right)A_3^0.
\end{aligned}
\tag{4.69}
$$

with $d\Phi_{ijk} = d\Phi_3/\Phi_2$. $C(\epsilon, 1)$ is the integral normalisation which is here given as

$$C(\epsilon, 1) = 8\pi^2 (4\pi)^{-\epsilon} e^{\epsilon \gamma_E}. \tag{4.70}$$

This integral is quite complex and spans across all A_3^0 terms. Accordingly, we are not going to compute this integral directly: instead we are going to use IBP reduction to reduce the expression to a linear combination of master integrals, compute those master integrals and then use the result to compute the integrated antenna. This process is explained in Subsection 2.5.2.

After applying IBP identities and reducing the integral to a linear combination of this master integral, we reach the following expression,

$$\mathcal{A}_3^0 = \frac{C(\epsilon, 1)}{\Phi_2} \frac{2(2 - 6\epsilon + 5\epsilon^2 - 2\epsilon^3)}{q^2 \epsilon^2} R_3 \tag{4.71}$$

This antenna has a single master, R_3, discussed in Section 4.1. This master integral is the integral over phase-space itself,

$$R_3 = \int d\Phi_3 = \frac{2^{-7+4\epsilon} \pi^{-3+2\epsilon} \Gamma^3(1-\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)} \left(q^2\right)^{1-2\epsilon}. \tag{4.72}$$

IBP Reduction of an Example Term

As an example of the IBP reduction method, we will consider the eikonal term of the unintegrated antenna in Eq. 4.68,

$$\mathcal{S}_{12}^{(3)} = \frac{2 s_{12}}{s_{13} s_{23}}. \tag{4.73}$$

We define,

$$\begin{gathered} y_{13} = \frac{s_{13}}{q^2}, \quad y_{23} = \frac{s_{23}}{q^2}, \quad y_{12} = \frac{s_{12}}{q^2} = 1 - y_{13} - y_{23} \\ J(a, b, c) = \int_\Delta dy_{13} dy_{23}\, y_{13}^{a-1} y_{23}^{b-1} y_{12}^{c-1}, \quad \Delta : y_{13}, y_{23}, y_{12} \geq 0, \;\; y_{13} + y_{23} + y_{12} = 1 \end{gathered} \tag{4.74}$$

and with these,

$$\begin{aligned} \Phi_3 &\equiv \int d\Phi_3 = R_3 \\ &= \mathcal{K}\left(q^2, \epsilon\right) \int_\Delta (y_{13} y_{23} y_{12})^{-\epsilon} dy_{13} dy_{23} \\ &= \mathcal{K}\left(q^2, \epsilon\right) J(1-\epsilon, 1-\epsilon, 1-\epsilon), \quad \mathcal{K}\left(q^2, \epsilon\right) = \frac{2^{-7+4\epsilon} \pi^{-3+2\epsilon}}{\Gamma(2-2\epsilon)} \left(q^2\right)^{1-2\epsilon}. \end{aligned} \tag{4.75}$$

We are also able to express the integral over the eikonal term using a J function as,

$$\int d\Phi_3\, \mathcal{S}_{12}^{(3)} = \frac{2\mathcal{K}}{q^2} J(-\epsilon, -\epsilon, 2-\epsilon). \tag{4.76}$$

The IBP identity we use to reduce the integral is implemented as,

$$\begin{aligned} 0 &= \int_\Delta dy_{13} dy_{23} \frac{\partial}{\partial y_{13}} \left(y_{13}^a y_{23}^{b-1} y_{12}^{c-1} \right) \\ &= aJ(a, b, c) - (c-1) J(a+1, b, c-1) \end{aligned} \tag{4.77}$$

such that, for y_{13},

$$J(-\epsilon,-\epsilon,2-\epsilon) = -\frac{1-\epsilon}{\epsilon} J(1-\epsilon,-\epsilon,1-\epsilon) \tag{4.78}$$

and for y_{23},

$$J(1-\epsilon,-\epsilon,2-\epsilon) = -\frac{1-\epsilon}{\epsilon} J(1-\epsilon,1-\epsilon,1-\epsilon). \tag{4.79}$$

By the symmetry $y_{13} \leftrightarrow y_{12}$,

$$J(2-\epsilon,-\epsilon,1-\epsilon) = J(1-\epsilon,-\epsilon,2-\epsilon). \tag{4.80}$$

And hence,

$$J(-\epsilon,-\epsilon,2-\epsilon) = \frac{(1-\epsilon)(2-3\epsilon)}{\epsilon^2} J(1-\epsilon,1-\epsilon,1-\epsilon) \tag{4.81}$$

which finally makes,

$$\int d\Phi_3 \, \frac{2s_{12}}{s_{13}s_{23}} = \frac{2(1-\epsilon)(2-3\epsilon)}{\epsilon^2 q^2} R_3. \tag{4.82}$$

Master Integral Substitution and Laurent Expansion

Substituting Eq. 4.72 into Eq. 4.71 gives

$$\begin{aligned} \mathcal{A}_3^0 &= \frac{C(\epsilon,1)}{\Phi_2} \left(q^2\right)^{-2\epsilon} (4\pi)^{-3+2\epsilon} (2-6\epsilon+5\epsilon^2-2\epsilon^3) \frac{\Gamma^3(1-\epsilon)}{\epsilon^2\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)} \\ &= \left(q^2\right)^{-\epsilon} e^{\epsilon\gamma_E} (1-2\epsilon)\big(2+\epsilon(\epsilon+2)\big) \frac{\Gamma^2(-\epsilon)}{\Gamma(3-3\epsilon)} \end{aligned} \tag{4.83}$$

and, after expanding the integrated expression as a Laurent series in ϵ, we finally arrive at the infrared pole structure,

$$\begin{aligned} \mathcal{A}_3^0 = \left(q^2\right)^{-\epsilon} \Bigg[& \frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{57-7\pi^2}{12} + \epsilon \left(\frac{84-21\pi^2-8\psi^{(2)}(1)+108\psi^{(2)}(3)}{24} \right) \\ & + \epsilon^2 \left(\frac{35640-3990\pi^2-71\pi^4-720\psi^{(2)}(1)+9720\psi^{(2)}(3)}{1440} \right) \Bigg] \end{aligned}$$

$$= \left(q^2\right)^{-\epsilon} \left[\frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12} + \epsilon \left(\frac{109}{8} - \frac{7\pi^2}{8} - \frac{25\zeta_3}{3} \right) + \epsilon^2 \left(\frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440} - \frac{25\zeta_3}{2} \right) \right] \tag{4.84}$$

where the usual polygamma identities, $\psi^{(2)}(1) = -2\zeta_3$ and $\psi^{(2)}(3) = -2\zeta_3 + \frac{9}{4}$, were employed.

Implementation in *AntCalc*

Using *AntCalc*, building the antenna is very simple, since the function `BuildAntenna[type, numFinalParticles, numLoops]` already encodes the entire logical chain and is able to return all intermediate steps using the appropriate options.

First, by running the function naively we get the final result in Eq. 4.68,

```
In[2]:= BuildAntenna[A,3,0]//Collect[#,Epsilon]&
```

```
Out[2]:=
```

$$\frac{2s_{12}}{q^2 s_{13}}+\frac{2s_{12}}{q^2 s_{23}}+\frac{2s_{12}^2}{q^2 s_{13}s_{23}}+\frac{s_{13}}{q^2 s_{23}}+\frac{s_{23}}{q^2 s_{13}}+\epsilon\left(\frac{-2}{q^2}-\frac{s_{13}}{q^2 s_{23}}-\frac{s_{23}}{q^2 s_{13}}\right)$$

We can also use the `ReturnRecord` option to access all intermediate steps, and check that *AntCalc* goes through the same steps as the by-hand derivation in Section 4.2.1. We can see that with,

```
In[3]:=
BuildAntenna[A,3,0,ReturnRecord->True]["IntermediateSteps"]
Out[3]:=
   Amplitude:
```

$$T^{a_3}_{i_1 i_2}\left[\frac{\bar{u}(k_1)\,\not{\epsilon}^*(k_3)\,(\not{k}_1+\not{k}_3)\,\not{\epsilon}(p)\,v(k_2)}{(k_1+k_3)^2}-\frac{\bar{u}(k_1)\,\not{\epsilon}(p)\,(\not{k}_2+\not{k}_3)\,\not{\epsilon}^*(k_3)\,v(k_2)}{(k_2+k_3)^2}\right]$$

```
   Interference:
```

$$\frac{4(\epsilon-1)(N^2-1)\Big(-2s_{12}^2-2s_{12}(s_{13}+s_{23})+(\epsilon-1)s_{13}^2+2\epsilon\, s_{13}s_{23}+(\epsilon-1)s_{23}^2\Big)}{s_{13}s_{23}}$$

```
   Result:
```

$$-\frac{2\epsilon}{q^2}+\frac{2s_{12}^2}{q^2 s_{13}s_{23}}+\frac{2s_{12}}{q^2 s_{13}}+\frac{2s_{12}}{q^2 s_{23}}+\frac{s_{13}}{q^2 s_{23}}-\frac{\epsilon\, s_{13}}{q^2 s_{23}}+\frac{s_{23}}{q^2 s_{13}}-\frac{\epsilon\, s_{23}}{q^2 s_{13}}$$

where `Amplitude`, `Interference` and `Result` respectively refer to the expressions in Eqs. 4.56, 4.64 and 4.68.

Integrating antennae using *AntCalc* is also simple. Every step in Section 4.2.1 is handled by a single function: `IntegrateAntenna[antenna]`.

Before going over how this function is used, a small clarification needs to be made: the `IntegrateAntenna` function is not compatible with the default output of the `BuildAntenna` function described earlier. This is so that the built antenna is immediately useful and easily readable. As described in Chapter 3, however, the IBP reduction steps are handled by the LiteRed2 package, which does not expect Mandelstam invariants, but rather, momenta. Accordingly, to use `BuildAntenna`'s output in `IntegrateAntenna` we must toggle the relevant flag: `BuildAntenna[A,3,0,IntegrableForm -> True]`. This will return a large association with all required information for integrating the antenna.

We may now proceed to the integration. The whole process goes as follows,

```
In[4]:= intA30 = BuildAntenna[A,3,0,IntegrableForm -> True];
In[5]:= IntegrateAntenna[intA30,ExpansionOrder->2]
Out[5]:=
```

$$\frac{19}{4}+\epsilon^{-2}+\frac{3}{2\epsilon}-\frac{7\pi^2}{12}+\epsilon^2\left(\frac{639}{16}-\frac{133\pi^2}{48}-\frac{71\pi^4}{1440}-\frac{25\zeta_3}{2}\right)+\epsilon\left(\frac{109}{8}-\frac{7\pi^2}{8}-\frac{25\zeta_3}{3}\right)$$

Much like with `BuildAntenna`, `IntegrateAntenna` also allows us to inspect its intermediate steps. Here, however, the available options show the master integral linear combination expression both with R_3 and with it substituted by the appropriate expression, that is, the equivalent to Eqs. 4.71 and 4.83 before normalisation. The code and results are,

```
In[6]:= resA30 = IntegrateAntenna[intA30,ExpansionOrder->2,
ReturnRecord->True];
In[7]:= resA30["MasterCombination"]//Simplify//Collect[#,R3]&
Out[7]:=
```

$$\frac{-4(d-3)\left(4+d(\epsilon-2)-4\epsilon\right)}{(d-4)^2 q^2} R_3$$

```
In[8]:= resA30["MasterSubstitutedExpression"]//Simplify
Out[8]:=
```

$$-\left(q^2\right)^{-2\epsilon}\frac{2^{-5+4\epsilon}(d-3)\left(4+d(\epsilon-2)-4\epsilon\right)\pi^{-3+2\epsilon}\Gamma^3(1-\epsilon)}{(d-4)^2\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)}$$

The results here are given as a combination of d and ϵ. This is a result of the way the function processes each step. Since these are snapshots of what is happening inside the function itself, they show the expressions before dimension regularisation is applied.

Finally, *AntCalc* also enables us to forgo the work we did building and the integrating the antenna with the use of the `BuildAndIntegrateAntenna[type, numFinalParticles,numLoops]` function. It goes over the whole process, as we did, and outputs the ϵ Laurent series,

```
In[9]:= BuildAndIntegrateAntenna[A,3,0,ExpansionOrder->2]
Out[9]:=
```

$$\frac{19}{4}+\epsilon^{-2}+\frac{3}{2\epsilon}-\frac{7\pi^2}{12}+\epsilon^2\left(\frac{639}{16}-\frac{133\pi^2}{48}-\frac{71\pi^4}{1440}-\frac{25\zeta_3}{2}\right)+\epsilon\left(\frac{109}{8}-\frac{7\pi^2}{8}-\frac{25\zeta_3}{3}\right)$$

The agreement between the analytic derivation and the corresponding *AntCalc* outputs validates both the construction and the integration workflow for A_3^0. Its local soft and collinear behaviour is validated separately in the following subsection.

4.2.2 Local Validation of A_3^0

To be usable as a subtraction term, an antenna function's unintegrated form must also reproduce the singular behaviour of the corresponding matrix element locally in phase space. In particular, A_3^0 must reproduce the universal eikonal factor when the gluon becomes soft and the Altarelli-Parisi splitting function in the relevant collinear limits. These two quantities provide local validation of the results obtained analytically and using *AntCalc*.

Soft Limit and the Eikonal Factor

When a gluon becomes soft, gauge theory amplitudes factorise universally to the amplitude without the soft gluon times a universal soft current as [7],

$$|\mathcal{M}_{n+1}(\ldots,i,j,k,\ldots)|^2 \xrightarrow{j\to 0} -g_s^2\sum_{i,k} T_i\cdot T_k \mathcal{S}_{ik}^{(j)}|\mathcal{M}_n(\ldots,i,k,\ldots)|^2 \tag{4.85}$$

where T_i are colour-charge operators and,

$$\mathcal{S}_{ik}^{(j)} = \frac{2s_{ik}}{s_{ij}s_{jk}} \tag{4.86}$$

is the eikonal factor.

Since this property is universal, we can use it to validate our results. For the A_3^0 antenna, the eikonal factor may be computed as,

$$\mathcal{S}_{12}^{(3)} = \lim_{k_3 \to 0} A_3^0(1_q, 3_g, 2_{\bar{q}}). \tag{4.87}$$

After building the antenna using *AntCalc*, the limit may be computed using *Wolfram Mathematica* as follows,

```
In[2]:= A30 = BuildAntenna[A,3,0];

softSubs = {s13 -> lambda * s13, s23 -> lambda * s23};
A30Soft = A30 /.  softSubs;

A30SoftSeries = Series[A30Soft, {lambda, 0, 0}];

eikonalFactor = SeriesCoefficient[A30SoftSeries, -2] /.  q2 -> s12
// Simplify;
Out[2]:=
```

$$\frac{2s_{12}}{s_{13}s_{23}}$$

This result agrees with the expected eikonal factor.

Collinear Limit and the Altarelli–Parisi Splitting Function

In the same manner, when two partons i and j become collinear, the amplitude factorises to [7, 38],

$$|\mathcal{M}_{n+1}(..., i, j, ...)|^2 \xrightarrow{i||j} \frac{8\pi\alpha_s}{s_{ij}} P_{IJ\to K}(z)|\mathcal{M}_n(..., K, ...)|^2 \tag{4.88}$$

where K is the parent parton in which i and j recombine, z is the momentum fraction carried by one of the daughter partons, and $P_{IJ\to K}$ is the splitting function. This function is universal but channel-dependent and, since it carries the spin and helicity structure of the collinear splitting, it depends on the partons involved. For A_3^0, where a gluon may become collinear with a quark, the function appears as,

$$P_{qg\to q}(z) = C_F \frac{1+z^2}{1-z}. \tag{4.89}$$

Noting that the building process already normalises the fundamental Casimir C_F, using *AntCalc* in *Wolfram Mathematica*, the AP splitting function may be computed as,

```
In[3]:= A30 = BuildAntenna[A,3,0];

collinearSubs = {s13 -> lambda, s12 -> z q2, s23 -> (1 - z) q2,
Epsilon -> 0};
A30Collinear = A30 /.  collinearSubs;

A30CollinearSeries = Series[A30Collinear, {lambda, 0, 0}];

apSplittingFunc = SeriesCoefficient[A30CollinearSeries, -1]
```

```
// Simplify;
Out[3]:=
```

$$\frac{1+z^2}{1-z}$$

This result agrees with the expected colour-stripped $qg \to q$ AP kernel.

4.2.3 The Virtual A_2^1 Contribution

To compute A_2^1, we may follow the same steps. This is a loop-antenna, however, which means that its build result has been PaVe reduced. It is given in terms of the B_0 and C_0 scalar one-loop function. The unintegrated antenna is computed, using *AntCalc*, as,

```
In[2]:= BuildAntenna[A,2,1]
Out[2]:=
```

$$-\frac{1}{2}\Big[(3+2\epsilon)B_0\Big(q^2;0,0\Big)\Big] - q^2 C_0\Big(0,0,q^2;0,0,0\Big)$$

The integrated final result is, then [4],

```
In[3]:= BuildAndIntegrateAntenna[A,2,1,ExpansionOrder->2]
Out[3]:=
```

$$-\frac{1}{\epsilon^2} - \frac{3}{2\epsilon} - 4 + \frac{7\pi^2}{12} + \epsilon\left(-8 + \frac{7\pi^2}{8} + \frac{7\zeta_3}{3}\right) + \epsilon^2\left(-16 + \frac{7\pi^2}{3} - \frac{73\pi^4}{1440} + \frac{7\zeta_3}{2}\right)$$

Note that, exclusively for this antenna, the same PaVe scalar functions appear in both the build- and integration-side reductions. No phase-space IBP reduction is required.

4.3 NNLO Antenna Results

4.3.1 Two-Parton Contributions: the A_2^2 Set

The A_2^2 route is computed similarly to A_2^1. This route, however, yields four different antenna components and is the longest-running calculation supported by *AntCalc* in its current version. The unintegrated antennae are computed using,

```
In[2]:= BuildAntenna[A,2,2]
Out[2]:=
```

$$\Bigg\{\frac{11\Big[(3+2\epsilon)B_0(s_{12};0,0) + 2s_{12}C_0(0,0,s_{12};0,0,0)\Big]}{12\epsilon} + \ldots,$$
$$24\epsilon^{14}\Big(391\pi^{12} - 1149918\pi^8\zeta_3 - 62009136\pi^4\zeta_3^2 - 823011840\zeta_3^3\Big) + \ldots,$$
$$-\frac{1}{6}\frac{(3+2\epsilon)B_0(s_{12};0,0) + 2s_{12}C_0(0,0,s_{12};0,0,0)}{\epsilon} + \ldots,$$
$$80(13-42\epsilon)\frac{\zeta_3}{1920\epsilon\Gamma^2(2-2\epsilon)} + \ldots\Bigg\}$$

The expressions have been truncated here. These four components represent, respectively, the leading, subleading, quark-loop and self-interference terms, or, A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$ and $\breve{A}_2^2$, in that order. Their complete unintegrated forms are lengthy and are, therefore collected in Appendix A. We may then obtain the integrated expressions as,

```
In[3]:= BuildAndIntegrateAntenna[A,2,2,ExpansionOrder->0]
Out[3]:=
```

$$
\left\{\frac{1}{4\epsilon^4}+\frac{17}{8\epsilon^3}+\frac{433-72\pi^2}{144\epsilon^2}+\frac{121350-44820\pi^2+15120\zeta_3}{25920\epsilon}+\frac{-45415-64590\pi^2+4734\pi^4+37440\zeta_3}{25920},\right.
$$
$$
-\frac{1}{4\epsilon^4}-\frac{3}{4\epsilon^3}+\frac{-1476+312\pi^2}{576\epsilon^2}+\frac{-3978+864\pi^2+1536\zeta_3}{576\epsilon}+\frac{-10359+2850\pi^2-118\pi^4+4176\zeta_3}{576},
$$
$$
-\frac{1}{4\epsilon^3}-\frac{1}{9\epsilon^2}+\frac{65+9\pi^2}{216\epsilon}+\frac{4085-546\pi^2+72\zeta_3}{1296},
$$
$$
\left.\frac{1}{4\epsilon^4}+\frac{3}{4\epsilon^3}+\frac{1230-20\pi^2}{480\epsilon^2}+\frac{-60\pi^2+560(6+\zeta_3)}{480\epsilon}+\frac{-205\pi^2-7\pi^4-240(-36+7\zeta_3)}{480}\right\}
$$

The integrated Laurent series agree with the established $\mathcal{A}_2^2$ set results [35].

4.3.2 Three-Parton Contributions: the A_3^1 Set

Computing the A_3^1 antenna set is very similar to the work done for the A_2^2 set, with the big differentiator being that this set does not include a self-interference term. The unintegrated antennae are computed using,

```
In[2]:= BuildAntenna[A,3,1]
Out[2]:=
```

$$
\left\{\epsilon(1+3\epsilon)s_{13}s_{23}+\ldots,\right.
$$
$$
\frac{2s_{12}\left[(-1+\epsilon)s_{13}+\epsilon s_{23}\right]}{2(-1+\epsilon)q^2s_{13}}D_0\left(0,0,0;q^2,s_{12},s_{23};0,0,0,0\right)+\ldots,
$$
$$
\left.\frac{2}{3}\frac{s_{12}^2}{\epsilon q^2 s_{13}s_{23}}+\ldots\right\}
$$

The expressions have been truncated here. These three components represent, respectively, the leading, subleading and quark-loop terms, or, A_3^1, $\tilde{A}_3^1$, and $\hat{A}_3^1$, in that order. Their complete unintegrated forms are lengthy and are, therefore collected in Appendix A. We may then obtain the integrated expressions as,

```
In[3]:= BuildAndIntegrateAntenna[A,3,1,ExpansionOrder->0]
Out[3]:=
```

$$\Big\{-\frac{1}{4\epsilon^4}-\frac{31}{12\epsilon^3}+\frac{-9540+660\pi^2}{1440\epsilon^2}+\frac{-38820+3520\pi^2+11040\zeta_3}{1440\epsilon}+\frac{-156930+12240\pi^2-123\pi^4+55120\zeta_3}{1440}+O(\epsilon),$$
$$\frac{-15+4\pi^2}{24\epsilon^2}+\frac{-19+\pi^2+28\zeta_3}{4\epsilon}+\frac{1215\pi^2+56\pi^4+540(-35+18\zeta_3)}{720},$$
$$\frac{1}{3\epsilon^3}+\frac{1}{2\epsilon^2}+\frac{114-14\pi^2}{72\epsilon}+\frac{327-21\pi^2-200\zeta_3}{72}+O(\epsilon)\Big\}$$

The integrated Laurent series agree with the established $\mathcal{A}_3^1$ set results [4, 35].

4.3.3 Four-Parton Contributions

The A_4^0 Antenna Set

The double real-radiation antenna set is A_4^0, and it has two components, A_4^0 and $\tilde{A}_4^0$. It represents the $\gamma^* \to qgg\bar{q}$ process.

Building the unintegrated antenna follows the same usual process,

```
In[2]:= BuildAntenna[A,4,0]
Out[2]:=
```

$$\Big\{-\frac{3s_{12}}{2(-1+\epsilon)\left(q^2\right)^3}+\ldots,$$
$$-\frac{\epsilon}{(-1+\epsilon)q^2 s_{13}}+\ldots\Big\}$$

and so does computing the integrated antenna functions,

```
In[3]:= BuildAndIntegrateAntenna[A,4,0,ExpansionOrder->0]
Out[3]:=
```

$$\Big\{\frac{3}{4\epsilon^4}+\frac{65}{24\epsilon^3}-\frac{1}{\epsilon^2}\left(\frac{217}{18}-\frac{13\pi^2}{12}\right)+\frac{1}{\epsilon}\left(\frac{43224}{864}-\frac{589\pi^2}{144}-\frac{71\zeta_3}{4}\right)+\frac{1076717}{5184}-\frac{7955\pi^2}{432}+\frac{373\pi^4}{1440}-\frac{1327\zeta_3}{18}+O(\epsilon),$$
$$\frac{1}{\epsilon^4}+\frac{3}{\epsilon^3}+\frac{1}{\epsilon^2}\left(13-\frac{3\pi^2}{2}\right)+\frac{1}{\epsilon}\left(\frac{845}{16}-\frac{9\pi^2}{2}-\frac{80\zeta_3}{3}\right)+\frac{6921}{32}-\frac{473\pi^2}{24}+\frac{17\pi^4}{72}-80\zeta_3+O(\epsilon)\Big\}$$

where for each case, the lists are $\{A_4^0, \tilde{A}_4^0\}$ and $\{\mathcal{A}_4^0, \tilde{\mathcal{A}}_4^0\}$, respectively.

The complete unintegrated expressions are collected in Appendix A. The integrated results agree with the established $\mathcal{A}_4^0$ antenna results [4].

The B_4^0 and C_4^0 Antennae

Finally, the $\gamma^* \to q\bar{q}q\bar{q}$ process is represented by the B_4^0 and C_4^0 antennae. The B_4^0 antenna describes non-identical quark flavours, while C_4^0 gives the identical-flavour contribution.

The antenna building process is completed as usual using *AntCalc*,

```
In[2]:= BuildAntenna[B,4,0]
Out[2]:=
```

$$-\frac{\epsilon s_{12}}{(-1+\epsilon)q^2 s_{134}^2} + \ldots$$

```
In[3]:= BuildAntenna[C,4,0]
Out[3]:=
```

$$-\frac{1}{2}\frac{\epsilon s_{12}}{(-1+\epsilon)q^2 s_{123}s_{134}} + \ldots$$

The integration process follows in the same way,

```
In[4]:= BuildAndIntegrateAntenna[B,4,0,ExpansionOrder->0]
Out[4]:=
```

$$-\frac{1}{12\epsilon^3} - \frac{7}{18\epsilon^2} + \frac{1}{\epsilon}\left(-\frac{407}{216} + \frac{11\pi^2}{72}\right) - \frac{11753}{1296} + \frac{77\pi^2}{108} + \frac{67\zeta_3}{18} + O(\epsilon)$$

```
In[5]:= BuildAndIntegrateAntenna[C,4,0,ExpansionOrder->0]
Out[5]:=
```

$$\frac{1}{\epsilon}\left(-\frac{13}{32} + \frac{\pi^2}{16} - \frac{\zeta_3}{4}\right) - \frac{339}{64} + \frac{17\pi^2}{48} - \frac{\pi^4}{45} + \frac{21\zeta_3}{8} + O(\epsilon)$$

The complete unintegrated expressions for both antennae are collected in Appendix A. The integrated results of both antennae agree with the established $\mathcal{B}_4^0$ and $\mathcal{C}_4^0$ results [4].

4.4 Summary of the Massless Antenna Set

The massless results presented in this chapter demonstrate that, using *AntCalc*, we are able to build and integrate all quark-antiquark final-final standard-model (SMQCD) antenna functions through NNLO. The selected case studies cover tree-level NLO and NNLO, one-, and two-loop contributions, showcasing the programme's ability to handle these different cases. Due to the size of the unintegrated antennae, some have been truncated. The complete results are shown in Appendix A. The resulting integrated antennae agree with the established results.

The more detailed A_3^0 explanation shows a method of local validation of the unintegrated result based on computing the eikonal factor and the AP splitting function using the antenna. This method might also be used with other antenna functions whenever the corresponding unresolved limit is present. A global validation is presented in Chapter 5, where we compute the massless hadronic R-ratio using all necessary integrated antennae.

Having established the massless sector, the following section considers the exploratory massive extension of the A_3^0 route.

4.5 Exploratory Massive Extension: A_3^0

The massive A_3^0 route is currently the only massive route supported in *AntCalc*. Its inner workflow is similar to that of the massless A_3^0 route with the added complexity of the quarks being massive.

Kinematically, instead of considering only the centre of mass energy as a scale, the quark mass must also be considered. Hence, q^2 and m_q^2 are scales, and they combine to create the relevant threshold variable r,

$$r = 1 - \frac{4m_q^2}{q^2}. \tag{4.90}$$

The phase space has $k_1^2 = k_2^2 = m_q^2$ and $k_3^2 = 0$. The cut families also shift the massless cuts by a mass term as,

$$\begin{aligned} k_{1,2}^2 &\to m_q^2 - k_{1,2}^2, \\ k_3^2 &\to k_3^2. \end{aligned} \tag{4.91}$$

This also implies using massive Mandelstam invariants which, in this case, with k_3 presenting the massless gluon's momentum, become,

$$s_{ij} = (q - k_k)^2 - m_i^2 - m_j^2, \quad \{i, j, k\} = \{1, 2, 3\}. \tag{4.92}$$

LiteRed2, then, has to be updated with these new families before recognising the topology. It will then generate two master integrals. In the current *AntCalc* implementation, the IBP reduction step produces two MI which do not exactly match the $I_1^{(m_q,0,m_q)}$ and $I_2^{(m_q,0,m_q)}$ integrals described in Subsection 4.1.3. Instead, it produces J_1 and J_2, shown below using *LiteRed2*'s $j[...]$ notation, where the complete basis, named MX30Basis123, uses the denominators $\left\{m_q^2 - k_1^2, m_q^2 - k_2^2, k_3^2, -s_{13}, -s_{23}\right\}$ in that order,

$$\begin{aligned} J_1^{(m_q,0,m_q)} &= j[\text{MX30Basis123}, 1, 1, 1, 0, 0] = \mathcal{N}_{\text{LiteRed2}} \int \text{Cut}\left[\frac{1}{D_1 D_2 D_3}\right], \\ J_2^{(m_q,0,m_q)} &= j[\text{MX30Basis123}, 2, 1, 1, 0, 0] = \mathcal{N}_{\text{LiteRed2}} \int \text{Cut}\left[\frac{1}{D_1^2 D_2 D_3}\right]. \end{aligned} \tag{4.93}$$

Here, $\mathcal{N}_{\text{LiteRed2}}$ represents the normalisation implicit in the cut-integral convention used for the family in *LiteRed2*. These two MI do not directly equal the two literature masters, but they can be mapped to them. We need here to introduce a conversion factor of $-1/4$ between the literature and *AntCalc*'s conventions. Using it, we can complete the mapping as,

$$\begin{aligned} I_1^{(m_q,0,m_q)} &= -\frac{1}{4} J_1^{(m_q,0,m_q)}, \\ I_2^{(m_q,0,m_q)} &= -\frac{1}{4}\left(a J_1^{(m_q,0,m_q)} + b J_2^{(m_q,0,m_q)}\right). \end{aligned} \tag{4.94}$$

with,

$$\begin{aligned} a &= \frac{(1-2\epsilon)m_q^2 + (1-\epsilon)q^2}{3(1-\epsilon)}, \\ b &= \frac{m_q^2\left(4m_q^2 - q^2\right)}{3(1-\epsilon)}. \end{aligned} \tag{4.95}$$

4.5.1 Build and Integrate Results

To build and integrate the massive A_3^0 in *AntCalc*, we use the same commands as for the massless A_3^0 route, as discussed in Subsection 4.2.1, with the added option `quarkMass -> mQ`, where `mQ` is the quark mass, so far written here as m_q. The building process is completed as,

```
In[2]:= BuildAntenna[A,3,0,quarkMass->mQ]
Out[2]:=
```

$$\left[-4(-2+\epsilon)m_q^4\left(s_{13}^2+s_{23}^2\right)\right]+\ldots$$

The integration process follows in the same usual way, noting that much like in all other integrated cases, the default output is given with $q^2 \to 1$, making here $r \to 1-4m_q^2$,

```
In[3]:= BuildAndIntegrateAntenna[A,3,0,quarkMass->mQ,
ExpansionOrder->0]
Out[3]:=
```

$$\frac{1}{2100m_q^2\left(1+2m_q^2\right)^2\pi^2}\left[\frac{1}{\epsilon}\left(70m_q^2\left(-1+2m_q^2+8m_q^4\right){}_2F_1\left(\frac{1}{2},2;\frac{7}{2};1-4m_q^2\right)+\ldots\right)\right.$$
$$\left.+\left(10\left(-1+2m_q^2+8m_q^4\right){}_2F_1\left(\frac{1}{2},2;\frac{7}{2};1-4m_q^2\right)+\ldots\right)\right]+O(\epsilon)$$

These expressions agree with the established literature results [37]. The full unintegrated and integrated antenna expressions were truncated in this section. They are shown completely in Appendix A.

4.6 Overview of Implemented Routes

Having covered both the massless antenna set and the exploratory massive extension, this section collects every implemented route in one place. Table 4.2 summarises how each route is constructed and integrated; Table 4.3 summarises the corresponding validation evidence and status. Together, they replace the individual validation statements given throughout this chapter with a single point of reference.

Route	**Process / Scope**	**API Call**[2]	**Construction Check**	**Integration Route**
A_2^0	$\gamma^* \to q\bar{q}$, LO	`[A,2,0]`	Self-interference, Born-normalised reference	Trivial (reference, $A_2^0 \equiv 1$)
A_3^0	$\gamma^* \to qg\bar{q}$, NLO real	`[A,3,0]`	Self-interference; colour-coefficient extraction; matched to by-hand derivation	IBP / *LiteRed2*, `X30` basis; MI: R_3
A_2^1	$\gamma^* \to q\bar{q}$, NLO virtual	`[A,2,1]`	Tree–loop interference; PaVe-reduced scalar extraction	PaVe / *FeynHelpers*; MI: B_0, C_0
A_3^1 set (A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$)	$\gamma^* \to qg\bar{q}$, NNLO, one loop	`[A,3,1]`	Tree–loop interference; colour-coefficient extraction (leading/subleading/quark-loop)	IBP / *LiteRed2*, `A31` basis; build-side MI are PaVe scalars (B_0, C_0, D_0, Table 4.1); integration MI: $V_{5,a}, V_{5,b}, V_8$

Table 4.2 continued.

Route	Process / Scope	API Call	Construction Check	Integration Route
A_2^2 set (A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$, $\breve{A}_2^2$)	$\gamma^* \to q\bar{q}$, NNLO, two loops	`[A,2,2]`	Two-loop tree interference and one-loop self-interference (two source contributions)	IBP / *LiteRed2*, `A22TwoLoopTree`/ `A22OneLoopSelf` bases; MI: $A_{22,\mathrm{LO}}$, A_3, A_4 (also A_6 for $\tilde{A}_2^2$)
A_4^0, $\tilde{A}_4^0$	$\gamma^* \to qgg\bar{q}$, NNLO, double-real	`[A,4,0]`	Colour-ordered amplitude; colour-coefficient extraction	IBP / *LiteRed2*, `X40` basis; MI: R_4, R_6, $R_{8,a}$, $R_{8,b}$
B_4^0	$\gamma^* \to q\bar{q}q'\bar{q}'$, NNLO	`[B,4,0]`	Sector self-interference (direct amplitude only)	IBP / *LiteRed2*, `X40` basis; MI: R_4, R_6, $R_{8,a}$, $R_{8,b}$
C_4^0	$\gamma^* \to q\bar{q}q\bar{q}$, NNLO	`[C,4,0]`	Sector symmetrised interference (direct and exchange)	IBP / *LiteRed2*, `X40` basis; MI: R_4, R_6, $R_{8,a}$, $R_{8,b}$
Massive A_3^0	$\gamma^* \to q(m_q)g\bar{q}(m_q)$, exploratory	`[A,3,0, quarkMass-> mQ]`	Same route as massless A_3^0, with the `quarkMass` option	IBP / *LiteRed2*, massive family; MI mapped from *LiteRed2*'s J_1, J_2 to the literature $I_1^{(m_q,0,m_q)}$, $I_2^{(m_q,0,m_q)}$

Table 4.2: Construction and integration overview of every antenna route implemented in *AntCalc*.

Route	Literature Reference (Order Checked)	Local Validation	Global Validation	Status
A_2^0	— (reference normalisation)	Not applicable	LO normalisation of the R-ratio	By construction
A_3^0	[4] (NLO)	Eikonal factor and AP splitting function reproduced	Feeds r_{NLO}	Pass
A_2^1	[4] (NLO)	Not applicable	Feeds r_{NLO}	Pass
A_3^1 set	[4, 35] (NNLO)	Not performed	Feeds r_{NNLO}	Pass
A_2^2 set	[35] (NNLO)	Not applicable	Feeds r_{NNLO}	Pass
A_4^0, $\tilde{A}_4^0$	[4] (NNLO)	Not performed	Feeds r_{NNLO}	Pass
B_4^0	[4] (NNLO)	Not performed	Feeds r_{NNLO}	Pass
C_4^0	[4] (NNLO)	Not performed	Feeds r_{NNLO}	Pass
Massive A_3^0	[37]	Not performed	Not applicable (does not feed the massless R-ratio)	Literature agreement; no global observable validation yet

Table 4.3: Validation evidence and status of every antenna route implemented in *AntCalc*. "Not applicable" marks routes with no real unresolved parton to take a soft/collinear limit of (A_2^0, A_2^1, A_2^2); "Not performed" marks routes where such a limit exists but was not checked in this work — see Section 4.4 for why A_3^0 was chosen to demonstrate the method rather than repeating it on every route. The massive A_3^0 route is deliberately marked as less mature than the massless set: it agrees with the literature locally, but has not yet been checked against any global observable.

Chapter 5

Global Validation: The Massless Hadronic R-Ratio

5.1 R-Ratio Definition

The hadronic R-ratio is usually defined as the ratio of total cross-section for e^+e^- annihilation into hadrons to the point-like cross-section for $e^+e^- \to \mu^+\mu^-$. It is mathematically described as follows,

$$\begin{aligned} R_{\text{hadr}} &= \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)} \\ &= \frac{3q^2}{4\pi\alpha_{\text{em}}^2}\,\sigma(e^+e^- \to \text{hadrons}). \end{aligned} \tag{5.1}$$

Looking closely, the $\sigma(e^+e^- \to \text{hadrons})$ term is what QCD predicts for the $\gamma^* \to$ hadrons, computed via the optical theorem [21]. What makes the R-ratio a clean QCD observable, and a good benchmark for *AntCalc* is that the ratio features terms for higher perturbative orders as a perturbative series in α_s,

$$R_{\text{hadr}} = \left(c_{\text{LO}} + \frac{\alpha_s}{\pi}c_{\text{NLO}} + \left(\frac{\alpha_s}{\pi}\right)^2 c_{\text{NNLO}} + O\left(\alpha_s^3\right)\right) \tag{5.2}$$

here the perturbative coefficients, c_{LO}, c_{NLO} and c_{NNLO}, respectively receive contributions from the multiparticle final states at the relative orders via their respective antennae, as described in Chapter 2. For example, A_3^0 as described in Chapter 4 contributes to c_{NLO}. Moreover, the leading-order term $R^{(0)}$ simply counts the number of quark colours and flavours weighted by their electric charges,

$$R^{(0)} = N_c \sum_q e_q^2 \tag{5.3}$$

since $R_{\text{hadr}} = R^{(0)}\left(1 + O(\alpha_s)\right) = R^{(0)}R(\alpha_s)$.

5.2 Global Validation with *AntCalc*

As discussed, the hadronic R-ratio is a very useful quantity to compute using the integrated antenna functions we computed using *AntCalc*. As such, the programme provides an R-ratio calculator. It is the longest-running function in the whole package. Its result gives,

```
In[2]:= BuildRRatio[SMQCD]
Out[2]:=
```

$$1 + \left(\frac{\alpha_s}{2\pi}\right)\frac{3}{2}C_F + \left(\frac{\alpha_s}{2\pi}\right)^2 C_F\left[N\left(\frac{243}{16} - 11\zeta_3\right) + \frac{3}{16N} + N_f\left(-\frac{11}{4} + 2\zeta_3\right)\right]$$

As expected, the results for each perturbative order are finite. This validates the mutual normalisation and assembly of the integrated antennae: at each perturbative order, their infrared poles cancel in the inclusive sum, leaving a finite coefficient.

5.2.1 Assembly from Integrated Antennae

Using *AntCalc*, we are also able to compute the R-ratio step-by-step, without having to resort to the automated function by just computing the intermediate steps ourselves. As described in Subsection 2.4.3, antenna functions are purely kinematic. By taking all components, we are able to construct a process-wide kinematics- and colour-conscious object. These are the $\mathcal{T}$-objects. These objects are defined, before integration and hence notated by T, conceptually as [1] ,

$$T^{2(k+1)}_{\text{topology}} = \text{(colour/flavour weight)} \times T^2_{q\bar{q}} \times \text{(colour-weighted sum of antennae at } k\text{-order)}. \tag{5.4}$$

This structure creates a very elegant structure for the A-type antenna relative T-objects. It goes like,

$$T^{2(k+1)}_{\text{topology}} = 2C_F T^2_{q\bar{q}} \sum_c F_c A^l_{n,(k),c} \tag{5.5}$$

where c denotes the component (leading, subleading, fermion loop, etc), F_c is the colour weight, that is N, $1/N$, N_f, etc, and $A^l_{n,(k),c}$ is the relative component antenna with n final-state particles and l loops. For this antenna type, this definition needs two extensions at NNLO that do not cleanly fit into an algorithmic formula:

- The A^1_3 antenna set represents a particularly interesting process. Unlike in lower order loop processes, here two very distinct kinds of Feynman diagram appear: those in which the loop and the emission of the final-state gluon are completely separate events, and those in which the loop involves all three external legs. The first kind of diagram maps exactly a A^1_2 antenna topology followed by a A^0_3 topology: first the two hard quarks exchange a gluon and then one of the hard quarks emits a final-state gluon[2]. Since this behaviour is not singular to the A^1_3 antenna, it must be subtracted from the bare result to retain the antenna's singular behaviour. When we then construct back the T-object, the $A^1_2 A^0_3$ term has to be added back on such that the process is considered in its totality [4].
- The A^2_2 antenna set, also at NNLO, is the first time we find a double-loop topology. Here the extension required is not a subtraction of some kind of behaviour, but rather an addition of a new one. This set is, in reality, composed of two different processes: the first follows the standard rule closely, as it produces a leading, a subleading and a fermion loop component. The second, however, has a singular antenna which describes the self-interference of the one-loop correction with itself, now that the system presents with two loops. This antenna is notated by $\breve{A}^2_2$. It is also the only genuinely square (not cross) term at NNLO. This means that it requires a normalisation of $(2C_F)^2$ [4].

As mentioned before Eq. 5.5, the equation is only valid for T-objects constructed from A-type antenna functions. In this work we consider all quark-antiquark antennae up to NNLO. This includes the B^0_4 and C^0_4 antennae introduced in Subsection 2.4.4 of Chapter 2. These antennae populate the $T^6_{q\bar{q}q'\bar{q}'}$ and $T^6_{q\bar{q}q\bar{q}}$ objects, respectively.

$T^6_{q\bar{q}q'\bar{q}'}$ is built in the same way as other single-topology antenna T-objects, like $T^4_{q\bar{q}g}$, with the only difference being that here all quark flavours are summed over, requiring the inclusion of a $(N_f - 1)$ term, constructing,

$$T^6_{q\bar{q}q'\bar{q}'} = 2C_F T^2_{q\bar{q}} (N_f - 1) B^0_4. \tag{5.6}$$

The identical-flavour quark-antiquark-quark-antiquark final state is described by the sum of antennae for non-identical-flavour and for identical-flavour-only [4]. Using Eq. 2.39 in the interference term, with the identity piece vanishing by tracelessness, we can describe the identical-flavour T-object as,

$$T^6_{q\bar{q}q\bar{q}} = 2C_F T^2_{q\bar{q}} \left[B^0_4 - \frac{1}{N} C^0_4 \right] \tag{5.7}$$

importantly without the inclusion of the quark-flavour sum term $(N_f - 1)$.

With this definition of the integrated antenna $\mathcal{X}^l_n$ (Eq. 2.47 of Chapter 2), we are now also able to introduce the often used integrated T-object as,

$$\mathcal{T}^{2(k+1)}_{\text{topology}} = 2C_F T_{q\bar{q}} \sum_c F_c \mathcal{X}^l_{n,(k),c} \tag{5.8}$$

written schematically from Eq. 5.5, though the structure may change with the antenna type used to define the relevant T-object. This equation describes that the integrated T-object, $\mathcal{T}^{2(k+1)}_{\text{topology}}$ is defined using the integrated antennae $\mathcal{X}^l_n$, integrated over their respective phase-spaces rather than blindly integrating the unintegrated T-object directly [4, 35].

[1]The T-objects are commonly notated by T^2, T^4 and T^6, depending on their perturbative order: LO, NLO and NNLO, respectively. That is the same as the k-index, introduced in Subsection 2.4.2 of Chapter 2, incremented by one and then doubled, $2(k+1)$. This is the notation used in [35] and the one that will be used for the rest of the present work.

[2]This is merely a conceptual, diagrammatic interpretation, as the actual process is much less sequential than this description suggests.

The R-ratio is assembled using these $\mathcal{T}$-objects. We simply sum the normalised objects order-by-order as,

$$
\begin{aligned}
R = \frac{R_{\text{hadr}}}{R^{(0)}} &= R_{\text{LO}} + R_{\text{NLO}} + R_{\text{NNLO}} \\
&= r_{\text{LO}} + \frac{\alpha_s}{2\pi} r_{\text{NLO}} + \left(\frac{\alpha_s}{2\pi}\right)^2 r_{\text{NNLO}} \\
&= 1 + \frac{\alpha_s}{2\pi} \frac{\mathcal{T}^4_{q\bar{q}} + \mathcal{T}^4_{q\bar{q}g}}{\mathcal{T}^2_{q\bar{q}}} + \left(\frac{\alpha_s}{2\pi}\right)^2 \frac{\mathcal{T}^6_{q\bar{q}} + \mathcal{T}^6_{q\bar{q}g} + \mathcal{T}^6_{q\bar{q}q\bar{q}} + \mathcal{T}^6_{q\bar{q}q'\bar{q}'} + \mathcal{T}^6_{q\bar{q}gg}}{\mathcal{T}^2_{q\bar{q}}} + O\left(\alpha_s^3\right).
\end{aligned}
\tag{5.9}
$$

Expanding the $\mathcal{T}$-objects, we are able to describe the R-ratio formula using the integrated antenna functions as,

$$
\begin{aligned}
r_{\text{LO}} &= 1 \\
r_{\text{NLO}} &= 2C_F \left(\mathcal{A}^1_2 + \mathcal{A}^0_3\right) \\
r_{\text{NNLO}} &= 2C_F \Bigg[N\left(\mathcal{A}^1_2\mathcal{A}^0_3 + \mathcal{A}^1_3 + \mathcal{A}^0_4 + \mathcal{A}^2_2\right) - \frac{1}{N}\left(\mathcal{A}^1_2\mathcal{A}^0_3 + \tilde{\mathcal{A}}^1_3 + \tilde{\mathcal{A}}^0_4 + C^0_4 - \tilde{\mathcal{A}}^2_2\right) \\
&\qquad + N_f\left(\mathcal{B}^0_4 + \hat{\mathcal{A}}^1_3 + \hat{\mathcal{A}}^2_2\right) + 2C_F\check{\mathcal{A}}^2_2\Bigg].
\end{aligned}
\tag{5.10}
$$

Assembling these pieces together, using the integrated antenna results from *AntCalc*, we finally get,

$$
R = 1 + \left(\frac{\alpha_s}{2\pi}\right)\frac{3}{2}C_F + \left(\frac{\alpha_s}{2\pi}\right)^2 C_F\left[N\left(\frac{243}{16} - 11\zeta_3\right) + \frac{3}{16N} + N_f\left(-\frac{11}{4} + 2\zeta_3\right)\right] \tag{5.11}
$$

which is the exact expression for the massless hadronic R-ratio at NNLO [35] and is also exactly the expression we get by running `BuildRRatio[SMQCD]`.

Chapter 6

Future Work

In its current version, *AntCalc* is implemented entirely in *Wolfram Mathematica*. This is the right language to build a trial-stage programme of its nature due to its forgiving mathematical environment and its general adoption throughout the field. But this choice of language comes at a cost: efficiency. Though convenient, *Mathematica* is slow and memory-hungry. In its current state, running the full `BuildRRatio` call takes one and a half to two hours and, on machines with less RAM, may cause the calculation to crash halfway through. It is then apparent that the next step in *AntCalc*'s development is taking it mostly out of the *Mathematica* environment and making it a truly modular, multi-language programme that exploits only the best characteristics of each tool, including *Mathematica*, keeping the current high-level workflow. The current plan for this step is to make the orchestrator a *Python* programme and to store the profiles in plain, machine-readable files, such as those in the `JSON` format. Then, retain *FeynArts* and *FeynCalc* for diagram generation and amplitude manipulation, respectively, but hand the algebraic manipulations required to obtain the antenna, or its components, over to *FORM*. Then, lastly, make IBP reduction faster and more stable by replacing *LiteRed2* with *Kira*, written in C++, keeping *Mathematica* as an algebraic simplifier.

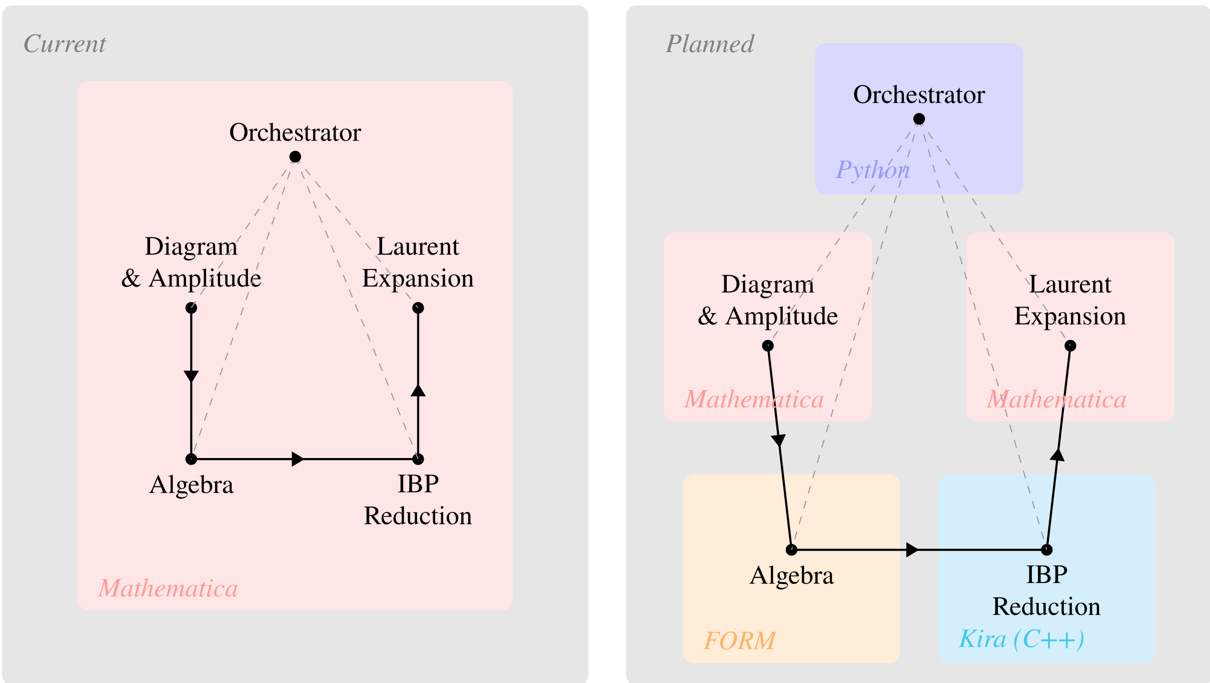

Figure 6.1: Diagram showing the current and planned workflow steps relative to their programming languages.

This step will then allow us to keep expanding *AntCalc*'s profile library without fear of hitting a computational bottleneck. The first point of expansion would be to the rest of the massive quark-antiquark final-final antenna functions, especially A_4^0, B_4^0 and C_4^0 at first, and then A_3^1 and A_2^2. After these are completed, expanding the scope to other relevant antenna types, particularly the quark-gluon (D and E) and the gluon-gluon (F, G and H) antennae, both in the massless and in the massive regimes becomes the logical next step. Lastly, extensions to kinematics other than final-final, namely initial-initial and initial-final configurations, and higher perturbative orders, particularly N^{3}LO, antennae would also be relevant.

A further, more structural, direction is to generalise *AntCalc* from a validated library of known antenna functions into a construction tool for antenna functions that have not yet been computed at all. The profile-driven architecture described in Chapter 3 already separates the process-specific description of a route, its profile, from the shared build and integration workflows; letting a user supply their own process description, rather than only selecting an existing route key, is a natural continuation of that design rather than a new one. Such user-defined routes would not, by definition, have a published result to validate against. This is precisely the gap that the amplitude-level Ward-identity check described in Section 3.7.1 is suited to: since it certifies gauge invariance directly from the generated amplitude, independently of any external reference, it, and checks like it, could be run automatically on a newly constructed antenna as a first, literature-independent certificate of correctness, before any attempt is made to compare it with an established result.

Connections to other established tools would also be pertinent. A connection to a Feynman diagram generation tool such as *MadGraph* [39] and one to a parton-level event generator such as *NNLOJet* [40] could allow us to generate more complex antennae, from more complex topologies and also enable the numerical validation of the results.

Chapter 7

Conclusion

This thesis developed the *Mathematica* package *AntCalc* to remove a friction that recurs at the start of most new antenna-subtraction calculations: before the intended physics can even be pursued, a group must first reconstruct and integrate the antenna functions it needs from scratch, much as Feynman-diagram algebra once had to be carried out by hand before tools such as *FeynCalc* became standard. *AntCalc* removes this initial barrier, providing a validated, ready-to-use library of antenna functions that can be loaded directly into a *Mathematica* notebook, making their construction and everyday use easier, more standardised, and streamlined.

The resulting framework was implemented completely within *Mathematica*, using the *FeynCalc* environment (including *FeynArts* and *FeynHelpers*) to handle the building stage and *LiteRed2* for the IBP reduction in the integration stage. That framework was shown to produce all massless quark-antiquark final-final antennae through NNLO. It was also shown to be able to be extended to the massive regime by implementing the massive A_3^0 route in its entirety. It begins by using *FeynArts* to produce one or more Feynman diagrams relevant for the process under study and generates their amplitudes. It then develops the amplitude into one antenna or multiple antenna components using *FeynCalc*. The unintegrated antenna is then handed to *LiteRed2* for IBP reduction and, finally, after the pre-computed MI are identified and substituted, the integrated antenna is Laurent-expanded around the dimensional regularisation parameter ϵ.

These unintegrated and integrated antennae were shown to be in agreement with established results in the literature. Furthermore, the massless A_3^0 antenna, whose workflow was shown in great detail, was locally validated by computing the eikonal factor and the Altarelli–Parisi splitting function, using a method that may be used to validate other appropriate antennae. All massless integrated antenna results were also validated when we computed the massless hadronic R-ratio.

The package is beginning to reach the computational limits of its current workflow, running completely within Mathematica. As such, the most pressing piece of future work is the overhaul to a new, truly modular and multi-language, workflow, that is able to harness the best characteristics of different tools, like Kira for IBP reduction and FORM for algebraic manipulation, without compromising efficiency. Once this overhaul is done, *AntCalc* could be expanded to handle much more complex cases, particularly those in the massive regime, other antenna types, such as quark-gluon, other kinematic configurations, such as initial-final antennae, and those in higher perturbative orders, such as N^3LO.

Appendix A

Complete Antenna Expressions

This appendix includes the complete expressions of the antennae computed using *AntCalc*. It shows the unintegrated and integrated antennae, along with the commands used to compute them.

A note on conventions: the unintegrated antenna functions below are given exactly as *AntCalc* returns them, in terms of s_{ij} and q^2. The integrated antennae, however, are quoted at $q^2 = 1$, matching `IntegrateAntenna`'s default convention. This does not break generality as established in Subsection 2.5.3 and Eq. 2.104, a quantity of this kind scales with q^2 as a single power fixed by its mass dimension and ϵ-scaling, so the complete q^2-dependent result is recovered by multiplying the expression given here by $(q^2)^{\lambda+\kappa\epsilon}$ for the appropriate λ and κ. The explicit case of $\mathcal{A}_3^0$, which scales as $(q^2)^{-\epsilon}$, is worked out in full in Eq. 4.83. We must also here introduce the shorthand,

$$C_\epsilon \equiv \frac{(4\pi)^\epsilon e^{-\gamma_E \epsilon}}{8\pi^2} = \frac{1}{C(\epsilon, 1)} \tag{A.1}$$

as the one-loop normalisation factor.

A.1 LO Antenna

A.1.1 A_2^0

Running `BuildAntenna[A,2,0]`, we get the built result,

$$A_2^0 = 1. \tag{A.2}$$

By running `BuildAndIntegrateAntenna[A,2,0]` we get the integrated result,

$$\mathcal{A}_2^0 = \frac{1}{4} + \epsilon\frac{9}{8} - \epsilon^2\left(\frac{9}{16} - \frac{\pi^2}{48}\right). \tag{A.3}$$

A.2 NLO Antennae

A.2.1 A_2^1

Running `BuildAntenna[A,2,0]`, we get the built result,

$$A_2^1 = -\frac{1}{2}(3+2\epsilon)B_0\left(q^2;0,0\right) - q^2 C_0\left(0,0,q^2;0,0,0\right). \tag{A.4}$$

By running `BuildAndIntegrateAntenna[A,2,1]` we get the integrated result,

$$\mathcal{A}_2^1 = -\frac{1}{\epsilon^2} - \frac{3}{2\epsilon} - 4 + \frac{7\pi^2}{12} + \epsilon\left(\frac{7\zeta_3}{3} - 8 + \frac{7\pi^2}{8}\right) + \epsilon^2\left(\frac{7\zeta_3}{2} - 16 + \frac{7\pi^2}{3} - \frac{73\pi^4}{1440}\right) + O\left(\epsilon^3\right). \tag{A.5}$$

A.2.2 A_3^0

Running `BuildAntenna[A,3,0]`, we get the built result,

$$A_3^0 = \frac{2s_{12}^2 - 2s_{12}(s_{13}+s_{23})}{q^2 s_{13}s_{23}} + \epsilon\frac{2}{q^2} + (\epsilon-1)\frac{s_{13}^2+s_{23}^2}{q^2 s_{13}s_{23}}. \tag{A.6}$$

By running `BuildAndIntegrateAntenna[A,3,0]` we get the integrated result,

$$\begin{aligned}\mathcal{A}_3^0 = &\frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12} + \epsilon\left(-\frac{25\zeta_3}{3} + \frac{109}{8} - \frac{7\pi^2}{8}\right)\\ &+\epsilon^2\left(-\frac{25\zeta_3}{2} + \frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440}\right) + O\left(\epsilon^3\right).\end{aligned} \tag{A.7}$$

A.3 NNLO Antennae

A.3.1 A_2^2

Running `BuildAntenna[A,2,2,Component->Leading]`, we get the built result,

$$A_2^2 = A_2^{2,(0)} + (-s_{12})^{-\epsilon}C_\epsilon A_2^{2,(-\epsilon)} + s_{12}^{-2\epsilon}A_2^{2,(-2\epsilon)}. \tag{A.8}$$

where,

$$A_2^{2,(0)} = \frac{1}{16\epsilon^4} - \frac{1}{\epsilon^2}\left(\frac{9}{32} - \frac{11\pi^2}{96}\right) - \frac{1}{\epsilon}\left(\frac{377}{64} - \frac{55\pi^2}{96}\right) + \frac{863}{384} - \frac{319\pi^2}{576} - \frac{13\zeta_3}{36} \tag{A.9}$$

$$A_2^{2,(-\epsilon)} = -\frac{11}{6\epsilon^3} + \frac{11}{4\epsilon^2} + \frac{22}{3\epsilon} + \frac{44}{3} + O(\epsilon) \tag{A.10}$$

$$\begin{aligned}A_2^{2,(-2\epsilon)} = &\frac{3}{16\epsilon^4} + \frac{7}{24\epsilon^3} + \frac{1}{\epsilon^2}\left(\frac{155}{288} - \frac{37\pi^2}{96}\right) + \frac{1}{\epsilon}\left(\frac{5597}{1728} - \frac{355\pi^2}{288} + \frac{7\zeta_3}{12}\right)\\ &-\frac{193531}{10368} - \frac{577\pi^2}{1728} + \frac{263\pi^4}{1440} + \frac{73\zeta_3}{12}.\end{aligned} \tag{A.11}$$

By running `BuildAndIntegrateAntenna[A,2,2,Component->Leading]` we get the integrated result,

$$\begin{aligned}\mathcal{A}_2^2 = &\frac{1}{4\epsilon^2} + \frac{17}{8\epsilon^3} + \frac{1}{\epsilon^2}\left(\frac{433}{144} - \frac{\pi^2}{2}\right) + \frac{1}{\epsilon}\left(\frac{4045}{864} - \frac{83\pi^2}{48} + \frac{7\zeta_3}{12}\right)\\ &-\frac{9083}{5184} - \frac{2153\pi^2}{864} + \frac{263\pi^4}{1440} + \frac{13\zeta_3}{9} + O(\epsilon).\end{aligned} \tag{A.12}$$

A.3.2 $\tilde{A}_2^2$

Running `BuildAntenna[A,2,2,Component->Subleading]`, we get the built result,

$$\tilde{A}_2^2 = \tilde{A}_2^{2,(0)} + s_{12}^{-2\epsilon}\tilde{A}_2^{2,(-2\epsilon)}. \tag{A.13}$$

where,

$$\begin{aligned}\tilde{A}_2^{2,(0)} = &-\frac{3}{16\epsilon^4} - \frac{9}{16\epsilon^3} - \frac{1}{\epsilon^2}\left(-\frac{89}{32} + \frac{11\pi^2}{32}\right) + \frac{1}{\epsilon}\left(\frac{533}{64} - \frac{11\pi^2}{6} - \frac{13\zeta_3}{12}\right)\\ &-\frac{1803}{128} - \frac{341\pi^2}{192} + \frac{79\pi^4}{2880} - \frac{39\zeta_3}{8}.\end{aligned} \tag{A.14}$$

$$\begin{aligned}\tilde{A}_2^{2,(-2\epsilon)} = &-\frac{1}{16\epsilon^4} - \frac{3}{16\epsilon^3} + \frac{1}{\epsilon^2}\left(\frac{7}{32} + \frac{19\pi^2}{96}\right) + \frac{1}{\epsilon}\left(-\frac{975}{64} + \frac{10\pi^2}{3} + \frac{15\zeta_3}{4}\right)\\ &-\frac{499}{128} + \frac{1291\pi^2}{192} - \frac{233\pi^4}{960} + \frac{97\zeta_3}{8}.\end{aligned} \tag{A.15}$$

By running `BuildAndIntegrateAntenna[A,2,2,Component->Subleading]` we get the integrated result,

$$\tilde{\mathcal{A}}_2^2 = -\frac{1}{4\epsilon^4} - \frac{3}{4\epsilon^3} - \frac{1}{\epsilon^2}\left(\frac{41}{16} - \frac{13\pi^2}{24}\right) + \frac{1}{\epsilon}\left(-\frac{221}{32} + \frac{3\pi^2}{2} + \frac{8\zeta_3}{3}\right) - \frac{1151}{64} + \frac{475\pi^2}{96} - \frac{59\pi^4}{288} + \frac{29\zeta_3}{4} + O(\epsilon). \tag{A.16}$$

A.3.3 $\hat{A}_2^2$

Running `BuildAntenna[A,2,2,Component->Nf]`, we get the built result,

$$\hat{A}_2^2 = (-s_{12})^{-\epsilon} C_\epsilon \hat{A}_2^{2,(-\epsilon)} + s_{12}^{-2\epsilon} \hat{A}_2^{2,(-2\epsilon)}. \tag{A.17}$$

where,

$$\hat{A}_2^{2,(-\epsilon)} = \frac{1}{3\epsilon^3} - \frac{1}{2\epsilon^2} - \frac{4}{3\epsilon} - \frac{8}{3} \tag{A.18}$$

$$\hat{A}_2^{2,(-2\epsilon)} = \frac{1}{12\epsilon^3} + \frac{7}{18\epsilon^2} + \frac{1}{\epsilon}\left(\frac{353}{216} - \frac{11\pi^2}{72}\right) + \frac{7541}{1296} - \frac{77\pi^2}{108} - \frac{13\zeta_3}{18}. \tag{A.19}$$

By running `BuildAndIntegrateAntenna[A,2,2,Component->Nf]` we get the integrated result,

$$\hat{\mathcal{A}}_2^2 = -\frac{1}{4\epsilon^3} - \frac{1}{9\epsilon^2} + \frac{1}{\epsilon}\left(\frac{65}{216} + \frac{\pi^2}{24}\right) + \frac{4085}{1296} - \frac{91\pi^2}{216} + \frac{\zeta_3}{18} + O(\epsilon). \tag{A.20}$$

A.3.4 $\breve{A}_2^2$

Running `BuildAntenna[A,2,2,Component->Breve]`, we get the built result,

$$\breve{A}_2^2 = \frac{(-2+\epsilon-2\epsilon^2)\Gamma^6(1-\epsilon)\Gamma^2(1+\epsilon)\left(120-20\pi^2\epsilon^2+80(13-42\epsilon)\zeta_3\epsilon^3+\pi^4\epsilon^4\right)}{1920\epsilon^4\Gamma^2(2-2\epsilon)}. \tag{A.21}$$

By running `BuildAndIntegrateAntenna[A,2,2,Component->Breve]` we get the integrated result,

$$\breve{\mathcal{A}}_2^2 = \frac{1}{4\epsilon^4} + \frac{3}{4\epsilon^3} + \frac{1}{\epsilon^2}\left(\frac{41}{16} - \frac{\pi^2}{24}\right) + \frac{1}{\epsilon}\left(7 - \frac{\pi^2}{8} + \frac{7\zeta_3}{6}\right) + 18 - \frac{41\pi^2}{96} - \frac{7\pi^4}{480} - \frac{7\zeta_3}{2} + O(\epsilon). \tag{A.22}$$

A.3.5 A_3^1

Running `BuildAntenna[A,3,1,Component->Leading]`, we get the built result,

$$A_3^1 = \frac{1}{2(\epsilon-1)q^2}\Bigg((-s_{13})^{-\epsilon}C_\epsilon A_3^{1,(s_{13})} + (-s_{23})^{-\epsilon}C_\epsilon A_3^{1,(s_{23})} + \left(-q^2\right)^{-\epsilon} C_\epsilon A_3^{1,(q^2)} + \left[\frac{\pi^2 q^2}{4} + \frac{1}{4}\log^2\left(\frac{s_{13}}{s_{23}}\right) + \mathrm{Li}_2\left(1-\frac{q^2}{s_{13}}\right) + \mathrm{Li}_2\left(1-\frac{q^2}{s_{23}}\right)\right] A_3^{1,(\text{Trans.})}\Bigg). \tag{A.23}$$

where,

$$A_3^{1,(s_{13})} = \mathcal{F}_1(s_{12}; s_{13}, s_{23}), \quad A_3^{1,(s_{23})} = \mathcal{F}_1(s_{12}; s_{23}, s_{13}) \tag{A.24}$$

$$\mathcal{F}_1(s_{12};x,y) = \frac{1}{4\left(q^2\right)^2(s_{12}+y)^2}\Bigg[\frac{1}{\epsilon}\left(10s_{12}^2 + 4s_{12}(x+4y) + 3y(x+2y)\right) - \frac{1}{x}\left(s_{12}^3 - s_{12}^2(23x-3y) - s_{12}\left(13x^2+36xy-3y^2\right) - 10x^2y - 13xy^2 + y^3\right)\Bigg] \tag{A.25}$$

$$A_3^{1,(q^2)} = \frac{\mathcal{P}\left(20s_{12}^2\left(q^2\right)^2 + 28s_{12}q^2s_{13}s_{23} + 11s_{13}^2s_{23}^2\right)}{12\epsilon\left(q^2\right)^2 s_{13}s_{23}\left(s_{12}q^2 + 13s_{23}\right)^2} + O(\epsilon), \quad \mathcal{P} = \left(q^2\right)^2 + s_{12}^2 - 2s_{13}s_{23} \tag{A.26}$$

$$A_3^{1,(\text{Trans.})} = -\frac{1}{2q^2}A_3^0. \tag{A.27}$$

By running `BuildAndIntegrateAntenna[A,3,1,Component->Leading]` we get the integrated result,

$$\begin{aligned}\mathcal{A}_3^1 = -\frac{1}{4\epsilon^4} - \frac{31}{12\epsilon^3} - \frac{1}{\epsilon^2}\left(\frac{53}{8} - \frac{11\pi^2}{24}\right) - \frac{1}{\epsilon}\left(\frac{647}{24} - \frac{22\pi^2}{9} - \frac{23\zeta_3}{3}\right)\\ -\frac{5231}{48} + \frac{17\pi^2}{2} - \frac{41\pi^4}{480} + \frac{689\zeta_3}{18} + O(\epsilon).\end{aligned} \tag{A.28}$$

A.3.6 $\tilde{A}_3^1$

Running `BuildAntenna[A,3,1,Component->Subleading]`, we get the built result,

$$\begin{aligned}\tilde{A}_3^1 = \frac{\mathcal{R}(\epsilon, s_{13}, s_{23})}{2(\epsilon-1)q^2}\Bigg[&(-s_{12}^{-\epsilon})\tilde{A}_3^{1,(s_{12})} + (-s_{13})^{-\epsilon}\tilde{A}_3^{1,(s_{13})} + (-s_{23})^{-\epsilon}\tilde{A}_3^{1,(s_{23})} +\\ &+ \frac{(-q^2)^{-\epsilon}}{(s_{12}+s_{13})^3(s_{12}+s_{23})^3}\tilde{A}_3^{1,(q^2)} + \epsilon^2(2\epsilon-1)\left(\frac{\pi^2}{4} + \mathrm{Li}_2\left(1 - \frac{q^2}{s_{12}}\right)\tilde{A}_3^{1,(\text{Trans. } s_{12})}\right.\\ &\left.+\left(\mathrm{Li}_2\left(1-\frac{q^2}{s_{13}}\right) + \frac{1}{4}\log^2\left(\frac{s_{12}}{s_{13}}\right)\right)\tilde{A}_3^{1,(\text{Trans. } s_{13})} + \left(\mathrm{Li}_2\left(1-\frac{q^2}{s_{23}}\right) + \frac{1}{4}\log^2\left(\frac{s_{12}}{s_{23}}\right)\right)\tilde{A}_3^{1,(\text{Trans. } s_{23})}\right)\Bigg]\end{aligned} \tag{A.29}$$

$$\mathcal{R}(\epsilon, s_{13}, s_{23}) = \frac{C_\epsilon}{\epsilon^2(2\epsilon-1)s_{13}s_{23}}$$

where, using $u = s_{13} + s_{23}$ and $v = s_{13}s_{23}$,

$$\begin{aligned}\tilde{A}_3^{1,(s_{12})} =& -4s_{12}^2u^2 + 4\epsilon\left(3s_{12}^2u^2 - s_{12}^2v + s_{12}u^3 - 2s_{12}uv + u^4 - u^2v\right)\\ &- 2\epsilon^2u\left(4s_{12}^2u + 4s_{12}u^2 - 4s_{12}v + 7u^3 - 6uv\right) + 2\epsilon^3u^2\left(5u^2 - 4v\right) + 4\epsilon^4u^4\end{aligned} \tag{A.30}$$

$$\tilde{A}_3^{1,(s_{13})} = \mathcal{G}(s_{12}, s_{13}, s_{23}), \quad \tilde{A}_3^{1,(s_{23})} = \mathcal{G}(s_{12}, s_{23}, s_{13}) \tag{A.31}$$

$$\begin{aligned}\mathcal{G}(s_{12}, x, y) =& -2(s_{12}+y)^2\left[2(s_{12}+y)^2 - 2y(s_{12}+y) + y^2\right] + \epsilon\Big(12(s_{12}+y)^4\\ &+ 4(s_{12}+y)^3(x-3y) + 4(s_{12}+y)^2(x^2 - xy + 2y^2) - 2xy(s_{12}+y)(x-2y) + x^2y^2\Big)\\ &- \epsilon^2\Big(8(s_{12}+y)^4 + (s_{12}+y)^3(8x-7y) + (s_{12}+y)^2\left(14x^2 + xy + 10y^2\right) + xy(s_{12}+y)(x+6y)\\ &+ x^2y^2\Big) + \epsilon^3(s_{12}+y)\Big(y(s_{12}+y)^2 + (s_{12}+y)^2 + (s_{12}+y)\left(10x^2 + 14xy + 5y^2\right)\\ &+ 5x^2y\Big) + \epsilon^4(s_{12}+y)(x+y)\Big((s_{12}+y)(4x-y) + 2xy\Big)\end{aligned} \tag{A.32}$$

$$\begin{aligned}\tilde{A}_3^{1,(q^2)} =& 2s_{12}^5\left(3u^2 - 2v\right) + 12s_{12}^4u\left(u^2 - v\right) + s_{12}^3\left(9u^4 - 10u^3v - 8v^2\right) + s_{12}^2u\left(3u^4 - 16v^2\right)\\ &+ 4s_{12}v\left(u^4 - 2u^2v - v^2\right) + 2uv^2\left(u^2 - 2v^2\right) - \epsilon u\Big(2s_{12}^5u + 4s_{12}^4(u^2 - v) + 2s_{12}^3u\left(3u^2 - 5v\right)\\ &+ 4s_{12}^2\left(u^4 - v^2\right) + s_{12}uv\left(9u^2 - 8v\right) + 2v^2\left(3u^2 - 2v\right)\Big)\end{aligned} \tag{A.33}$$

$$\tilde{A}_3^{1,(\text{Trans. } s_{12})} = 4s_{12}^2 + 2s_{12}(x+y) + x^2 + y^2 - 4\epsilon\left(s_{12}^2 + s_{12}(x+y) + x^2 + xy + y^2\right)$$

$$+\epsilon^2\left(4(x+y)^2-2xy\right)+\epsilon^3(x+y)^2 \tag{A.34}$$

$$\tilde{A}_3^{1,(\text{Trans. } s_{13})}=\mathcal{H}(s_{12},s_{13},s_{23}),\quad \tilde{A}_3^{1,(\text{Trans. } s_{23})}=\mathcal{H}(s_{12},s_{23},s_{13}) \tag{A.35}$$

$$\begin{aligned}\mathcal{H}(s_{12},x,y) = {} & s_{12}^2+(s_{12}+y)^2-2\epsilon\left(s_{12}^2+s_{12}(x+y)+x^2+xy+y^2\right)\\ & +\epsilon^2\left(3x(x+y)+y^2\right)+\epsilon^3 x(x+y).\end{aligned} \tag{A.36}$$

By running `BuildAndIntegrateAntenna[A,3,1,Component->Subleading]` we get the integrated result,

$$\tilde{\mathcal{A}}_3^1=-\frac{1}{\epsilon^2}\left(\frac{5}{8}-\frac{\pi^2}{6}\right)-\frac{1}{\epsilon}\left(\frac{19}{4}-\frac{\pi^2}{4}-7\zeta_3\right)-\frac{105}{4}+\frac{27\pi^2}{16}+\frac{7\pi^4}{90}+\frac{27\zeta_3}{2}+O(\epsilon). \tag{A.37}$$

A.3.7 $\hat{A}_3^1$

Running `BuildAntenna[A,3,1,Component->Nf]`, we get the built result,

$$\begin{aligned}\hat{A}_3^1 = {} & \frac{4s_{12}^2}{3s_{13}s_{23}}+\frac{4s_{12}}{3s_{13}}+\frac{4s_{12}}{3s_{23}}+\frac{4s_{23}}{3s_{13}}+\frac{4s_{13}}{3s_{23}}+\frac{4}{3}\\ & -\frac{1}{\epsilon}\left(\frac{4s_{12}^2}{3s_{13}s_{23}}+\frac{4s_{12}}{3s_{23}}+\frac{2s_{23}}{3s_{13}}+\frac{2s_{13}}{3s_{23}}\right)\\ & -\epsilon\left(\frac{2s_{13}}{3s_{23}}+\frac{2s_{23}}{3s_{13}}+\frac{4}{3}\right).\end{aligned} \tag{A.38}$$

By running `BuildAndIntegrateAntenna[A,3,1,Component->Nf]` we get the integrated result,

$$\hat{\mathcal{A}}_3^1=\frac{1}{3\epsilon^3}+\frac{1}{2\epsilon^2}+\frac{1}{\epsilon}\left(\frac{19}{12}-\frac{7\pi^2}{36}\right)+\frac{109}{24}-\frac{7\pi^2}{24}-\frac{25\zeta_3}{9}+O(\epsilon). \tag{A.39}$$

A.3.8 A_4^0

Running `BuildAntenna[A,4,0,Component->Leading]`, we get the built result,

$$A_4^0=\frac{1}{2(\epsilon-1)q^2}\left(A_4^{0,(0)}+\epsilon A_4^{0,(1)}+\epsilon^2 A_4^{0,(2)}+\epsilon^3 A_4^{0,(3)}\right) \tag{A.40}$$

where,

$$\begin{aligned}A_4^{0,(0)} = {} & \frac{2s_{12}s_{14}}{s_{13}s_{134}^2}+\frac{s_{14}s_{23}}{s_{13}s_{134}^2}-\frac{4s_{23}}{s_{13}s_{134}}-\frac{4s_{14}s_{24}}{s_{13}s_{134}^2}-\frac{s_{12}s_{24}}{s_{13}s_{134}s_{234}}-\frac{s_{24}}{s_{134}s_{234}}-\frac{5s_{24}}{s_{134}^2}\\ & -\frac{2s_{24}s_{34}}{s_{13}s_{134}^2}-\frac{s_{12}s_{34}}{s_{13}s_{134}s_{234}}-\frac{2s_{34}}{s_{13}s_{24}}-\frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}}-\frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}}-\frac{s_{12}s_{34}}{s_{134}s_{234}s_{24}}-\frac{4s_{12}^2s_{34}}{s_{13}s_{134}s_{234}s_{24}}\\ & -\frac{s_{12}s_{34}}{s_{234}^2s_{24}}-\frac{2s_{13}s_{34}}{s_{234}^2s_{24}}-\frac{s_{12}s_{34}}{s_{13}s_{134}^2}-\frac{2s_{12}}{s_{13}s_{134}}+\frac{2}{s_{134}}-\frac{s_{12}s_{14}}{s_{13}s_{134}s_{234}}+\frac{s_{14}s_{23}}{s_{13}s_{134}s_{234}}\\ & -\frac{s_{12}s_{23}}{s_{13}s_{134}s_{234}}+\frac{2s_{12}}{s_{134}s_{234}}-\frac{s_{13}}{s_{134}s_{234}}-\frac{2s_{12}^2}{s_{13}s_{134}s_{234}}+\frac{2}{s_{234}}-\frac{4s_{14}}{s_{13}s_{24}}-\frac{5s_{12}s_{14}}{s_{13}s_{134}s_{24}}\\ & +\frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}}-\frac{4s_{23}}{s_{13}s_{24}}-\frac{3s_{12}s_{23}}{s_{13}s_{134}s_{24}}-\frac{2s_{12}}{s_{13}s_{24}}-\frac{4s_{12}^2}{s_{13}s_{134}s_{24}}-\frac{2s_{12}}{s_{234}s_{24}}-\frac{3s_{12}s_{14}}{s_{13}s_{234}s_{24}}\\ & -\frac{s_{12}s_{14}}{s_{134}s_{234}s_{24}}-\frac{2s_{12}^2s_{14}}{s_{13}s_{134}s_{234}s_{24}}-\frac{4s_{14}}{s_{234}s_{24}}+\frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}}+\frac{6s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}}+\frac{s_{14}s_{23}}{s_{134}s_{234}s_{24}}\\ & -\frac{5s_{12}s_{23}}{s_{13}s_{234}s_{24}}-\frac{s_{12}s_{23}}{s_{134}s_{234}s_{24}}-\frac{2s_{12}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}}-\frac{4s_{12}^2}{s_{13}s_{234}s_{24}}-\frac{2s_{12}^2}{s_{134}s_{234}s_{24}}-\frac{s_{12}s_{13}}{s_{134}s_{234}s_{24}}\\ & -\frac{4s_{12}^3}{s_{13}s_{134}s_{234}s_{24}}-\frac{2s_{12}s_{23}}{s_{234}^2s_{24}}-\frac{4s_{13}s_{23}}{s_{234}^2s_{24}}+\frac{s_{14}s_{23}}{s_{234}^2s_{24}}-\frac{3s_{12}s_{14}^2}{s_{13}s_{134}^2s_{34}}-\frac{s_{24}^2}{s_{134}s_{234}s_{34}}\\ & -\frac{4s_{12}s_{14}}{s_{13}s_{134}s_{34}}-\frac{s_{14}}{s_{134}s_{34}}-\frac{6s_{12}s_{14}}{s_{134}^2s_{34}}+\frac{2s_{14}^2s_{23}}{s_{13}s_{134}^2s_{34}}-\frac{6s_{14}s_{23}}{s_{13}s_{134}s_{34}}-\frac{3s_{23}}{s_{134}s_{34}}\end{aligned}$$

$$
\begin{aligned}
&-\frac{4s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}}-\frac{6s_{14}s_{24}}{s_{134}^2s_{34}}+\frac{2s_{24}}{s_{134}s_{34}}-\frac{s_{12}s_{14}s_{24}}{s_{13}s_{134}s_{234}s_{34}}-\frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}}+\frac{s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}}\\
&-\frac{s_{23}s_{24}}{s_{134}s_{234}s_{34}}+\frac{s_{12}s_{24}}{s_{134}s_{234}s_{34}}-\frac{4s_{13}s_{24}}{s_{134}s_{234}s_{34}}-\frac{4s_{13}s_{24}}{s_{134}^2s_{34}}-\frac{3s_{12}s_{24}}{s_{234}^2s_{34}}-\frac{4s_{13}s_{24}}{s_{234}^2s_{34}}\\
&+\frac{4s_{12}}{s_{134}s_{34}}+\frac{s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{34}}+\frac{4s_{12}}{s_{234}s_{34}}+\frac{2s_{13}}{s_{234}s_{34}}-\frac{s_{12}s_{14}}{s_{13}s_{234}s_{34}}-\frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}}-\frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}}\\
&-\frac{3s_{14}}{s_{234}s_{34}}+\frac{s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{34}}+\frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{34}}+\frac{6s_{14}s_{23}}{s_{134}s_{234}s_{34}}-\frac{s_{12}s_{23}}{s_{13}s_{234}s_{34}}-\frac{s_{12}s_{23}}{s_{134}s_{234}s_{34}}\\
&-\frac{s_{13}s_{23}}{s_{134}s_{234}s_{34}}-\frac{s_{23}}{s_{234}s_{34}}-\frac{2s_{12}^2}{s_{13}s_{234}s_{34}}+\frac{8s_{12}^2}{s_{134}s_{234}s_{34}}-\frac{s_{13}^2}{s_{134}s_{234}s_{34}}+\frac{s_{12}s_{13}}{s_{134}s_{234}s_{34}}\\
&-\frac{4s_{14}^2}{s_{13}s_{24}s_{34}}-\frac{3s_{12}s_{14}^2}{s_{13}s_{134}s_{24}s_{34}}+\frac{3s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}}-\frac{4s_{23}^2}{s_{13}s_{24}s_{34}}-\frac{4s_{12}s_{14}}{s_{13}s_{24}s_{34}}-\frac{s_{12}s_{14}}{s_{134}s_{24}s_{34}}\\
&-\frac{6s_{12}^2s_{14}}{s_{13}s_{134}s_{24}s_{34}}+\frac{5s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}}-\frac{6s_{14}s_{23}}{s_{13}s_{24}s_{34}}+\frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{24}s_{34}}-\frac{4s_{12}s_{23}}{s_{13}s_{24}s_{34}}-\frac{s_{12}s_{23}}{s_{134}s_{24}s_{34}}\\
&-\frac{2s_{12}^2}{s_{134}s_{24}s_{34}}-\frac{2s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}}+\frac{5s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}}+\frac{4s_{12}s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}}+\frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}}\\
&-\frac{3s_{12}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}}+\frac{3s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}}+\frac{4s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}}+\frac{s_{14}^2s_{23}}{s_{134}s_{234}s_{24}s_{34}}-\frac{4s_{12}s_{23}}{s_{234}s_{24}s_{34}}\\
&+\frac{s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{24}s_{34}}+\frac{s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}}+\frac{s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}}+\frac{8s_{12}^2s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}}-\frac{6s_{14}s_{23}}{s_{234}s_{24}s_{34}}\\
&-\frac{6s_{12}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}}-\frac{s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{24}s_{34}}-\frac{3s_{12}s_{23}^2}{s_{234}^2s_{24}s_{34}}-\frac{4s_{13}s_{23}^2}{s_{234}^2s_{24}s_{34}}+\frac{2s_{14}s_{23}^2}{s_{234}^2s_{24}s_{34}}-\frac{3s_{12}s_{13}}{s_{134}^2s_{34}}\\
&-\frac{6s_{12}s_{23}}{s_{234}^2s_{34}}-\frac{6s_{13}s_{23}}{s_{234}^2s_{34}}+\frac{2}{s_{34}}-\frac{3s_{12}}{s_{134}^2}-\frac{3s_{12}}{s_{234}^2}-\frac{5s_{13}}{s_{234}^2}-\frac{3s_{14}^2}{s_{134}s_{34}^2}\\
&-\frac{5s_{12}s_{14}^2}{s_{134}^2s_{34}^2}-\frac{s_{14}s_{24}^2}{s_{134}s_{234}s_{34}^2}-\frac{3s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2}-\frac{s_{12}s_{24}^2}{s_{234}^2s_{34}^2}-\frac{s_{13}s_{24}^2}{s_{234}^2s_{34}^2}+\frac{6s_{12}s_{14}}{s_{134}s_{34}^2}\\
&-\frac{s_{13}s_{14}}{s_{134}s_{34}^2}-\frac{2s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2}+\frac{4s_{14}}{s_{34}^2}+\frac{3s_{14}^3s_{23}}{s_{13}s_{134}^2s_{34}^2}-\frac{4s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2}-\frac{3s_{14}s_{23}}{s_{134}s_{34}^2}+\frac{s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2}\\
&-\frac{s_{13}s_{23}}{s_{134}s_{34}^2}+\frac{4s_{23}}{s_{34}^2}-\frac{3s_{14}^2s_{24}}{s_{134}^2s_{34}^2}+\frac{4s_{14}s_{24}}{s_{134}s_{34}^2}+\frac{2s_{12}s_{24}}{s_{234}s_{34}^2}-\frac{2s_{12}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2}\\
&-\frac{s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{s_{14}s_{24}}{s_{234}s_{34}^2}+\frac{s_{14}^2s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}^2}+\frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{s_{23}s_{24}}{s_{234}s_{34}^2}\\
&-\frac{3s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{6s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{s_{13}^2s_{24}}{s_{134}^2s_{34}^2}-\frac{2s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2}+\frac{s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2}+\frac{2s_{12}s_{13}}{s_{134}s_{34}^2}\\
&+\frac{s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{34}^2}+\frac{s_{14}s_{23}^2}{s_{13}s_{234}s_{34}^2}+\frac{3s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2}-\frac{3s_{23}^2}{s_{234}s_{34}^2}+\frac{s_{14}^2s_{23}}{s_{13}s_{234}s_{34}^2}+\frac{3s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2}\\
&+\frac{6s_{12}s_{23}}{s_{234}s_{34}^2}+\frac{4s_{13}s_{23}}{s_{234}s_{34}^2}+\frac{2s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{34}^2}-\frac{2s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}+\frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{3s_{14}s_{23}}{s_{234}s_{34}^2}\\
&-\frac{s_{13}^2s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{2s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{34}^2}+\frac{3s_{14}^2s_{23}^2}{s_{13}s_{134}s_{24}s_{34}^2}-\frac{4s_{14}s_{23}^2}{s_{13}s_{24}s_{34}^2}+\frac{s_{14}s_{23}^2}{s_{134}s_{24}s_{34}^2}+\frac{3s_{14}^3s_{23}}{s_{13}s_{134}s_{24}s_{34}^2}\\
&-\frac{4s_{14}^2s_{23}}{s_{13}s_{24}s_{34}^2}+\frac{6s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}^2}+\frac{s_{14}^2s_{23}}{s_{134}s_{24}s_{34}^2}-\frac{8s_{12}s_{14}s_{23}}{s_{13}s_{24}s_{34}^2}+\frac{2s_{12}s_{14}s_{23}}{s_{134}s_{24}s_{34}^2}\\
&-\frac{2s_{14}^2s_{23}^3}{s_{13}s_{134}s_{234}s_{24}s_{34}^2}+\frac{3s_{14}s_{23}^3}{s_{13}s_{234}s_{24}s_{34}^2}-\frac{2s_{14}^3s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2}+\frac{3s_{14}^2s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2}-\frac{4s_{12}s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2}\\
&+\frac{s_{14}^2s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2}+\frac{6s_{12}s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2}+\frac{s_{13}s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2}-\frac{4s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2}+\frac{3s_{14}s_{23}^3}{s_{234}^2s_{24}s_{34}^2}\\
&-\frac{s_{12}s_{13}^2}{s_{134}^2s_{34}^2}-\frac{5s_{12}s_{23}^2}{s_{234}^2s_{34}^2}-\frac{3s_{13}s_{23}^2}{s_{234}^2s_{34}^2}
\end{aligned}
\tag{A.41}
$$

$$
\begin{aligned}
A_4^{0,(1)} = {} & \frac{4s_{12}^3}{s_{13}s_{134}s_{234}s_{24}} + \frac{2s_{12}^2}{s_{13}s_{134}s_{234}} + \frac{4s_{12}^2}{s_{13}s_{134}s_{24}} + \frac{2s_{12}^2s_{14}}{s_{13}s_{134}s_{234}s_{24}} + \frac{2s_{12}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}} + \frac{4s_{12}^2}{s_{13}s_{234}s_{24}} + \frac{2s_{12}^2}{s_{134}s_{234}s_{24}} \\
& + \frac{2s_{12}^2}{s_{13}s_{234}s_{34}} - \frac{8s_{12}^2}{s_{134}s_{234}s_{34}} + \frac{6s_{12}^2s_{14}}{s_{13}s_{134}s_{24}s_{34}} + \frac{2s_{12}^2}{s_{134}s_{24}s_{34}} - \frac{8s_{12}^2s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{6s_{12}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} - \frac{s_{12}s_{14}}{s_{13}s_{134}^2} \\
& + \frac{2s_{12}s_{24}}{s_{13}s_{134}s_{234}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}s_{14}s_{34}}{s_{13}s_{134}s_{234}s_{24}} - \frac{2s_{12}s_{23}s_{34}}{s_{13}s_{134}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{234}^2s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}^2} \\
& + \frac{2s_{12}}{s_{13}s_{134}} + \frac{s_{12}s_{14}}{s_{13}s_{134}s_{234}} + \frac{6s_{12}}{s_{134}s_{234}} + \frac{5s_{12}s_{14}}{s_{13}s_{134}s_{24}} + \frac{5s_{12}s_{23}}{s_{13}s_{134}s_{24}} + \frac{2s_{12}}{s_{13}s_{24}} + \frac{5s_{12}s_{14}}{s_{13}s_{234}s_{24}} \\
& + \frac{5s_{12}s_{23}}{s_{13}s_{234}s_{24}} + \frac{s_{12}s_{23}}{s_{134}s_{234}s_{24}} + \frac{2s_{12}s_{13}}{s_{134}s_{234}s_{24}} + \frac{2s_{12}}{s_{234}s_{24}} - \frac{s_{12}s_{23}}{s_{234}^2s_{24}} + \frac{3s_{12}s_{14}^2}{s_{13}s_{134}^2s_{34}} + \frac{4s_{12}s_{14}}{s_{13}s_{134}s_{34}} \\
& + \frac{4s_{12}s_{14}}{s_{134}^2s_{34}} + \frac{s_{12}s_{14}s_{24}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{12}s_{24}}{s_{134}s_{234}s_{34}} + \frac{5s_{12}s_{24}}{s_{234}^2s_{34}} - \frac{4s_{12}}{s_{134}s_{34}} + \frac{s_{12}s_{14}}{s_{13}s_{234}s_{34}} \\
& - \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{12}s_{23}}{s_{13}s_{234}s_{34}} - \frac{s_{12}s_{23}}{s_{134}s_{234}s_{34}} + \frac{s_{12}s_{13}}{s_{134}s_{234}s_{34}} - \frac{4s_{12}}{s_{234}s_{34}} \\
& + \frac{3s_{12}s_{14}^2}{s_{13}s_{134}s_{24}s_{34}} + \frac{4s_{12}s_{14}}{s_{13}s_{24}s_{34}} + \frac{s_{12}s_{14}}{s_{134}s_{24}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{24}s_{34}} + \frac{4s_{12}s_{23}}{s_{13}s_{24}s_{34}} + \frac{s_{12}s_{23}}{s_{134}s_{24}s_{34}} \\
& - \frac{4s_{12}s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{3s_{12}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} - \frac{4s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{24}s_{34}} - \frac{s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} \\
& + \frac{s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{24}s_{34}} + \frac{4s_{12}s_{23}}{s_{234}s_{24}s_{34}} + \frac{3s_{12}s_{23}^2}{s_{234}^2s_{24}s_{34}} + \frac{5s_{12}s_{13}}{s_{134}^2s_{34}} + \frac{4s_{12}s_{23}}{s_{234}^2s_{34}} + \frac{5s_{12}}{s_{134}^2} \\
& + \frac{5s_{12}}{s_{234}^2} + \frac{6s_{12}s_{14}^2}{s_{134}^2s_{34}^2} + \frac{2s_{12}s_{24}^2}{s_{234}^2s_{34}^2} - \frac{6s_{12}s_{14}}{s_{134}s_{34}^2} + \frac{8s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{12}s_{24}}{s_{234}s_{34}^2} \\
& - \frac{2s_{12}s_{13}}{s_{134}s_{34}^2} - \frac{2s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{34}^2} + \frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{6s_{12}s_{23}}{s_{234}s_{34}^2} - \frac{6s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}^2} + \frac{8s_{12}s_{14}s_{23}}{s_{13}s_{24}s_{34}^2} \\
& - \frac{2s_{12}s_{14}s_{23}}{s_{134}s_{24}s_{34}^2} + \frac{4s_{12}s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} - \frac{6s_{12}s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2} + \frac{2s_{12}s_{13}^2}{s_{134}^2s_{34}^2} + \frac{6s_{12}s_{23}^2}{s_{234}^2s_{34}^2} \\
& + \frac{8s_{14}}{s_{13}s_{134}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}^2} + \frac{8s_{23}}{s_{13}s_{134}} + \frac{8s_{14}s_{24}}{s_{13}s_{134}^2} + \frac{4s_{24}}{s_{134}s_{234}} + \frac{14s_{24}}{s_{134}^2} \\
& + \frac{6s_{24}s_{34}}{s_{13}s_{134}^2} + \frac{4s_{34}}{s_{13}s_{134}} + \frac{6s_{34}}{s_{13}s_{24}} + \frac{4s_{34}}{s_{234}s_{24}} + \frac{6s_{13}s_{34}}{s_{234}^2s_{24}} + \frac{4}{s_{134}} + \frac{2s_{14}}{s_{134}s_{234}} \\
& + \frac{2s_{23}}{s_{134}s_{234}} + \frac{4s_{13}}{s_{134}s_{234}} + \frac{4}{s_{234}} + \frac{8s_{14}}{s_{13}s_{24}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}} + \frac{8s_{23}}{s_{13}s_{24}} + \frac{2s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}} \\
& + \frac{8s_{14}}{s_{234}s_{24}} + \frac{2s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}} - \frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}} + \frac{8s_{23}}{s_{234}s_{24}} + \frac{8s_{13}s_{23}}{s_{234}^2s_{24}} - \frac{2s_{14}s_{23}}{s_{234}^2s_{24}} \\
& + \frac{8s_{14}^2}{s_{13}s_{134}s_{34}} + \frac{2s_{24}^2}{s_{134}s_{234}s_{34}} + \frac{7s_{14}}{s_{134}s_{34}} + \frac{s_{14}^2s_{23}}{s_{13}s_{134}^2s_{34}} + \frac{6s_{14}s_{23}}{s_{13}s_{134}s_{34}} + \frac{s_{14}s_{23}}{s_{134}^2s_{34}} \\
& + \frac{5s_{23}}{s_{134}s_{34}} + \frac{8s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}} + \frac{11s_{14}s_{24}}{s_{134}^2s_{34}} - \frac{2s_{24}}{s_{134}s_{34}} + \frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}} \\
& + \frac{6s_{13}s_{24}}{s_{134}s_{234}s_{34}} + \frac{2s_{24}}{s_{234}s_{34}} + \frac{11s_{13}s_{24}}{s_{134}^2s_{34}} + \frac{11s_{13}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{13}}{s_{134}s_{34}} - \frac{2s_{13}}{s_{234}s_{34}} \\
& + \frac{5s_{14}}{s_{234}s_{34}} - \frac{s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{2s_{14}s_{23}}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{23}}{s_{134}s_{234}s_{34}} + \frac{7s_{23}}{s_{234}s_{34}} + \frac{2s_{13}^2}{s_{134}s_{234}s_{34}} \\
& + \frac{8s_{14}^2}{s_{13}s_{24}s_{34}} - \frac{5s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} + \frac{8s_{23}^2}{s_{13}s_{24}s_{34}} - \frac{5s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}} + \frac{6s_{14}s_{23}}{s_{13}s_{24}s_{34}} \\
& - \frac{5s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} - \frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}} + \frac{8s_{23}^2}{s_{234}s_{24}s_{34}} - \frac{5s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} - \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} \\
& + \frac{6s_{14}s_{23}}{s_{234}s_{24}s_{34}} + \frac{8s_{13}s_{23}^2}{s_{234}^2s_{24}s_{34}} + \frac{s_{14}s_{23}^2}{s_{234}^2s_{24}s_{34}} + \frac{11s_{13}s_{23}}{s_{234}^2s_{34}} + \frac{s_{14}s_{23}}{s_{234}^2s_{34}} - \frac{2}{s_{34}} \\
& + \frac{14s_{13}}{s_{234}^2} + \frac{3s_{14}^2}{s_{134}s_{34}^2} + \frac{s_{14}s_{24}^2}{s_{134}s_{234}s_{34}^2} + \frac{5s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} + \frac{3s_{13}s_{24}^2}{s_{234}^2s_{34}^2} + \frac{s_{13}s_{14}}{s_{134}s_{34}^2}
\end{aligned}
$$

$$
\begin{aligned}
& -\frac{4s_{14}}{s_{34}^2}-\frac{3s_{14}^3s_{23}}{s_{13}s_{134}^2s_{34}^2}+\frac{4s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2}+\frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2}+\frac{3s_{14}s_{23}}{s_{134}s_{34}^2}-\frac{3s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2}\\
& +\frac{s_{13}s_{23}}{s_{134}s_{34}^2}-\frac{4s_{23}}{s_{34}^2}+\frac{3s_{14}^2s_{24}}{s_{134}^2s_{34}^2}-\frac{4s_{14}s_{24}}{s_{134}s_{34}^2}-\frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2}-\frac{s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2}\\
& +\frac{s_{14}s_{24}}{s_{234}s_{34}^2}-\frac{s_{14}^2s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}^2}-\frac{4s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}+\frac{s_{23}s_{24}}{s_{234}s_{34}^2}+\frac{5s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2}\\
& +\frac{3s_{13}^2s_{24}}{s_{134}^2s_{34}^2}-\frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2}-\frac{3s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2}-\frac{s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{34}^2}-\frac{s_{14}s_{23}^2}{s_{13}s_{234}s_{34}^2}-\frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2}\\
& +\frac{3s_{23}^2}{s_{234}s_{34}^2}-\frac{s_{14}^2s_{23}}{s_{13}s_{234}s_{34}^2}-\frac{s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{4s_{13}s_{23}}{s_{234}s_{34}^2}-\frac{4s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}+\frac{3s_{14}s_{23}}{s_{234}s_{34}^2}\\
& +\frac{s_{13}^2s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{3s_{14}^2s_{23}^2}{s_{13}s_{134}s_{24}s_{34}^2}+\frac{4s_{14}s_{23}^2}{s_{13}s_{24}s_{34}^2}-\frac{s_{14}s_{23}^2}{s_{134}s_{24}s_{34}^2}-\frac{3s_{14}^3s_{23}}{s_{13}s_{134}s_{24}s_{34}^2}+\frac{4s_{14}^2s_{23}}{s_{13}s_{24}s_{34}^2}\\
& -\frac{s_{14}^2s_{23}}{s_{134}s_{24}s_{34}^2}+\frac{2s_{14}^2s_{23}^3}{s_{13}s_{134}s_{234}s_{24}s_{34}^2}-\frac{3s_{14}s_{23}^3}{s_{13}s_{234}s_{24}s_{34}^2}+\frac{2s_{14}^3s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2}-\frac{3s_{14}^2s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2}\\
& -\frac{s_{14}^2s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2}-\frac{s_{13}s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2}+\frac{4s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2}-\frac{3s_{14}s_{23}^3}{s_{234}^2s_{24}s_{34}^2}+\frac{3s_{13}s_{23}^2}{s_{234}^2s_{34}^2}+\frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2}
\end{aligned}
\tag{A.42}
$$

$$
\begin{aligned}
A_4^{0,(2)} = & -\frac{s_{13}^2}{s_{134}s_{234}s_{34}}-\frac{2s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{2s_{13}^2s_{24}}{s_{134}^2s_{34}^2}-\frac{s_{12}s_{13}^2}{s_{134}^2s_{34}^2}-\frac{6s_{13}s_{34}}{s_{234}^2s_{24}}-\frac{7s_{13}}{s_{134}s_{234}}-\frac{s_{12}s_{13}}{s_{134}s_{234}s_{24}}\\
& -\frac{4s_{13}s_{23}}{s_{234}^2s_{24}}-\frac{12s_{13}s_{24}}{s_{134}s_{234}s_{34}}-\frac{7s_{13}s_{24}}{s_{134}^2s_{34}}-\frac{7s_{13}s_{24}}{s_{234}^2s_{34}}-\frac{s_{13}}{s_{134}s_{34}}+\frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}}-\frac{2s_{13}s_{23}}{s_{134}s_{234}s_{34}}\\
& -\frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}}+\frac{s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}}-\frac{4s_{13}s_{23}^2}{s_{234}^2s_{24}s_{34}}-\frac{2s_{12}s_{13}}{s_{134}^2s_{34}}-\frac{5s_{13}s_{23}}{s_{234}^2s_{34}}-\frac{11s_{13}}{s_{234}^2}\\
& -\frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2}-\frac{2s_{13}s_{24}^2}{s_{234}^2s_{34}^2}+\frac{2s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2}+\frac{2s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2}+\frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2}+\frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2}\\
& +\frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{2s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2}+\frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2}+\frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}+\frac{2s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{7s_{24}}{s_{134}s_{234}}\\
& -\frac{11s_{24}}{s_{134}^2}-\frac{2s_{34}}{s_{134}s_{234}}+\frac{s_{12}s_{34}}{s_{134}s_{234}s_{24}}-\frac{8s_{34}}{s_{234}s_{24}}-\frac{3s_{12}s_{34}}{s_{234}^2s_{24}}-\frac{11}{s_{134}}-\frac{4s_{14}}{s_{134}s_{234}}\\
& -\frac{4s_{23}}{s_{134}s_{234}}+\frac{2s_{12}}{s_{134}s_{234}}-\frac{11}{s_{234}}-\frac{s_{12}}{s_{134}s_{24}}+\frac{s_{12}s_{14}}{s_{134}s_{234}s_{24}}-\frac{4s_{14}}{s_{234}s_{24}}-\frac{s_{14}s_{23}}{s_{134}s_{234}s_{24}}\\
& -\frac{2s_{12}s_{23}}{s_{134}s_{234}s_{24}}-\frac{8s_{23}}{s_{234}s_{24}}+\frac{3s_{12}s_{23}}{s_{234}^2s_{24}}+\frac{3s_{14}s_{23}}{s_{234}^2s_{24}}-\frac{s_{24}^2}{s_{134}s_{234}s_{34}}-\frac{5s_{14}}{s_{134}s_{34}}+\frac{2s_{12}s_{14}}{s_{134}^2s_{34}}\\
& -\frac{s_{14}s_{23}}{s_{134}^2s_{34}}-\frac{2s_{23}}{s_{134}s_{34}}-\frac{5s_{14}s_{24}}{s_{134}^2s_{34}}-\frac{2s_{14}s_{24}}{s_{134}s_{234}s_{34}}+\frac{s_{23}s_{24}}{s_{134}s_{234}s_{34}}-\frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}}\\
& -\frac{s_{24}}{s_{234}s_{34}}-\frac{2s_{12}s_{24}}{s_{234}^2s_{34}}+\frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}}-\frac{2s_{14}}{s_{234}s_{34}}+\frac{4s_{14}s_{23}}{s_{134}s_{234}s_{34}}+\frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}}\\
& -\frac{5s_{23}}{s_{234}s_{34}}+\frac{s_{14}s_{23}}{s_{134}s_{24}s_{34}}+\frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}}-\frac{8s_{23}^2}{s_{234}s_{24}s_{34}}-\frac{s_{14}^2s_{23}}{s_{134}s_{234}s_{24}s_{34}}-\frac{3s_{14}s_{23}^2}{s_{234}^2s_{24}s_{34}}\\
& +\frac{2s_{12}s_{23}}{s_{234}^2s_{34}}-\frac{s_{14}s_{23}}{s_{234}^2s_{34}}-\frac{4s_{12}}{s_{134}^2}-\frac{4s_{12}}{s_{234}^2}-\frac{s_{12}s_{14}^2}{s_{134}^2s_{34}^2}-\frac{s_{12}s_{24}^2}{s_{234}^2s_{34}^2}\\
& -\frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2}+\frac{2s_{12}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2}+\frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}+\frac{2s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2}+\frac{2s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2}-\frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2}\\
& -\frac{2s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{2s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{s_{12}s_{23}^2}{s_{234}^2s_{34}^2}-\frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2}-\frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{234}s_{24}}-\frac{8s_{14}}{s_{13}s_{134}}\\
& +\frac{3s_{12}s_{14}}{s_{13}s_{134}^2}+\frac{3s_{14}s_{23}}{s_{13}s_{134}^2}-\frac{4s_{23}}{s_{13}s_{134}}-\frac{4s_{14}s_{24}}{s_{13}s_{134}^2}-\frac{s_{12}s_{24}}{s_{13}s_{134}s_{234}}-\frac{6s_{24}s_{34}}{s_{13}s_{134}^2}\\
& -\frac{8s_{34}}{s_{13}s_{134}}+\frac{s_{12}s_{34}}{s_{13}s_{134}s_{234}}+\frac{s_{12}s_{34}}{s_{13}s_{134}s_{24}}+\frac{s_{12}s_{34}}{s_{13}s_{234}s_{24}}+\frac{2s_{14}s_{23}s_{34}}{s_{13}s_{134}s_{234}s_{24}}-\frac{6s_{34}}{s_{13}s_{24}}
\end{aligned}
$$

$$
\begin{aligned}
&-\frac{3s_{12}s_{34}}{s_{13}s_{134}^2}-\frac{s_{12}}{s_{13}s_{234}}-\frac{2s_{12}s_{14}}{s_{13}s_{134}s_{234}}-\frac{s_{14}s_{23}}{s_{13}s_{134}s_{234}}+\frac{s_{12}s_{23}}{s_{13}s_{134}s_{234}}+\frac{3s_{12}s_{14}}{s_{13}s_{134}s_{24}}\\
&-\frac{4s_{14}}{s_{13}s_{24}}-\frac{s_{14}s_{23}}{s_{13}s_{134}s_{24}}-\frac{2s_{12}s_{23}}{s_{13}s_{134}s_{24}}-\frac{4s_{23}}{s_{13}s_{24}}-\frac{2s_{12}s_{14}}{s_{13}s_{234}s_{24}}+\frac{3s_{12}s_{23}}{s_{13}s_{234}s_{24}}\\
&-\frac{s_{14}s_{23}}{s_{13}s_{234}s_{24}}-\frac{8s_{14}^2}{s_{13}s_{134}s_{34}}-\frac{3s_{14}^2s_{23}}{s_{13}s_{134}^2s_{34}}-\frac{4s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}}+\frac{s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}}-\frac{s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{34}}\\
&+\frac{2s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{34}}+\frac{s_{14}s_{23}}{s_{13}s_{234}s_{34}}-\frac{4s_{14}^2}{s_{13}s_{24}s_{34}}+\frac{2s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}}-\frac{4s_{23}^2}{s_{13}s_{24}s_{34}}-\frac{3s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}}\\
&-\frac{3s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}}+\frac{2s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}}
\end{aligned}
\tag{A.43}
$$

$$
\begin{aligned}
A_4^{0,(3)}=&\frac{2s_{12}s_{34}}{s_{13}s_{134}s_{234}}+\frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}}+\frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}}+\frac{2s_{12}s_{34}}{s_{134}s_{234}s_{24}}+\frac{2s_{12}s_{34}}{s_{234}^2s_{24}}+\frac{2s_{12}s_{34}}{s_{13}s_{134}^2}+\frac{4s_{12}}{s_{134}s_{234}}\\
&+\frac{2s_{12}}{s_{134}^2}+\frac{2s_{12}}{s_{234}^2}-\frac{2s_{14}s_{23}}{s_{13}s_{134}^2}+\frac{2s_{24}}{s_{134}s_{234}}+\frac{2s_{24}}{s_{134}^2}+\frac{2s_{24}s_{34}}{s_{13}s_{134}^2}+\frac{4s_{34}}{s_{13}s_{134}}\\
&+\frac{4s_{34}}{s_{134}s_{234}}+\frac{2s_{34}}{s_{13}s_{24}}+\frac{4s_{34}}{s_{234}s_{24}}+\frac{2s_{13}s_{34}}{s_{234}^2s_{24}}+\frac{2}{s_{134}}-\frac{2s_{14}s_{23}}{s_{13}s_{134}s_{234}}+\frac{2s_{13}}{s_{134}s_{234}}\\
&+\frac{2}{s_{234}}-\frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}}-\frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}}-\frac{2s_{14}s_{23}}{s_{134}s_{234}s_{24}}-\frac{2s_{14}s_{23}}{s_{234}^2s_{24}}+\frac{2s_{13}}{s_{234}^2}.
\end{aligned}
\tag{A.44}
$$

By running `BuildAndIntegrateAntenna[A,4,0,Component->Leading]` we get the integrated result,

$$
\begin{aligned}
\mathcal{A}_4^0=&\frac{3}{4\epsilon^4}+\frac{65}{24\epsilon^3}+\frac{434-39\pi^2}{36\epsilon^2}-\frac{1}{\epsilon}\left(-\frac{43223}{864}+\frac{589\pi^2}{144}+\frac{71\zeta_3}{4}\right)\\
&+\frac{1076717}{5184}-\frac{7955\pi^2}{432}+\frac{373\pi^4}{1440}-\frac{1327\zeta_3}{18}+O(\epsilon)\,.
\end{aligned}
\tag{A.45}
$$

A.3.9 $\tilde{A}_4^0$

Running `BuildAntenna[A,4,0,Component->Subleading]`, we get the built result,

$$
\tilde{A}_4^0=\frac{1}{2(\epsilon-1)q^2}\left(\tilde{A}_4^{0,(0)}+\epsilon\tilde{A}_4^{0,(1)}+\epsilon^2\tilde{A}_4^{0,(2)}+\epsilon^3\tilde{A}_4^{0,(3)}\right)
\tag{A.46}
$$

where,

$$
\begin{aligned}
\tilde{A}_4^{0,(0)}=&-\frac{2}{s_{14}s_{23}\mathcal{D}_{4,gg}}\Bigg[-\frac{2s_{12}s_{34}^2}{s_{13}s_{134}^2s_{14}}+\frac{4s_{14}^3s_{23}}{s_{13}s_{134}^2s_{34}^2}\\
&-\frac{2s_{14}^2\left(2s_{12}-3s_{23}+2s_{24}\right)}{s_{13}s_{134}^2s_{34}}-\frac{2\left(3s_{12}s_{14}+3s_{12}s_{34}-3s_{14}s_{23}+2s_{14}s_{24}-s_{23}s_{34}+s_{24}s_{34}\right)}{s_{13}s_{134}^2}\\
&-\frac{2s_{12}s_{34}(s_{12}+s_{24})}{s_{13}s_{134}s_{14}s_{23}}-\frac{2s_{12}s_{34}(s_{12}+s_{23})}{s_{13}s_{134}s_{14}s_{24}}\\
&-\frac{2s_{12}^2\left(2s_{12}+s_{14}+s_{24}+2s_{34}\right)}{s_{13}s_{134}s_{23}s_{234}}-\frac{2s_{12}\left(2s_{12}+2s_{24}+s_{34}\right)}{s_{13}s_{134}s_{23}}\\
&-\frac{2s_{14}^2s_{23}^2\left(2s_{12}+s_{14}+s_{23}\right)}{s_{13}s_{134}s_{234}s_{24}s_{34}^2}+\frac{2s_{14}s_{23}\left(4s_{12}^2+2s_{12}s_{14}+2s_{12}s_{23}-s_{14}s_{23}\right)}{s_{13}s_{134}s_{234}s_{24}s_{34}}\\
&-\frac{2s_{12}\left(2s_{12}^2+s_{12}s_{14}+s_{12}s_{23}+2s_{12}s_{34}-3s_{14}s_{23}\right)}{s_{13}s_{134}s_{234}s_{24}}-\frac{2s_{14}^2s_{23}\left(2s_{12}+s_{14}-s_{24}\right)}{s_{13}s_{134}s_{234}s_{34}^2}\\
&+\frac{2s_{14}\left(4s_{12}^2+2s_{12}s_{14}+2s_{12}s_{23}+s_{23}^2+s_{23}s_{24}\right)}{s_{13}s_{134}s_{234}s_{34}}-\frac{2\left(2s_{12}^2-2s_{12}s_{14}+s_{12}s_{23}+s_{12}s_{24}+s_{12}s_{34}-s_{14}s_{23}\right)}{s_{13}s_{134}s_{234}}\\
&+\frac{4s_{14}^2s_{23}\left(2s_{12}+s_{14}+s_{23}\right)}{s_{13}s_{134}s_{24}s_{34}^2}-\frac{2s_{14}\left(4s_{12}^2+2s_{12}s_{14}-2s_{12}s_{23}-3s_{14}s_{23}-3s_{23}^2\right)}{s_{13}s_{134}s_{24}s_{34}}\\
&-\frac{2\left(4s_{12}^2+3s_{12}s_{14}+2s_{12}s_{23}+s_{12}s_{34}-s_{14}s_{23}-s_{23}^2\right)}{s_{13}s_{134}s_{24}}-\frac{8s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2}
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{4s_{14}(s_{12}-2s_{23}+s_{24})}{s_{13}s_{134}s_{34}}+\frac{2(s_{12}+s_{14}-2s_{23}+s_{24})}{s_{13}s_{134}}\\
&-\frac{4s_{12}^3}{s_{13}s_{14}s_{23}s_{24}}-\frac{2s_{12}^2}{s_{13}s_{14}s_{23}}-\frac{2s_{12}^2}{s_{13}s_{14}s_{24}}-\frac{2s_{12}}{s_{13}s_{14}}\\
&-\frac{2s_{12}s_{34}(s_{12}+s_{14})}{s_{13}s_{23}s_{234}s_{24}}-\frac{2s_{12}(2s_{12}+2s_{14}+s_{34})}{s_{13}s_{23}s_{234}}-\frac{2s_{12}^2}{s_{13}s_{23}s_{24}}\\
&+\frac{4s_{14}s_{23}^2(2s_{12}+s_{14}+s_{23})}{s_{13}s_{234}s_{24}s_{34}^2}-\frac{2s_{23}\left(4s_{12}^2-2s_{12}s_{14}+2s_{12}s_{23}-3s_{14}^2-3s_{14}s_{23}\right)}{s_{13}s_{234}s_{24}s_{34}}\\
&-\frac{2\left(4s_{12}^2+2s_{12}s_{14}+3s_{12}s_{23}+s_{12}s_{34}-s_{14}^2-s_{14}s_{23}\right)}{s_{13}s_{234}s_{24}}+\frac{4s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{234}s_{34}^2}\\
&-\frac{4\left(2s_{12}^2+s_{12}s_{23}-s_{14}^2-s_{14}s_{23}\right)}{s_{13}s_{234}s_{34}}-\frac{2(2s_{12}-s_{14})}{s_{13}s_{234}}\\
&-\frac{6s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{24}s_{34}^2}+\frac{2\left(4s_{12}^2-2s_{14}^2-3s_{14}s_{23}-2s_{23}^2\right)}{s_{13}s_{24}s_{34}}\\
&-\frac{2(s_{12}+2s_{14}+2s_{23}+s_{34})}{s_{13}s_{24}}+\frac{2s_{14}s_{23}}{s_{13}s_{34}^2}+\frac{2s_{23}}{s_{13}s_{34}}\\
&+\frac{4s_{13}^3s_{24}}{s_{134}^2s_{14}s_{34}^2}-\frac{2s_{13}^2(2s_{12}+2s_{23}-3s_{24})}{s_{134}^2s_{14}s_{34}}\\
&-\frac{2\left(3s_{12}s_{13}+3s_{12}s_{34}+2s_{13}s_{23}-3s_{13}s_{24}+s_{23}s_{34}-s_{24}s_{34}\right)}{s_{134}^2s_{14}}\\
&-\frac{4\left(2s_{12}s_{13}^2+4s_{12}s_{13}s_{14}+2s_{12}s_{14}^2+s_{13}^2s_{23}-s_{13}^2s_{24}+s_{13}s_{14}s_{23}+s_{13}s_{14}s_{24}-s_{14}^2s_{23}+s_{14}^2s_{24}\right)}{s_{134}^2s_{34}^2}\\
&-\frac{10\left(2s_{12}s_{13}+2s_{12}s_{14}+s_{13}s_{23}+s_{14}s_{24}\right)}{s_{134}^2s_{34}}-\frac{2(8s_{12}+s_{23}+s_{24})}{s_{134}^2}\\
&-\frac{2s_{13}^2s_{24}^2(2s_{12}+s_{13}+s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}^2}+\frac{2s_{13}s_{24}\left(4s_{12}^2+2s_{12}s_{13}+2s_{12}s_{24}-s_{13}s_{24}\right)}{s_{134}s_{14}s_{23}s_{234}s_{34}}\\
&-\frac{2s_{12}\left(2s_{12}^2+s_{12}s_{13}+s_{12}s_{24}+2s_{12}s_{34}-3s_{13}s_{24}\right)}{s_{134}s_{14}s_{23}s_{234}}+\frac{4s_{13}^2s_{24}(2s_{12}+s_{13}+s_{24})}{s_{134}s_{14}s_{23}s_{34}^2}\\
&-\frac{2s_{13}\left(4s_{12}^2+2s_{12}s_{13}-2s_{12}s_{24}-3s_{13}s_{24}-3s_{24}^2\right)}{s_{134}s_{14}s_{23}s_{34}}\\
&-\frac{2\left(4s_{12}^2+3s_{12}s_{13}+2s_{12}s_{24}+s_{12}s_{34}-s_{13}s_{24}-s_{24}^2\right)}{s_{134}s_{14}s_{23}}\\
&-\frac{2s_{12}^2(2s_{12}+s_{13}+s_{23}+2s_{34})}{s_{134}s_{14}s_{234}s_{24}}-\frac{2s_{13}^2s_{24}(2s_{12}+s_{13}-s_{23})}{s_{134}s_{14}s_{234}s_{34}^2}\\
&+\frac{2s_{13}\left(4s_{12}^2+2s_{12}s_{13}+2s_{12}s_{24}+s_{23}s_{24}+s_{24}^2\right)}{s_{134}s_{14}s_{234}s_{34}}\\
&-\frac{2\left(2s_{12}^2-2s_{12}s_{13}+s_{12}s_{23}+s_{12}s_{24}+s_{12}s_{34}-s_{13}s_{24}\right)}{s_{134}s_{14}s_{234}}\\
&-\frac{2s_{12}(2s_{12}+2s_{23}+s_{34})}{s_{134}s_{14}s_{24}}-\frac{8s_{13}^2s_{24}}{s_{134}s_{14}s_{34}^2}\\
&+\frac{4s_{13}(s_{12}+s_{23}-2s_{24})}{s_{134}s_{14}s_{34}}+\frac{2(s_{12}+s_{13}+s_{23}-2s_{24})}{s_{134}s_{14}}\\
&-\frac{2s_{13}s_{24}^2(2s_{12}-s_{14}+s_{24})}{s_{134}s_{23}s_{234}s_{34}^2}+\frac{2s_{24}\left(4s_{12}^2+2s_{12}s_{13}+2s_{12}s_{24}+s_{13}^2+s_{13}s_{14}\right)}{s_{134}s_{23}s_{234}s_{34}}\\
&-\frac{2\left(2s_{12}^2+s_{12}s_{13}+s_{12}s_{14}-2s_{12}s_{24}+s_{12}s_{34}-s_{13}s_{24}\right)}{s_{134}s_{23}s_{234}}\\
&+\frac{4s_{13}s_{24}(2s_{12}+s_{13}+s_{24})}{s_{134}s_{23}s_{34}^2}-\frac{4\left(2s_{12}^2+s_{12}s_{13}-s_{13}s_{24}-s_{24}^2\right)}{s_{134}s_{23}s_{34}}-\frac{2(2s_{12}-s_{24})}{s_{134}s_{23}}
\end{aligned}
$$

$$- \frac{2s_{14}s_{23}^2(2s_{12} - s_{13} + s_{23})}{s_{134}s_{234}s_{24}s_{34}^2} + \frac{2s_{23}\left(4s_{12}^2 + 2s_{12}s_{14} + 2s_{12}s_{23} + s_{13}s_{14} + s_{14}^2\right)}{s_{134}s_{234}s_{24}s_{34}}$$

$$- \frac{2\left(2s_{12}^2 + s_{12}s_{13} + s_{12}s_{14} - 2s_{12}s_{23} + s_{12}s_{34} - s_{14}s_{23}\right)}{s_{134}s_{234}s_{24}}$$

$$- \frac{2\left(4s_{12}s_{13}s_{23} + 6s_{12}s_{13}s_{24} + 6s_{12}s_{14}s_{23} + 4s_{12}s_{14}s_{24} - s_{13}s_{14}s_{23} - s_{13}s_{14}s_{24} - s_{13}s_{23}s_{24} - s_{14}s_{23}s_{24}\right)}{s_{134}s_{234}s_{34}^2}$$

$$+ \frac{2}{s_{134}s_{234}s_{34}}\Big(8s_{12}^2 - 2s_{12}s_{13} - 2s_{12}s_{14} - 2s_{12}s_{23} - 2s_{12}s_{24} - s_{13}^2 - 2s_{13}s_{14} + s_{13}s_{24}$$

$$- s_{14}^2 + s_{14}s_{23} - s_{23}^2 - 2s_{23}s_{24} - s_{24}^2\Big)$$

$$- \frac{2(2s_{12} + s_{13} + s_{14} + s_{23} + s_{24})}{s_{134}s_{234}}$$

$$+ \frac{4s_{14}s_{23}(2s_{12} + s_{14} + s_{23})}{s_{134}s_{24}s_{34}^2} - \frac{4\left(2s_{12}^2 + s_{12}s_{14} - s_{14}s_{23} - s_{23}^2\right)}{s_{134}s_{24}s_{34}} - \frac{2(2s_{12} - s_{23})}{s_{134}s_{24}}$$

$$+ \frac{4\left(4s_{12}s_{13} + 4s_{12}s_{14} - s_{13}^2 - 2s_{13}s_{14} + s_{13}s_{23} - s_{13}s_{24} - s_{14}^2 - s_{14}s_{23} + s_{14}s_{24}\right)}{s_{134}s_{34}^2}$$

$$+ \frac{2(8s_{12} - s_{13} - s_{14} + s_{23} + s_{24})}{s_{134}s_{34}} + \frac{4}{s_{134}}$$

$$- \frac{2s_{12}s_{34}(s_{12} + s_{13})}{s_{14}s_{23}s_{234}s_{24}} + \frac{4s_{13}s_{24}^2(2s_{12} + s_{13} + s_{24})}{s_{14}s_{23}s_{234}s_{34}^2}$$

$$- \frac{2s_{24}\left(4s_{12}^2 - 2s_{12}s_{13} + 2s_{12}s_{24} - 3s_{13}^2 - 3s_{13}s_{24}\right)}{s_{14}s_{23}s_{234}s_{34}}$$

$$- \frac{2\left(4s_{12}^2 + 2s_{12}s_{13} + 3s_{12}s_{24} + s_{12}s_{34} - s_{13}^2 - s_{13}s_{24}\right)}{s_{14}s_{23}s_{234}} - \frac{2s_{12}^2}{s_{14}s_{23}s_{24}}$$

$$- \frac{6s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{14}s_{23}s_{34}^2} + \frac{2\left(4s_{12}^2 - 2s_{13}^2 - 3s_{13}s_{24} - 2s_{24}^2\right)}{s_{14}s_{23}s_{34}}$$

$$- \frac{2(s_{12} + 2s_{13} + 2s_{24} + s_{34})}{s_{14}s_{23}} - \frac{2s_{12}(2s_{12} + 2s_{13} + s_{34})}{s_{14}s_{234}s_{24}}$$

$$+ \frac{4s_{13}s_{24}(2s_{12} + s_{13} + s_{24})}{s_{14}s_{234}s_{34}^2} - \frac{4\left(2s_{12}^2 + s_{12}s_{24} - s_{13}^2 - s_{13}s_{24}\right)}{s_{14}s_{234}s_{34}}$$

$$- \frac{2(2s_{12} - s_{13})}{s_{14}s_{234}} + \frac{2s_{13}s_{24}}{s_{14}s_{34}^2} + \frac{2s_{24}}{s_{14}s_{34}}$$

$$- \frac{2s_{12}s_{34}^2}{s_{23}s_{234}^2s_{24}} + \frac{4s_{13}s_{24}^3}{s_{23}s_{234}^2s_{34}^2}$$

$$- \frac{2s_{24}^2(2s_{12} - 3s_{13} + 2s_{14})}{s_{23}s_{234}^2s_{34}} - \frac{2\left(3s_{12}s_{24} + 3s_{12}s_{34} - 3s_{13}s_{24} - s_{13}s_{34} + 2s_{14}s_{24} + s_{14}s_{34}\right)}{s_{23}s_{234}^2}$$

$$- \frac{8s_{13}s_{24}^2}{s_{23}s_{234}s_{34}^2} + \frac{4s_{24}(s_{12} - 2s_{13} + s_{14})}{s_{23}s_{234}s_{34}}$$

$$+ \frac{2(s_{12} - 2s_{13} + s_{14} + s_{24})}{s_{23}s_{234}} - \frac{2s_{12}}{s_{23}s_{24}} + \frac{2s_{13}s_{24}}{s_{23}s_{34}^2} + \frac{2s_{13}}{s_{23}s_{34}}$$

$$+ \frac{4s_{14}s_{23}^3}{s_{234}^2s_{24}s_{34}^2} - \frac{2s_{23}^2(2s_{12} + 2s_{13} - 3s_{14})}{s_{234}^2s_{24}s_{34}}$$

$$- \frac{2\left(3s_{12}s_{23} + 3s_{12}s_{34} + 2s_{13}s_{23} + s_{13}s_{34} - 3s_{14}s_{23} - s_{14}s_{34}\right)}{s_{234}^2s_{24}}$$

$$- \frac{4\left(2s_{12}s_{23}^2 + 4s_{12}s_{23}s_{24} + 2s_{12}s_{24}^2 + s_{13}s_{23}^2 + s_{13}s_{23}s_{24} - s_{13}s_{24}^2 - s_{14}s_{23}^2 + s_{14}s_{23}s_{24} + s_{14}s_{24}^2\right)}{s_{234}^2s_{34}^2}$$

$$- \frac{10\left(2s_{12}s_{23} + 2s_{12}s_{24} + s_{13}s_{23} + s_{14}s_{24}\right)}{s_{234}^2s_{34}} - \frac{2(8s_{12} + s_{13} + s_{14})}{s_{234}^2}$$

$$
\begin{aligned}
&- \frac{8s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2} + \frac{4s_{23}(s_{12}+s_{13}-2s_{14})}{s_{234}s_{24}s_{34}} \\
&+ \frac{2(s_{12}+s_{13}-2s_{14}+s_{23})}{s_{234}s_{24}} \\
&+ \frac{4\left(4s_{12}s_{23}+4s_{12}s_{24}+s_{13}s_{23}-s_{13}s_{24}-s_{14}s_{23}+s_{14}s_{24}-s_{23}^2-2s_{23}s_{24}-s_{24}^2\right)}{s_{234}s_{34}^2} \\
&+ \frac{2(8s_{12}+s_{13}+s_{14}-s_{23}-s_{24})}{s_{234}s_{34}} + \frac{4}{s_{234}} \\
&+ \frac{2s_{14}s_{23}}{s_{24}s_{34}^2} + \frac{2s_{14}}{s_{24}s_{34}} - \frac{4(2s_{12}-s_{13}-s_{14}-s_{23}-s_{24})}{s_{34}^2} - \frac{4}{s_{34}}\Bigg]
\end{aligned}
\tag{A.47}
$$

$$
\begin{aligned}
\tilde{A}_4^{0,(1)} = \frac{2}{s_{14}s_{23}\mathcal{D}_{4,gg}} &\Bigg[\frac{4s_{12}s_{34}^2}{s_{13}s_{134}^2s_{14}} - \frac{4s_{14}^3s_{23}}{s_{13}s_{134}^2s_{34}^2} + \frac{2s_{14}^2(2s_{12}-3s_{23}+4s_{24})}{s_{13}s_{134}^2s_{34}} \\
&+ \frac{2(3s_{12}s_{14}+6s_{12}s_{34}-6s_{14}s_{23}+4s_{14}s_{24}-2s_{23}s_{34}+3s_{24}s_{34})}{s_{13}s_{134}^2} \\
&+ \frac{2s_{12}s_{34}(s_{12}+2s_{24}+s_{34})}{s_{13}s_{134}s_{14}s_{23}} + \frac{2s_{12}s_{34}(s_{12}+2s_{23}+s_{34})}{s_{13}s_{134}s_{14}s_{24}} \\
&+ \frac{2s_{12}(2s_{12}^2+s_{12}s_{14}+s_{12}s_{24}+2s_{14}s_{24}+s_{14}s_{34}+s_{24}s_{34}+2s_{34}^2)}{s_{13}s_{134}s_{23}s_{234}} \\
&+ \frac{4s_{12}(s_{12}+2s_{24}+s_{34})}{s_{13}s_{134}s_{23}} + \frac{2s_{14}^2s_{23}^2(2s_{12}+s_{14}+s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} \\
&- \frac{4s_{12}s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}} \\
&+ \frac{2(2s_{12}^3+s_{12}^2s_{14}+s_{12}^2s_{23}-s_{12}s_{14}s_{34}-s_{12}s_{23}s_{34}+s_{14}^2s_{23}+s_{14}s_{23}^2)}{s_{13}s_{134}s_{234}s_{24}} \\
&+ \frac{2s_{14}^2s_{23}(2s_{12}+s_{14}-s_{24})}{s_{13}s_{134}s_{234}s_{34}^2} \\
&- \frac{2s_{14}(4s_{12}^2+2s_{12}s_{14}+2s_{12}s_{23}+3s_{14}s_{23}+2s_{14}s_{24}+s_{23}^2+s_{23}s_{24})}{s_{13}s_{134}s_{234}s_{34}} \\
&+ \frac{2(2s_{12}^2+3s_{12}s_{14}+s_{12}s_{23}+s_{12}s_{24}+2s_{12}s_{34}-s_{14}^2-2s_{14}s_{23}-s_{14}s_{24}-2s_{14}s_{34})}{s_{13}s_{134}s_{234}} \\
&- \frac{4s_{14}^2s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{134}s_{24}s_{34}^2} \\
&+ \frac{2s_{14}(4s_{12}^2+2s_{12}s_{14}-2s_{12}s_{23}-3s_{14}s_{23}-5s_{23}^2)}{s_{13}s_{134}s_{24}s_{34}} \\
&+ \frac{2(4s_{12}^2+3s_{12}s_{14}+4s_{12}s_{23}+2s_{12}s_{34}-2s_{14}s_{23}-2s_{23}^2-s_{23}s_{34})}{s_{13}s_{134}s_{24}} \\
&+ \frac{8s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2} + \frac{4s_{14}(-s_{12}+2s_{14}+2s_{23}-2s_{24})}{s_{13}s_{134}s_{34}} \\
&+ \frac{2(-s_{12}+2s_{14}+4s_{23}-2s_{24}+s_{34})}{s_{13}s_{134}} + \frac{4s_{12}(s_{12}^2+s_{12}s_{34}+s_{34}^2)}{s_{13}s_{14}s_{23}s_{24}} \\
&+ \frac{2s_{12}(s_{12}+s_{34})}{s_{13}s_{14}s_{23}} + \frac{2s_{12}(s_{12}+s_{34})}{s_{13}s_{14}s_{24}} + \frac{2s_{12}}{s_{13}s_{14}} \\
&+ \frac{2s_{12}s_{34}(s_{12}+2s_{14}+s_{34})}{s_{13}s_{23}s_{234}s_{24}} + \frac{4s_{12}(s_{12}+2s_{14}+s_{34})}{s_{13}s_{23}s_{234}} \\
&+ \frac{2s_{12}(s_{12}+s_{34})}{s_{13}s_{23}s_{24}} + \frac{4s_{12}}{s_{13}s_{23}} \\
&- \frac{4s_{14}s_{23}^2(2s_{12}+s_{14}+s_{23})}{s_{13}s_{234}s_{24}s_{34}^2} \\
&+ \frac{2s_{23}(4s_{12}^2-2s_{12}s_{14}+2s_{12}s_{23}-5s_{14}^2-3s_{14}s_{23})}{s_{13}s_{234}s_{24}s_{34}}
\end{aligned}
$$

$$+\frac{2(4s_{12}^2+4s_{12}s_{14}+3s_{12}s_{23}+2s_{12}s_{34}-2s_{14}^2-2s_{14}s_{23}-s_{14}s_{34})}{s_{13}s_{234}s_{24}}$$
$$-\frac{4s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{234}s_{34}^2}+\frac{4(2s_{12}^2+s_{12}s_{23}-2s_{14}^2-s_{14}s_{23})}{s_{13}s_{234}s_{34}}$$
$$+\frac{4(s_{12}-s_{14})}{s_{13}s_{234}}+\frac{6s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{24}s_{34}^2}$$
$$+\frac{8(-s_{12}^2+s_{14}^2+s_{14}s_{23}+s_{23}^2)}{s_{13}s_{24}s_{34}}+\frac{2(-2s_{12}+3s_{14}+3s_{23}+s_{34})}{s_{13}s_{24}}$$
$$-\frac{2s_{14}s_{23}}{s_{13}s_{34}^2}-\frac{2(2s_{14}+s_{23})}{s_{13}s_{34}}-\frac{2}{s_{13}}$$
$$-\frac{4s_{13}^3s_{24}}{s_{134}^2s_{14}s_{34}^2}+\frac{2s_{13}^2(2s_{12}+4s_{23}-3s_{24})}{s_{134}^2s_{14}s_{34}}$$
$$+\frac{2(3s_{12}s_{13}+6s_{12}s_{34}+4s_{13}s_{23}-6s_{13}s_{24}+3s_{23}s_{34}-2s_{24}s_{34})}{s_{134}^2s_{14}}$$
$$+\frac{4(2s_{12}s_{13}^2+4s_{12}s_{13}s_{14}+2s_{12}s_{14}^2+s_{13}^2s_{23}-s_{13}^2s_{24}+s_{13}s_{14}s_{23}+s_{13}s_{14}s_{24}-s_{14}^2s_{23}+s_{14}^2s_{24})}{s_{134}^2s_{34}^2}$$
$$+\frac{2(10s_{12}s_{13}+10s_{12}s_{14}+9s_{13}s_{23}+2s_{13}s_{24}+2s_{14}s_{23}+9s_{14}s_{24})}{s_{134}^2s_{34}}$$
$$+\frac{4(5s_{12}+s_{23}+s_{24})}{s_{134}^2}$$
$$+\frac{2s_{13}^2s_{24}^2(2s_{12}+s_{13}+s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}^2}$$
$$-\frac{4s_{12}s_{13}s_{24}(2s_{12}+s_{13}+s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}}$$
$$+\frac{2(2s_{12}^3+s_{12}^2s_{13}+s_{12}^2s_{24}-s_{12}s_{13}s_{34}-s_{12}s_{24}s_{34}+s_{13}^2s_{24}+s_{13}s_{24}^2)}{s_{134}s_{14}s_{23}s_{234}}$$
$$-\frac{4s_{13}^2s_{24}(2s_{12}+s_{13}+s_{24})}{s_{134}s_{14}s_{23}s_{34}^2}$$
$$+\frac{2s_{13}(4s_{12}^2+2s_{12}s_{13}-2s_{12}s_{24}-3s_{13}s_{24}-5s_{24}^2)}{s_{134}s_{14}s_{23}s_{34}}$$
$$+\frac{2(4s_{12}^2+3s_{12}s_{13}+4s_{12}s_{24}+2s_{12}s_{34}-2s_{13}s_{24}-2s_{24}^2-s_{24}s_{34})}{s_{134}s_{14}s_{23}}$$
$$+\frac{2s_{12}(2s_{12}^2+s_{12}s_{13}+s_{12}s_{23}+2s_{13}s_{23}+s_{13}s_{34}+s_{23}s_{34}+2s_{34}^2)}{s_{134}s_{14}s_{234}s_{24}}$$
$$+\frac{2s_{13}^2s_{24}(2s_{12}+s_{13}-s_{23})}{s_{134}s_{14}s_{234}s_{34}^2}$$
$$-\frac{2s_{13}(4s_{12}^2+2s_{12}s_{13}+2s_{12}s_{24}+2s_{13}s_{23}+3s_{13}s_{24}+s_{23}s_{24}+s_{24}^2)}{s_{134}s_{14}s_{234}s_{34}}$$
$$+\frac{2(2s_{12}^2+3s_{12}s_{13}+s_{12}s_{23}+s_{12}s_{24}+2s_{12}s_{34}-s_{13}^2-s_{13}s_{23}-2s_{13}s_{24}-2s_{13}s_{34})}{s_{134}s_{14}s_{234}}$$
$$+\frac{4s_{12}(s_{12}+2s_{23}+s_{34})}{s_{134}s_{14}s_{24}}+\frac{8s_{13}^2s_{24}}{s_{134}s_{14}s_{34}^2}$$
$$+\frac{4s_{13}(-s_{12}+2s_{13}-2s_{23}+2s_{24})}{s_{134}s_{14}s_{34}}+\frac{2(-s_{12}+2s_{13}-2s_{23}+4s_{24}+s_{34})}{s_{134}s_{14}}$$
$$+\frac{2s_{13}s_{24}^2(2s_{12}-s_{14}+s_{24})}{s_{134}s_{23}s_{234}s_{34}^2}$$
$$-\frac{2s_{24}(4s_{12}^2+2s_{12}s_{13}+2s_{12}s_{24}+s_{13}^2+s_{13}s_{14}+3s_{13}s_{24}+2s_{14}s_{24})}{s_{134}s_{23}s_{234}s_{34}}$$
$$+\frac{2(2s_{12}^2+s_{12}s_{13}+s_{12}s_{14}+3s_{12}s_{24}+2s_{12}s_{34}-2s_{13}s_{24}-s_{14}s_{24}-s_{24}^2-2s_{24}s_{34})}{s_{134}s_{23}s_{234}}$$

$$
\begin{aligned}
&- \frac{4s_{13}s_{24}(2s_{12}+s_{13}+s_{24})}{s_{134}s_{23}s_{34}^2} + \frac{4(2s_{12}^2+s_{12}s_{13}-s_{13}s_{24}-2s_{24}^2)}{s_{134}s_{23}s_{34}} \\
&+ \frac{4(s_{12}-s_{24})}{s_{134}s_{23}} + \frac{2s_{14}s_{23}^2(2s_{12}-s_{13}+s_{23})}{s_{134}s_{234}s_{24}s_{34}^2} \\
&- \frac{2s_{23}(4s_{12}^2+2s_{12}s_{14}+2s_{12}s_{23}+s_{13}s_{14}+2s_{13}s_{23}+s_{14}^2+3s_{14}s_{23})}{s_{134}s_{234}s_{24}s_{34}} \\
&+ \frac{2(2s_{12}^2+s_{12}s_{13}+s_{12}s_{14}+3s_{12}s_{23}+2s_{12}s_{34}-s_{13}s_{23}-2s_{14}s_{23}-s_{23}^2-2s_{23}s_{34})}{s_{134}s_{234}s_{24}} \\
&+ \frac{2(4s_{12}s_{13}s_{23}+6s_{12}s_{13}s_{24}+6s_{12}s_{14}s_{23}+4s_{12}s_{14}s_{24}-s_{13}s_{14}s_{23}-s_{13}s_{14}s_{24}-s_{13}s_{23}s_{24}-s_{14}s_{23}s_{24})}{s_{134}s_{234}s_{34}^2} \\
&+ \frac{2}{s_{134}s_{234}s_{34}}\Big((-8s_{12}^2+2s_{12}s_{13}+2s_{12}s_{14}+2s_{12}s_{23}+2s_{12}s_{24}+s_{13}^2+2s_{13}s_{14}+2s_{13}s_{23}+s_{14}^2 \\
&\qquad +2s_{14}s_{24}+s_{23}^2+2s_{23}s_{24}+s_{24}^2)\Big) \\
&+ \frac{2(16s_{12}+3s_{13}+3s_{14}+3s_{23}+3s_{24})}{s_{134}s_{234}} \\
&- \frac{4s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{134}s_{24}s_{34}^2} + \frac{4(2s_{12}^2+s_{12}s_{14}-s_{14}s_{23}-2s_{23}^2)}{s_{134}s_{24}s_{34}} \\
&+ \frac{4(s_{12}-s_{23})}{s_{134}s_{24}} \\
&+ \frac{4(-4s_{12}s_{13}-4s_{12}s_{14}+s_{13}^2+2s_{13}s_{14}-s_{13}s_{23}+s_{13}s_{24}+s_{14}^2+s_{14}s_{23}-s_{14}s_{24})}{s_{134}s_{34}^2} \\
&+ \frac{2(-8s_{12}+5s_{13}+5s_{14}-3s_{23}-3s_{24})}{s_{134}s_{34}} \\
&+ \frac{2s_{12}s_{34}(s_{12}+2s_{13}+s_{34})}{s_{14}s_{23}s_{234}s_{24}} \\
&- \frac{4s_{13}s_{24}^2(2s_{12}+s_{13}+s_{24})}{s_{14}s_{23}s_{234}s_{34}^2} \\
&+ \frac{2s_{24}(4s_{12}^2-2s_{12}s_{13}+2s_{12}s_{24}-5s_{13}^2-3s_{13}s_{24})}{s_{14}s_{23}s_{234}s_{34}} \\
&+ \frac{2(4s_{12}^2+4s_{12}s_{13}+3s_{12}s_{24}+2s_{12}s_{34}-2s_{13}^2-2s_{13}s_{24}-s_{13}s_{34})}{s_{14}s_{23}s_{234}} \\
&+ \frac{2s_{12}(s_{12}+s_{34})}{s_{14}s_{23}s_{24}} + \frac{6s_{13}s_{24}(2s_{12}+s_{13}+s_{24})}{s_{14}s_{23}s_{34}^2} \\
&+ \frac{8(-s_{12}^2+s_{13}^2+s_{13}s_{24}+s_{24}^2)}{s_{14}s_{23}s_{34}} + \frac{2(-2s_{12}+3s_{13}+3s_{24}+s_{34})}{s_{14}s_{23}} \\
&+ \frac{4s_{12}(s_{12}+2s_{13}+s_{34})}{s_{14}s_{234}s_{24}} - \frac{4s_{13}s_{24}(2s_{12}+s_{13}+s_{24})}{s_{14}s_{234}s_{34}^2} \\
&+ \frac{4(2s_{12}^2+s_{12}s_{24}-2s_{13}^2-s_{13}s_{24})}{s_{14}s_{234}s_{34}} + \frac{4(s_{12}-s_{13})}{s_{14}s_{234}} \\
&+ \frac{4s_{12}}{s_{14}s_{24}} - \frac{2s_{13}s_{24}}{s_{14}s_{34}^2} - \frac{2(2s_{13}+s_{24})}{s_{14}s_{34}} - \frac{2}{s_{14}} \\
&+ \frac{4s_{12}s_{34}^2}{s_{23}s_{234}^2s_{24}} - \frac{4s_{13}s_{24}^3}{s_{23}s_{234}^2s_{34}^2} \\
&+ \frac{2s_{24}^2(2s_{12}-3s_{13}+4s_{14})}{s_{23}s_{234}^2s_{34}} \\
&+ \frac{2(3s_{12}s_{24}+6s_{12}s_{34}-6s_{13}s_{24}-2s_{13}s_{34}+4s_{14}s_{24}+3s_{14}s_{34})}{s_{23}s_{234}^2} \\
&+ \frac{8s_{13}s_{24}^2}{s_{23}s_{234}s_{34}^2} + \frac{4s_{24}(-s_{12}+2s_{13}-2s_{14}+2s_{24})}{s_{23}s_{234}s_{34}} \\
&+ \frac{2(-s_{12}+4s_{13}-2s_{14}+2s_{24}+s_{34})}{s_{23}s_{234}} + \frac{2s_{12}}{s_{23}s_{24}}
\end{aligned}
$$

$$
\begin{aligned}
&-\frac{2s_{13}s_{24}}{s_{23}s_{34}^2}-\frac{2(s_{13}+2s_{24})}{s_{23}s_{34}}-\frac{2}{s_{23}}\\
&-\frac{4s_{14}s_{23}^3}{s_{234}^2s_{24}s_{34}^2}+\frac{2s_{23}^2(2s_{12}+4s_{13}-3s_{14})}{s_{234}^2s_{24}s_{34}}\\
&+\frac{2(3s_{12}s_{23}+6s_{12}s_{34}+4s_{13}s_{23}+3s_{13}s_{34}-6s_{14}s_{23}-2s_{14}s_{34})}{s_{234}^2s_{24}}\\
&+\frac{4(2s_{12}s_{23}^2+4s_{12}s_{23}s_{24}+2s_{12}s_{24}^2+s_{13}s_{23}^2+s_{13}s_{23}s_{24}-s_{13}s_{24}^2-s_{14}s_{23}^2+s_{14}s_{23}s_{24}+s_{14}s_{24}^2)}{s_{234}^2s_{34}^2}\\
&+\frac{2(10s_{12}s_{23}+10s_{12}s_{24}+9s_{13}s_{23}+2s_{13}s_{24}+2s_{14}s_{23}+9s_{14}s_{24})}{s_{234}^2s_{34}}\\
&+\frac{4(5s_{12}+s_{13}+s_{14})}{s_{234}^2}+\frac{8s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2}\\
&+\frac{4s_{23}(-s_{12}-2s_{13}+2s_{14}+2s_{23})}{s_{234}s_{24}s_{34}}\\
&+\frac{2(-s_{12}-2s_{13}+4s_{14}+2s_{23}+s_{34})}{s_{234}s_{24}}\\
&+\frac{4(-4s_{12}s_{23}-4s_{12}s_{24}-s_{13}s_{23}+s_{13}s_{24}+s_{14}s_{23}-s_{14}s_{24}+s_{23}^2+2s_{23}s_{24}+s_{24}^2)}{s_{234}s_{34}^2}\\
&+\frac{2(-8s_{12}-3s_{13}-3s_{14}+5s_{23}+5s_{24})}{s_{234}s_{34}}\\
&-\frac{2s_{14}s_{23}}{s_{24}s_{34}^2}-\frac{2(s_{14}+2s_{23})}{s_{24}s_{34}}-\frac{2}{s_{24}}\\
&+\frac{4(2s_{12}-s_{13}-s_{14}-s_{23}-s_{24})}{s_{34}^2}-\frac{8}{s_{34}}\Bigg]
\end{aligned}
\tag{A.48}
$$

$$
\begin{aligned}
\tilde{A}_4^{0,(2)}=\frac{1}{s_{14}s_{23}\mathcal{D}_{4,gg}}\Bigg[&-\frac{4s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}}-\frac{2\,(3s_{12}s_{34}-3s_{14}s_{23}+2s_{14}s_{24}+3s_{24}s_{34})}{s_{13}s_{134}^2}-\frac{2s_{12}s_{34}(s_{24}+2s_{34})}{s_{13}s_{134}s_{14}s_{23}}\\
&-\frac{2s_{12}s_{34}(s_{23}+2s_{34})}{s_{13}s_{134}s_{14}s_{24}}\\
&-\frac{2s_{12}\left(2s_{14}s_{24}+s_{14}s_{34}+s_{24}s_{34}+3s_{34}^2\right)}{s_{13}s_{134}s_{23}s_{234}}-\frac{2s_{12}(2s_{24}+s_{34})}{s_{13}s_{134}s_{23}}\\
&-\frac{2s_{34}(s_{12}s_{34}-s_{14}s_{23})}{s_{13}s_{134}s_{234}s_{24}}+\frac{4s_{14}^2(s_{23}+s_{24})}{s_{13}s_{134}s_{234}s_{34}}\\
&-\frac{2\,(2s_{12}s_{14}+2s_{12}s_{34}-s_{14}^2-2s_{14}s_{23}-s_{14}s_{24}-3s_{14}s_{34})}{s_{13}s_{134}s_{234}}\\
&+\frac{4s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}}-\frac{2\left(2s_{12}s_{23}+3s_{12}s_{34}-3s_{14}s_{23}-s_{23}^2-2s_{23}s_{34}\right)}{s_{13}s_{134}s_{24}}\\
&-\frac{4s_{14}(2s_{14}-s_{24})}{s_{13}s_{134}s_{34}}-\frac{2(3s_{14}+2s_{23}-s_{24}+2s_{34})}{s_{13}s_{134}}\\
&-\frac{6s_{12}s_{34}^2}{s_{13}s_{14}s_{23}s_{24}}-\frac{2s_{12}s_{34}}{s_{13}s_{14}s_{23}}-\frac{2s_{12}s_{34}}{s_{13}s_{14}s_{24}}\\
&-\frac{2s_{12}s_{34}(s_{14}+2s_{34})}{s_{13}s_{23}s_{234}s_{24}}-\frac{2s_{12}(2s_{14}+s_{34})}{s_{13}s_{23}s_{234}}\\
&-\frac{2s_{12}s_{34}}{s_{13}s_{23}s_{24}}-\frac{4s_{12}}{s_{13}s_{23}}+\frac{4s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}}\\
&-\frac{2\,(2s_{12}s_{14}+3s_{12}s_{34}-s_{14}^2-3s_{14}s_{23}-2s_{14}s_{34})}{s_{13}s_{234}s_{24}}\\
&+\frac{4s_{14}^2}{s_{13}s_{234}s_{34}}+\frac{2s_{14}}{s_{13}s_{234}}-\frac{4(s_{14}^2+s_{23}^2)}{s_{13}s_{24}s_{34}}\\
&-\frac{2(s_{14}+s_{23})}{s_{13}s_{24}}+\frac{4s_{14}}{s_{13}s_{34}}+\frac{2}{s_{13}}
\end{aligned}
$$

$$
\begin{aligned}
&-\frac{4s_{13}^2 s_{23}}{s_{134}^2 s_{14} s_{34}}-\frac{2\left(3s_{12}s_{34}+2s_{13}s_{23}-3s_{13}s_{24}+3s_{23}s_{34}\right)}{s_{134}^2 s_{14}}\\
&-\frac{4\left(2s_{13}s_{23}+s_{13}s_{24}+s_{14}s_{23}+2s_{14}s_{24}\right)}{s_{134}^2 s_{34}}-\frac{2(2s_{12}+s_{23}+s_{24})}{s_{134}^2}\\
&-\frac{2s_{34}(s_{12}s_{34}-s_{13}s_{24})}{s_{134}s_{14}s_{23}s_{234}}+\frac{4s_{13}s_{24}^2}{s_{134}s_{14}s_{23}s_{34}}\\
&-\frac{2\left(2s_{12}s_{24}+3s_{12}s_{34}-3s_{13}s_{24}-s_{24}^2-2s_{24}s_{34}\right)}{s_{134}s_{14}s_{23}}\\
&-\frac{2s_{12}\left(2s_{13}s_{23}+s_{13}s_{34}+s_{23}s_{34}+3s_{34}^2\right)}{s_{134}s_{14}s_{234}s_{24}}\\
&+\frac{4s_{13}^2(s_{23}+s_{24})}{s_{134}s_{14}s_{234}s_{34}}-\frac{2\left(2s_{12}s_{13}+2s_{12}s_{34}-s_{13}^2-s_{13}s_{23}-2s_{13}s_{24}-3s_{13}s_{34}\right)}{s_{134}s_{14}s_{234}}\\
&-\frac{2s_{12}(2s_{23}+s_{34})}{s_{134}s_{14}s_{24}}-\frac{4s_{13}(2s_{13}-s_{23})}{s_{134}s_{14}s_{34}}\\
&-\frac{2(3s_{13}-s_{23}+2s_{24}+2s_{34})}{s_{134}s_{14}}+\frac{4s_{24}^2(s_{13}+s_{14})}{s_{134}s_{23}s_{234}s_{34}}\\
&-\frac{2\left(2s_{12}s_{24}+2s_{12}s_{34}-2s_{13}s_{24}-s_{14}s_{24}-s_{24}^2-3s_{24}s_{34}\right)}{s_{134}s_{23}s_{234}}\\
&+\frac{4s_{24}^2}{s_{134}s_{23}s_{34}}+\frac{2s_{24}}{s_{134}s_{23}}+\frac{4s_{23}^2(s_{13}+s_{14})}{s_{134}s_{234}s_{24}s_{34}}\\
&-\frac{2\left(2s_{12}s_{23}+2s_{12}s_{34}-s_{13}s_{23}-2s_{14}s_{23}-s_{23}^2-3s_{23}s_{34}\right)}{s_{134}s_{234}s_{24}}\\
&+\frac{8(s_{13}+s_{14})(s_{23}+s_{24})}{s_{134}s_{234}s_{34}}-\frac{4\left(4s_{12}+2s_{13}+2s_{14}+2s_{23}+2s_{24}+s_{34}\right)}{s_{134}s_{234}}\\
&+\frac{4s_{23}^2}{s_{134}s_{24}s_{34}}+\frac{2s_{23}}{s_{134}s_{24}}-\frac{4(2s_{13}+2s_{14}-s_{23}-s_{24})}{s_{134}s_{34}}-\frac{4}{s_{134}}\\
&-\frac{2s_{12}s_{34}(s_{13}+2s_{34})}{s_{14}s_{23}s_{234}s_{24}}+\frac{4s_{13}^2 s_{24}}{s_{14}s_{23}s_{234}s_{34}}\\
&-\frac{2\left(2s_{12}s_{13}+3s_{12}s_{34}-s_{13}^2-3s_{13}s_{24}-2s_{13}s_{34}\right)}{s_{14}s_{23}s_{234}}\\
&-\frac{2s_{12}s_{34}}{s_{14}s_{23}s_{24}}-\frac{4(s_{13}^2+s_{24}^2)}{s_{14}s_{23}s_{34}}-\frac{2(s_{13}+s_{24})}{s_{14}s_{23}}\\
&-\frac{2s_{12}(2s_{13}+s_{34})}{s_{14}s_{234}s_{24}}+\frac{4s_{13}^2}{s_{14}s_{234}s_{34}}+\frac{2s_{13}}{s_{14}s_{234}}\\
&-\frac{4s_{12}}{s_{14}s_{24}}+\frac{4s_{13}}{s_{14}s_{34}}+\frac{2}{s_{14}}\\
&-\frac{4s_{14}s_{24}^2}{s_{23}s_{234}^2 s_{34}}-\frac{2\left(3s_{12}s_{34}-3s_{13}s_{24}+2s_{14}s_{24}+3s_{14}s_{34}\right)}{s_{23}s_{234}^2}\\
&+\frac{4s_{24}(s_{14}-2s_{24})}{s_{23}s_{234}s_{34}}-\frac{2(2s_{13}-s_{14}+3s_{24}+2s_{34})}{s_{23}s_{234}}\\
&+\frac{4s_{24}}{s_{23}s_{34}}+\frac{2}{s_{23}}-\frac{4s_{13}s_{23}^2}{s_{234}^2 s_{24}s_{34}}\\
&-\frac{2\left(3s_{12}s_{34}+2s_{13}s_{23}+3s_{13}s_{34}-3s_{14}s_{23}\right)}{s_{234}^2 s_{24}}\\
&-\frac{4\left(2s_{13}s_{23}+s_{13}s_{24}+s_{14}s_{23}+2s_{14}s_{24}\right)}{s_{234}^2 s_{34}}-\frac{2(2s_{12}+s_{13}+s_{14})}{s_{234}^2}\\
&+\frac{4s_{23}(s_{13}-2s_{23})}{s_{234}s_{24}s_{34}}+\frac{2(s_{13}-2s_{14}-3s_{23}-2s_{34})}{s_{234}s_{24}}\\
&\left.+\frac{4(s_{13}+s_{14}-2s_{23}-2s_{24})}{s_{234}s_{34}}-\frac{4}{s_{234}}+\frac{4s_{23}}{s_{24}s_{34}}+\frac{2}{s_{24}}\right]
\end{aligned}
\tag{A.49}
$$

$$\tilde{A}_4^{0,(3)} = \frac{1}{s_{14}s_{23}\mathcal{D}_{4,gg}}\Bigg[-\frac{2s_{12}s_{34}^2}{s_{13}s_{134}^2s_{14}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{14}s_{23}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{14}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{23}s_{234}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{14}s_{23}s_{24}}$$
$$- \frac{2s_{12}s_{34}^2}{s_{13}s_{23}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{134}s_{14}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{14}s_{23}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{23}s_{234}^2s_{24}}$$
$$+ \frac{2s_{34}(s_{23}+s_{24})}{s_{13}s_{134}^2} + \frac{2s_{14}s_{34}}{s_{13}s_{134}s_{234}} + \frac{2s_{23}s_{34}}{s_{13}s_{134}s_{24}}$$
$$+ \frac{2s_{14}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{34}(s_{23}+s_{24})}{s_{134}^2s_{14}} + \frac{2s_{24}s_{34}}{s_{134}s_{14}s_{23}}$$
$$+ \frac{2s_{13}s_{34}}{s_{134}s_{14}s_{234}} + \frac{2s_{24}s_{34}}{s_{134}s_{23}s_{234}} + \frac{2s_{23}s_{34}}{s_{134}s_{234}s_{24}}$$
$$+ \frac{2s_{13}s_{34}}{s_{14}s_{23}s_{234}} + \frac{2s_{34}(s_{13}+s_{14})}{s_{23}s_{234}^2} + \frac{2s_{34}(s_{13}+s_{14})}{s_{234}^2s_{24}}$$
$$+ \frac{6s_{34}}{s_{13}s_{134}} + \frac{4s_{34}}{s_{13}s_{24}} + \frac{6s_{34}}{s_{134}s_{14}} + \frac{8s_{34}}{s_{134}s_{234}} + \frac{4s_{34}}{s_{14}s_{23}} + \frac{6s_{34}}{s_{23}s_{234}} + \frac{6s_{34}}{s_{234}s_{24}}\Bigg] \tag{A.50}$$

using,

$$\mathcal{D}_{4,gg} = s_{13}s_{24}s_{34}^2s_{134}^2s_{234}^2$$

By running `BuildAndIntegrateAntenna[A,4,0,Component->Subleading]` we get the integrated result,

$$\begin{aligned}\tilde{\mathcal{A}}_4^0 = \frac{1}{\epsilon^4} + \frac{3}{\epsilon^3} + \frac{1}{\epsilon^2}\left(13 - \frac{3\pi^2}{2}\right) + \frac{1}{\epsilon}\left(\frac{845}{16} - \frac{9\pi^2}{2} - \frac{80\zeta_3}{3}\right)\\ + \frac{6921}{32} - \frac{473\pi^2}{24} + \frac{17\pi^4}{72} - 80\zeta_3 + O(\epsilon).\end{aligned} \tag{A.51}$$

A.3.10 B_4^0

Running `BuildAntenna[B,4,0]`, we get the built result,

$$B_4^0 = \epsilon^2\Bigg(\frac{4s_{12}}{s_{134}s_{234}} + \frac{2s_{12}}{s_{134}^2} + \frac{2s_{12}}{s_{234}^2} - \frac{2s_{14}s_{23}}{s_{134}^2s_{34}} - \frac{2s_{13}s_{23}}{s_{134}^2s_{34}} - \frac{2s_{14}s_{24}}{s_{134}^2s_{34}}$$
$$- \frac{4s_{14}s_{24}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{13}s_{24}}{s_{134}^2s_{34}} - \frac{2s_{13}s_{24}}{s_{234}^2s_{34}} - \frac{2s_{14}s_{24}}{s_{234}^2s_{34}}$$
$$- \frac{4s_{14}s_{23}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{23}}{s_{134}s_{234}s_{34}} - \frac{2s_{13}s_{23}}{s_{234}^2s_{34}} - \frac{2s_{14}s_{23}}{s_{234}^2s_{34}}\Bigg)$$
$$+ \epsilon\Bigg(\frac{4s_{12}^2}{s_{134}s_{234}s_{34}} + \frac{4s_{12}}{s_{134}s_{234}} + \frac{2s_{12}s_{14}}{s_{134}^2s_{34}} + \frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}}$$
$$+ \frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{13}}{s_{134}^2s_{34}} + \frac{2s_{12}s_{23}}{s_{234}^2s_{34}} - \frac{2s_{12}}{s_{134}^2} - \frac{2s_{12}}{s_{234}^2}$$
$$+ \frac{4s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2} - \frac{4s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{4s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}}{s_{134}^2s_{34}} + \frac{4s_{13}s_{23}}{s_{134}^2s_{34}}$$
$$+ \frac{4s_{14}s_{24}}{s_{134}^2s_{34}} + \frac{4s_{14}s_{24}}{s_{134}s_{234}s_{34}} + \frac{2s_{13}s_{24}}{s_{134}^2s_{34}} + \frac{2s_{13}s_{24}}{s_{234}^2s_{34}} + \frac{4s_{14}s_{24}}{s_{234}^2s_{34}} + \frac{4s_{13}s_{23}}{s_{134}s_{234}s_{34}}$$
$$+ \frac{4s_{13}s_{23}}{s_{234}^2s_{34}} + \frac{2s_{14}s_{23}}{s_{234}^2s_{34}} - \frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{24}^2}{s_{234}^2s_{34}^2} - \frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2}$$
$$+ \frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2}$$
$$- \frac{2s_{13}^2s_{24}}{s_{134}^2s_{34}^2} + \frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} - \frac{2s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2}$$
$$+ \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2}\Bigg)$$
$$- \frac{4s_{12}}{s_{134}s_{234}} - \frac{2s_{12}s_{14}}{s_{134}^2s_{34}} - \frac{2s_{13}s_{23}}{s_{134}^2s_{34}} - \frac{2s_{14}s_{24}}{s_{134}^2s_{34}} - \frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{234}^2s_{34}}$$

$$
\begin{aligned}
&-\frac{2s_{14}s_{24}}{s_{234}^2s_{34}}-\frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}}-\frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}}-\frac{4s_{12}^2}{s_{134}s_{234}s_{34}}-\frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}}-\frac{2s_{12}s_{13}}{s_{134}^2s_{34}}\\
&-\frac{2s_{12}s_{23}}{s_{234}^2s_{34}}-\frac{2s_{13}s_{23}}{s_{234}^2s_{34}}+\frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2}+\frac{2s_{13}s_{24}^2}{s_{234}^2s_{34}^2}-\frac{4s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2}+\frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2}\\
&-\frac{2s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2}-\frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2}-\frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}\\
&+\frac{2s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2}+\frac{4s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2}+\frac{2s_{13}^2s_{24}}{s_{134}^2s_{34}^2}-\frac{4s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2}-\frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2}-\frac{2s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2}\\
&+\frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2}+\frac{2s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2}+\frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}+\frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2}.
\end{aligned}
\tag{A.52}
$$

By running `BuildAndIntegrateAntenna[B,4,0]` we get the integrated result,

$$
\mathcal{B}_4^0=-\frac{1}{12\epsilon^3}-\frac{7}{18\epsilon^2}-\frac{1}{\epsilon}\frac{11\left(37-3\pi^2\right)}{216}-\frac{11753}{1296}+\frac{77\pi^2}{108}+\frac{67\zeta_3}{18}+O(\epsilon).
\tag{A.53}
$$

A.3.11 C_4^0

Running `BuildAntenna[C,4,0]`, we get the built result,

$$
\begin{aligned}
C_4^0=\epsilon^3\Bigg(&\frac{s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}}+\frac{s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}}+\frac{s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}}+\frac{s_{13}s_{24}^2}{s_{23}s_{234}^2s_{34}}+\frac{s_{13}s_{24}}{s_{123}s_{134}s_{23}}\\
&-\frac{s_{12}s_{24}}{s_{123}s_{134}s_{23}}+\frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}}-\frac{s_{12}s_{24}}{s_{123}s_{23}s_{234}}-\frac{s_{12}s_{24}}{s_{134}s_{23}s_{234}}+\frac{s_{13}s_{24}}{s_{134}s_{23}s_{234}}\\
&-\frac{s_{14}s_{24}}{s_{123}s_{134}s_{34}}+\frac{s_{13}s_{24}}{s_{123}s_{134}s_{34}}+\frac{s_{13}s_{24}}{s_{123}s_{234}s_{34}}-\frac{s_{14}s_{24}}{s_{123}s_{234}s_{34}}-\frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}}\\
&+\frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}}+\frac{s_{13}s_{24}}{s_{234}^2s_{34}}-\frac{s_{14}s_{24}}{s_{234}^2s_{34}}-\frac{s_{12}s_{24}}{s_{23}s_{234}^2}+\frac{s_{13}s_{24}}{s_{23}s_{234}^2}\\
&+\frac{s_{14}}{s_{123}s_{134}}-\frac{s_{12}s_{34}}{s_{123}s_{134}s_{23}}-\frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}}-\frac{s_{12}s_{34}}{s_{134}s_{23}s_{234}}-\frac{s_{12}s_{34}}{s_{23}s_{234}^2}\\
&+\frac{2s_{13}}{s_{123}s_{134}}+\frac{s_{12}}{s_{123}s_{134}}+\frac{2s_{13}}{s_{123}s_{234}}+\frac{s_{14}}{s_{123}s_{234}}+\frac{s_{14}}{s_{134}s_{234}}\\
&+\frac{s_{12}}{s_{123}s_{234}}+\frac{s_{12}}{s_{134}s_{234}}+\frac{2s_{13}}{s_{134}s_{234}}-\frac{s_{14}s_{23}}{s_{123}s_{134}s_{34}}-\frac{s_{14}s_{23}}{s_{123}s_{234}s_{34}}\\
&-\frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}}-\frac{s_{14}s_{23}}{s_{234}^2s_{34}}+\frac{s_{12}}{s_{234}^2}+\frac{2s_{13}}{s_{234}^2}+\frac{s_{14}}{s_{234}^2}\Bigg)\\
+\epsilon^2\Bigg(&-\frac{s_{12}^2}{s_{123}s_{134}s_{23}}-\frac{s_{12}^2}{s_{134}s_{23}s_{234}}-\frac{3s_{12}s_{24}}{s_{123}s_{134}s_{23}}-\frac{2s_{12}s_{24}}{s_{123}s_{23}s_{234}}-\frac{2s_{12}s_{24}}{s_{134}s_{23}s_{234}}\\
&-\frac{2s_{12}s_{34}}{s_{123}s_{134}s_{23}}-\frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}}+\frac{2s_{12}s_{34}}{s_{23}s_{234}^2}+\frac{2s_{12}}{s_{123}s_{134}}+\frac{s_{12}s_{14}}{s_{123}s_{134}s_{23}}\\
&-\frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}}-\frac{2s_{12}}{s_{123}s_{234}}+\frac{s_{12}}{s_{134}s_{234}}-\frac{s_{12}s_{13}}{s_{123}s_{23}s_{234}}+\frac{s_{12}s_{14}}{s_{123}s_{23}s_{234}}\\
&-\frac{s_{12}s_{13}}{s_{134}s_{23}s_{234}}+\frac{s_{12}s_{14}}{s_{123}s_{134}s_{34}}+\frac{s_{12}s_{13}s_{24}}{s_{123}s_{134}s_{23}s_{34}}-\frac{2s_{12}s_{24}}{s_{123}s_{234}s_{34}}+\frac{s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}}\\
&-\frac{2s_{12}s_{24}}{s_{234}^2s_{34}}+\frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}}-\frac{2s_{12}}{s_{234}^2}+\frac{2s_{14}}{s_{123}s_{134}}+\frac{2s_{13}s_{24}}{s_{123}s_{134}s_{23}}\\
&+\frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}}-\frac{2s_{14}s_{24}}{s_{134}s_{23}s_{234}}-\frac{2s_{13}s_{24}}{s_{23}s_{234}^2}-\frac{2s_{14}s_{24}}{s_{23}s_{234}^2}-\frac{2s_{13}}{s_{123}s_{234}}\\
&+\frac{s_{14}}{s_{123}s_{234}}-\frac{2s_{14}}{s_{134}s_{234}}-\frac{2s_{13}}{s_{134}s_{234}}-\frac{s_{14}^2}{s_{123}s_{134}s_{34}}+\frac{3s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}}\\
&+\frac{2s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}}+\frac{2s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}}-\frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}}-\frac{2s_{14}s_{23}}{s_{123}s_{134}s_{34}}-\frac{3s_{14}s_{24}}{s_{123}s_{134}s_{34}}\\
&+\frac{2s_{13}s_{24}}{s_{123}s_{134}s_{34}}+\frac{s_{13}s_{14}s_{24}}{s_{123}s_{134}s_{23}s_{34}}+\frac{2s_{13}^2s_{24}}{s_{123}s_{134}s_{23}s_{34}}-\frac{2s_{14}s_{24}}{s_{123}s_{234}s_{34}}-\frac{2s_{14}s_{24}}{s_{134}s_{234}s_{34}}
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}}+\frac{s_{13}^2s_{24}}{s_{123}s_{23}s_{234}s_{34}}+\frac{s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}}+\frac{s_{13}^2s_{24}}{s_{134}s_{23}s_{234}s_{34}}-\frac{2s_{13}s_{24}}{s_{234}^2s_{34}}\\
&-\frac{s_{14}^2}{s_{123}s_{234}s_{34}}-\frac{s_{13}s_{14}}{s_{123}s_{234}s_{34}}-\frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}}-\frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}}+\frac{2s_{14}s_{23}}{s_{234}^2s_{34}}\\
&-\frac{4s_{13}}{s_{234}^2}-\frac{2s_{14}}{s_{234}^2}\Bigg)\\
+\epsilon\Bigg(&\frac{s_{12}^2}{s_{123}s_{134}s_{23}}-\frac{s_{12}^2}{s_{123}s_{23}s_{234}}+\frac{2s_{12}^2}{s_{134}s_{23}s_{234}}+\frac{2s_{12}s_{24}}{s_{123}s_{134}s_{23}}+\frac{s_{12}s_{24}}{s_{123}s_{23}s_{234}}\\
&+\frac{s_{12}s_{24}}{s_{134}s_{23}s_{234}}+\frac{3s_{12}s_{24}}{s_{23}s_{234}^2}+\frac{s_{12}s_{34}}{s_{123}s_{134}s_{23}}-\frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}}+\frac{2s_{12}s_{34}}{s_{134}s_{23}s_{234}}\\
&-\frac{s_{12}s_{34}}{s_{23}s_{234}^2}-\frac{s_{12}}{s_{123}s_{134}}-\frac{s_{12}s_{14}}{s_{123}s_{134}s_{23}}-\frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}}+\frac{s_{12}}{s_{134}s_{234}}\\
&+\frac{s_{12}s_{13}}{s_{123}s_{23}s_{234}}-\frac{2s_{12}s_{14}}{s_{123}s_{23}s_{234}}+\frac{s_{12}s_{14}}{s_{134}s_{23}s_{234}}+\frac{s_{12}s_{13}}{s_{134}s_{23}s_{234}}-\frac{s_{12}s_{14}}{s_{123}s_{134}s_{34}}\\
&-\frac{s_{12}s_{13}s_{24}}{s_{123}s_{134}s_{23}s_{34}}+\frac{2s_{12}s_{24}}{s_{123}s_{234}s_{34}}+\frac{s_{12}s_{13}s_{24}}{s_{123}s_{23}s_{234}s_{34}}-\frac{2s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}}+\frac{4s_{12}s_{24}}{s_{234}^2s_{34}}\\
&-\frac{2s_{12}s_{13}s_{14}}{s_{123}s_{134}s_{23}s_{34}}+\frac{s_{12}s_{14}}{s_{123}s_{234}s_{34}}-\frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}}+\frac{s_{12}}{s_{234}^2}-\frac{s_{14}}{s_{123}s_{134}}\\
&-\frac{s_{13}s_{24}}{s_{123}s_{134}s_{23}}+\frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}}+\frac{2s_{14}s_{24}}{s_{134}s_{23}s_{234}}+\frac{s_{13}s_{24}}{s_{23}s_{234}^2}+\frac{4s_{14}s_{24}}{s_{23}s_{234}^2}\\
&-\frac{4s_{13}}{s_{123}s_{134}}+\frac{s_{14}}{s_{123}s_{234}}+\frac{s_{14}^2}{s_{123}s_{134}s_{34}}-\frac{2s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}}-\frac{s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}}\\
&-\frac{s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}}-\frac{3s_{13}s_{24}^2}{s_{23}s_{234}^2s_{34}}-\frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}}+\frac{s_{14}s_{23}}{s_{123}s_{134}s_{34}}+\frac{2s_{14}s_{24}}{s_{123}s_{134}s_{34}}\\
&-\frac{s_{13}s_{24}}{s_{123}s_{134}s_{34}}-\frac{s_{13}s_{14}s_{24}}{s_{123}s_{134}s_{23}s_{34}}+\frac{s_{14}s_{24}}{s_{123}s_{234}s_{34}}+\frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}}+\frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}}\\
&-\frac{s_{13}^2s_{24}}{s_{123}s_{23}s_{234}s_{34}}-\frac{2s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}}+\frac{s_{13}s_{14}s_{24}}{s_{134}s_{23}s_{234}s_{34}}-\frac{s_{13}^2s_{24}}{s_{134}s_{23}s_{234}s_{34}}+\frac{s_{13}s_{24}}{s_{234}^2s_{34}}\\
&+\frac{3s_{14}s_{24}}{s_{234}^2s_{34}}+\frac{2s_{14}^2}{s_{123}s_{234}s_{34}}-\frac{s_{14}^2}{s_{134}s_{234}s_{34}}+\frac{s_{13}s_{14}}{s_{123}s_{234}s_{34}}+\frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}}\\
&+\frac{2s_{14}s_{23}}{s_{123}s_{234}s_{34}}-\frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}}-\frac{s_{14}s_{23}}{s_{234}^2s_{34}}+\frac{2s_{13}}{s_{234}^2}+\frac{s_{14}}{s_{234}^2}\Bigg)\\
&-\frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}}-\frac{s_{13}s_{24}}{s_{134}s_{23}s_{234}}-\frac{2s_{12}s_{24}}{s_{23}s_{234}^2}-\frac{2s_{14}s_{24}}{s_{23}s_{234}^2}+\frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}}\\
&-\frac{s_{12}s_{34}}{s_{134}s_{23}s_{234}}+\frac{2s_{13}}{s_{123}s_{134}}+\frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}}-\frac{s_{14}}{s_{123}s_{234}}+\frac{s_{14}}{s_{134}s_{234}}\\
&+\frac{s_{12}}{s_{123}s_{234}}-\frac{s_{12}}{s_{134}s_{234}}+\frac{s_{12}s_{14}}{s_{123}s_{23}s_{234}}-\frac{s_{12}s_{14}}{s_{134}s_{23}s_{234}}+\frac{s_{12}^2}{s_{123}s_{23}s_{234}}\\
&-\frac{s_{12}^2}{s_{134}s_{23}s_{234}}+\frac{2s_{13}s_{24}^2}{s_{23}s_{234}^2s_{34}}+\frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}}-\frac{s_{13}s_{24}}{s_{123}s_{234}s_{34}}-\frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}}\\
&-\frac{s_{12}s_{13}s_{24}}{s_{123}s_{23}s_{234}s_{34}}+\frac{s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}}-\frac{s_{13}s_{14}s_{24}}{s_{134}s_{23}s_{234}s_{34}}+\frac{s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}}-\frac{2s_{12}s_{24}}{s_{234}^2s_{34}}\\
&-\frac{2s_{14}s_{24}}{s_{234}^2s_{34}}+\frac{2s_{12}s_{13}s_{14}}{s_{123}s_{134}s_{23}s_{34}}-\frac{s_{14}^2}{s_{123}s_{234}s_{34}}+\frac{s_{14}^2}{s_{134}s_{234}s_{34}}-\frac{s_{12}s_{14}}{s_{123}s_{234}s_{34}}\\
&+\frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}}-\frac{s_{14}s_{23}}{s_{123}s_{234}s_{34}}+\frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}}. \qquad \text{(A.54)}
\end{aligned}
$$

By running `BuildAndIntegrateAntenna[C,4,0]` we get the integrated result,

$$
C_4^0=\frac{1}{\epsilon}\frac{\left(-13+2\pi^2-8\zeta_3\right)}{32}-\frac{339}{64}+\frac{17\pi^2}{48}-\frac{\pi^4}{45}+\frac{21\zeta_3}{8}+O(\epsilon). \qquad \text{(A.55)}
$$

A.4 Exploratory Massive A_3^0

Running `BuildAntenna[A,3,0,quarkMass->mQ]`, we get the built result,

$$
\begin{aligned}
A_3^{0,(m_q)} = &\frac{1}{s_{13}s_{23}(2m_q^2+q^2)}\Bigg[\frac{1}{\epsilon^2}\frac{m_q^2}{\left(2m_q^2+q^2\right)^2}\left(m_q^2\left(s_{13}^2+4s_{13}s_{23}+s_{23}^2\right)+s_{13}s_{23}s_{123}\right)\\
&-\frac{1}{\epsilon}\frac{1}{4(2m_q^2+q^2)}\left(2m_q^2\left(3s_{13}^2+4s_{13}s_{23}+3s_{23}^2\right)+s_{12}\left(s_{13}+s_{23}\right)^2+\left(s_{13}+s_{23}\right)^3\right)\\
&-\frac{1}{4s_{13}s_{23}}\Bigg(-8m_q^4\left(s_{13}^2+s_{23}^2\right)-2m_q^2\left(s_{12}\left(s_{13}^2-s_{13}s_{23}+s_{23}^2\right)+\left(s_{13}+s_{23}\right)^3\right)\\
&+2s_{12}s_{13}s_{23}s_{123}+s_{13}s_{23}\left(s_{13}^2+s_{23}^2\right)\Bigg)\Bigg]+O\left(\epsilon^3\right). \qquad \text{(A.56)}
\end{aligned}
$$

By running `BuildAndIntegrateAntenna[A,3,0,quarkMass->mQ]` we get the integrated result,

$$
\begin{aligned}
\mathcal{A}_3^{0,(m_q)} = &\frac{2^{-2(1+\epsilon)}(1-\epsilon)\pi^{\epsilon-2}\left(1-4m_q^2\right)^{1-2\epsilon}}{\epsilon(2\epsilon-1)m_q^2(\epsilon-2m_q^2-1)}\Bigg[-\frac{24\Gamma(4-3\epsilon)\Gamma(3-2\epsilon)}{\Gamma(8-6\epsilon)}(\epsilon-1)\left(4m_q^2-1\right)\\
&\cdot\left(\epsilon^2\left(8m_q^2-2\right)+\epsilon\left(3-4m_q^2\right)-4m_q^2-2\right){}_2F_1\left(\frac{1}{2},3-2\epsilon;\frac{9}{2}-3\epsilon;1-4m_q^2\right)\\
&-\frac{4\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)}{\Gamma(6-6\epsilon)}\Bigg(4m_q^4+\epsilon^3\left(8m_q^2-2\right)+6m_q^2+\epsilon^2\left(16m_q^4-12m_q^2+5\right)\\
&-\epsilon\left(16m_q^4+4m_q^2+5\right)+2\Bigg){}_2F_1\left(\frac{1}{2},2-2\epsilon;\frac{7}{2}-3\epsilon;1-4m_q^2\right)\Bigg]. \qquad \text{(A.57)}
\end{aligned}
$$

Appendix B

Supporting Derivations

B.1 The Massive Extension

Up until now every part of the theory has been explained generally, but with a clear preference for the massless regime. That is because the vast majority of the present work's ensuing chapters will focus on that regime. But a still significant part of it will explore an extension of the discussed methods into the massive regime.

The fundamental change is the massive on-shell condition; the remainder of this section traces its consequences for the antenna construction and integration. Since final-state particles now have mass, their squared momenta are not set to zero, they are set to their squared mass.

$$p_q^2 = p_{\bar{q}}^2 = 0 \xrightarrow{\text{Massive Regime}} p_q^2 = p_{\bar{q}}^2 = m_q^2. \tag{B.1}$$

where p_i represents the momentum and m_i represents the mass of particle i.

The antenna subtraction formalism and the reduction architecture that was developed and described throughout Chapter 2 are retained, however. What changes is the intermediate steps and kinematics: the massive on-shell condition changes the kinematics themselves and the unresolved limits, with soft radiation remaining singular but heavy-quark collinear behaviour becoming quasi-collinear. Massive integration is also different as the massive phase-scale is not equal to its massless counterpart. A new scale of m_q^2/q^2 is also introduced and leads to distinct cut-integral families and master integrals [37].

B.1.1 Massive Kinematics and Infrared Structure

In Subsection 2.2.1, the Mandelstam invariants were introduced for the massless regime. The general definition is,

$$\begin{aligned} s_{ij} = (p_i + p_j)^2 &= p_i^2 + 2p_i \cdot p_j + p_j^2 \\ &= m_i^2 + m_j^2 + 2p_i \cdot p_j. \end{aligned} \tag{B.2}$$

Through momentum conservation, which states that $q = \sum_i p_i$, we also have to reconsider the q^2 expression, as,

$$q^2 = \sum_i m_i^2 + 2\sum_{i<j} p_i \cdot p_j. \tag{B.3}$$

Soft radiation remains singular in the massive regime and therefore continues to generate explicit infrared poles after integration. A non-zero mass regulates collinear limits involving a massive parton, whereas collinear limits between two massless partons remain singular. If the two final-state collinear partons are massless, then the limit follows the massless collinear limit from the massless regime. If the two partons are massive, no strict collinear singularity exists. But if one of the collinear partons is massive, consider that particle i of mass m_i and energy E_i, and the other is massless, consider that particle j, such that $m_j = 0$, then,

$$p_i \cdot p_j = E_i E_j (1 - \beta_i \cos\theta), \quad \beta_i = \sqrt{1 - \frac{m_i^2}{E_i^2}} < 1. \tag{B.4}$$

As the angle between the partons θ tends towards zero, the mass prevents the invariant from reaching the collinear singular surface,

$$\lim_{\theta\to 0} p_i \cdot p_j = E_i E_j (1 - \beta_i) \neq 0. \tag{B.5}$$

This regulated collinear region still carries physically important information, particularly in the joint high-energy and small-angle region, $E_i >> m_i$ and $\theta << 1$,

$$p_i \cdot p_j \sim \frac{E_i E_j}{2}\left(\theta^2 + \frac{m_i^2}{E_i^2}\right) \tag{B.6}$$

and consequently since,

$$\int_0^{\theta_{\max}^2} d\theta^2 \, \frac{1}{p_i \cdot p_j} \sim \int_0^{\theta_{\max}^2} d\theta^2 \left(\theta^2 + \frac{m_i^2}{E_i^2}\right)^{-1} \sim \ln\left(\frac{E_i^2}{m_i^2}\right) \tag{B.7}$$

the massive-massless quasi-collinear region replaces the strict collinear pole by a mass logarithm. The quasi-collinear limit, then, is the limit in which both the mass m_i and the opening angle θ become small together. It is through this limit that the massive antennae recover the massless splitting behaviour, as $m_i \to 0$ [37].

Mass does not regulate UV divergences, as $\ell^2 >> m_i^2, p_i^2$. It is noteworthy, however, that in a theory with massive quarks, the quark mass and quark field must be renormalised as,

$$g_{s,0} = \mu^{\epsilon} Z_g g_s, \quad m_{q,0} = Z_m m_q, \quad \psi_{q,0} = \sqrt{Z_2^q}\psi_q. \tag{B.8}$$

B.1.2 Massive Phase-Space Integration and Reduction Methods

For final-state particles of momenta p_i and masses m_i, the phase-space becomes,

$$d\Phi_n(q; p_1, ..., p_n) = (2\pi)^d \delta^{(d)}\left(q - \sum_{i=1}^{n} p_i\right) \prod_{i=1}^{n} \frac{d^d p_i}{(2\pi)^{d-1}} \delta^+\left(p_i^2 - m_i^2\right). \tag{B.9}$$

At the level of the phase-space measure, the formal change from the massless case, as described in Subsubsection 2.5.2, is,

$$\delta^+(p_i^2) \;\to\; \delta^+(p_i^2 - m_i^2) \tag{B.10}$$

which physically imposes that $p_i^2 = m_i^2$ and $p_i^0 > 0$.

Phase-space factorisation still holds for the massive regime, but the momentum mapping must preserve the relevant mass-shell conditions. So,

$$d\Phi_{m+r}(\{p\}; q) = d\Phi_m(\{\tilde{p}\}; q) d\Phi_X. \tag{B.11}$$

For a final-final antenna with hard radiator i, k and unresolved particles between them, this is schematically,

$$d\Phi_{m+r} = d\Phi_m(..., \tilde{p}_I, \tilde{p}_K, ...; q) d\Phi_{X_{ijk}} \tag{B.12}$$

while $\tilde{p}_I^2 = m_i^2$, $\tilde{p}_K^2 = m_k^2$ and momentum conservation is satisfied,

$$\tilde{p}_I + \tilde{p}_K = p_i + \left(\sum_{\text{unresolved } j} p_j\right) + p_k. \tag{B.13}$$

Antenna integration is computed in the same manner as in the massless regime,

$$\mathcal{X}_n^l \sim \int d\Phi_X \left(\prod_{r=1}^{l} d^d \ell_r\right) X_n^l(\{p\}, \{\ell\}) \sim \left(q^2\right)^{\lambda + \kappa\epsilon} f\left(\left\{\frac{m^2}{q^2}\right\}, \epsilon\right). \tag{B.14}$$

PaVe and IBP reduction remain applicable in the massive regime, following the same principles introduced for the massless case. For IBP reduction, however, the massive case is significantly harder analytically: the phase-space in non-trivial and has mass-dependent thresholds; the master integrals are generally multi-scale, and can involve logarithms, polylogarithms and more complicated transcendental functions; and lastly, boundary conditions are less direct, although common choices are the massless limit at $m \to 0$ and the threshold limit at $\beta \to 0$ [37].

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